

# POLS 570 Notes

## Causal Inference with Measured Confounding

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The expected value [expectation] of an R.V.: the average realization over hypothetical random draws.

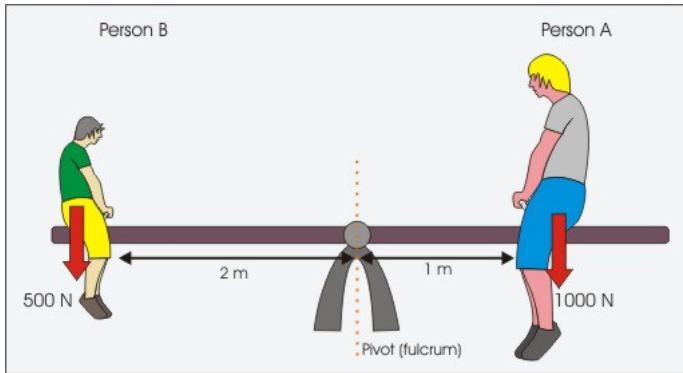
Discrete:

$$E[X] = \sum_{x \in \text{Supp}(X)} xf(x)$$

Continuous:

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx$$

What does this mean, physically?



What do we mean by random draws?

| $i$        | 1        | 2        | ...      | 625       |                                    |
|------------|----------|----------|----------|-----------|------------------------------------|
| $s$        | $Y_1$    | $Y_2$    | ...      | $Y_{625}$ | $\bar{Y}_{625}$                    |
| 1          | 1        | 1        | ...      | 0         | .68                                |
| 2          | 0        | 1        | ...      | 1         | .70                                |
| 3          | 0        | 0        | ...      | 1         | .75                                |
| 4          | 0        | 1        | ...      | 1         | .73                                |
| 5          | 0        | 1        | ...      | 1         | .69                                |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$                           |
| 1 mil      | 0        | 1        | ...      | 1         | .74                                |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$                           |
| $f(\cdot)$ | Be(.71)  | Be(.71)  |          | Be(.71)   | $\frac{\text{Bin}(625, .71)}{625}$ |
| E.(.)      | .71      | .71      |          | .71       | .71                                |

## Beyond binary

| $i$      | 1        | 2        | ...      | 625       |                 |
|----------|----------|----------|----------|-----------|-----------------|
| $s$      | $Y_1$    | $Y_2$    | ...      | $Y_{625}$ | $\bar{Y}_{625}$ |
| 1        | 35       | 50       | ...      | 22        | 45              |
| 2        | 28       | 19       | ...      | 65        | 47              |
| 3        | 33       | 27       | ...      | 45        | 42              |
| 4        | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$        |
| 5        | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$        |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$        |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$        |
| 1 mil    | 41       | 53       | ...      | 21        | 46              |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$        |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$        |
| $f.(.)$  | ?        | ?        |          | ?         | ?               |
| $E.(.)$  | 45       | 45       |          | 45        | ?               |

Important properties:

$$E[c] = c$$

$$E[aX + bY] = aE[X] + bE[Y]$$

What do these properties mean?

| $i$        | 1        | 2        | ...      | 625       |                                   |
|------------|----------|----------|----------|-----------|-----------------------------------|
| $s$        | $Y_1$    | $Y_2$    | ...      | $Y_{625}$ | $\frac{1}{2}Y_1 + \frac{1}{2}Y_2$ |
| 1          | 35       | 50       | ...      | 22        | $\frac{1}{2}35 + \frac{1}{2}50$   |
| 2          | 28       | 19       | ...      | 65        | $\frac{1}{2}28 + \frac{1}{2}19$   |
| 3          | 33       | 27       | ...      | 45        | $\frac{1}{2}33 + \frac{1}{2}27$   |
| 4          | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$                          |
| 5          | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$                          |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$                          |
| 1 mil      | 41       | 53       | ...      | 21        | $\frac{1}{2}41 + \frac{1}{2}53$   |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$                          |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$                          |
| $f(\cdot)$ | ?        | ?        |          | ?         | ?                                 |
| $E(\cdot)$ | 45       | 45       |          | 45        | $\frac{1}{2}45 + \frac{1}{2}45$   |

Higher moments about the mean. The  $j$ th *central* moment:

$$\mu_j = \text{E} \left[ (X - \text{E}[X])^j \right] .$$

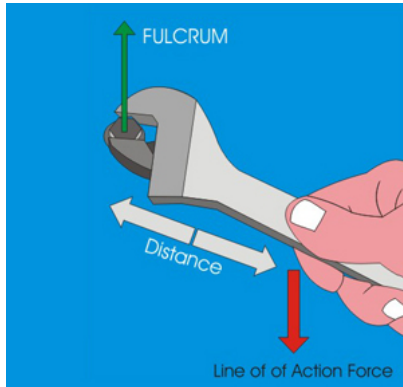


The 2nd *central* moment, the variance:

$$\begin{aligned} V(X) &= E \left[ (X - E[X])^2 \right] \\ &= E \left[ X^2 - 2XE[X] + E[X]^2 \right] \\ &= E \left[ X^2 \right] - 2E[XE[X]] + E \left[ E[X]^2 \right] \\ &= E \left[ X^2 \right] - 2E[X]E[X] + E[X]^2 \\ &= E \left[ X^2 \right] - 2E[X]^2 + E[X]^2 \\ &= E \left[ X^2 \right] - E[X]^2 \end{aligned}$$

- ▶ The average squared deviation from the expected value.
- ▶ Expectation of the square minus the square of the expectation.

What does this mean, physically?



What do we mean by random draws?

| $i$        | 1          | 2          | ...      | 625        |                             |
|------------|------------|------------|----------|------------|-----------------------------|
| $s$        | $Y_1$      | $Y_2$      | ...      | $Y_{625}$  | $\bar{Y}_{625}$             |
| 1          | 1          | 1          | ...      | 0          | .68                         |
| 2          | 0          | 1          | ...      | 1          | .70                         |
| 3          | 0          | 0          | ...      | 1          | .75                         |
| 4          | 0          | 1          | ...      | 1          | .73                         |
| 5          | 0          | 1          | ...      | 1          | .69                         |
| $\vdots$   | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$   | $\vdots$                    |
| 1 mil      | 0          | 1          | ...      | 1          | .74                         |
| $\vdots$   | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$   | $\vdots$                    |
| $f(\cdot)$ | Be(.71)    | Be(.71)    |          | Be(.71)    | $\frac{Bin(625, .71)}{625}$ |
| E.(.)      | .71        | .71        |          | .71        | .71                         |
| V.(.)      | .71(1-.71) | .71(1-.71) |          | .71(1-.71) | ?                           |

## Beyond binary

| $i$        | 1        | 2        | ...      | 625       |                 |
|------------|----------|----------|----------|-----------|-----------------|
| $s$        | $Y_1$    | $Y_2$    | ...      | $Y_{625}$ | $\bar{Y}_{625}$ |
| 1          | 35       | 50       | ...      | 22        | 45              |
| 2          | 28       | 19       | ...      | 65        | 47              |
| 3          | 33       | 27       | ...      | 45        | 42              |
| 4          | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$        |
| 5          | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$        |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$        |
| 1 mil      | 41       | 53       | ...      | 21        | 46              |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$        |
| $f(\cdot)$ | ?        | ?        |          | ?         | ?               |
| $V(\cdot)$ | 25       | 25       |          | 25        | ?               |

Important properties:

$$V[c] = 0$$

$$V[aX] = a^2 V[X]$$

$$V[c + aX] = a^2 V[X]$$

What do these properties mean?

| $i$        | 1                | 2                | ...      | 625              |                                    |
|------------|------------------|------------------|----------|------------------|------------------------------------|
| $s$        | $Y_1$            | $Y_2$            | ...      | $Y_{625}$        | $\bar{Y}_{625}$                    |
| 1          | 1                | 1                | ...      | 0                | .68                                |
| 2          | 0                | 1                | ...      | 1                | .70                                |
| 3          | 0                | 0                | ...      | 1                | .75                                |
| 4          | 0                | 1                | ...      | 1                | .73                                |
| 5          | 0                | 1                | ...      | 1                | .69                                |
| $\vdots$   | $\vdots$         | $\vdots$         | $\vdots$ | $\vdots$         | $\vdots$                           |
| 1 mil      | 0                | 1                | ...      | 1                | .74                                |
| $\vdots$   | $\vdots$         | $\vdots$         | $\vdots$ | $\vdots$         | $\vdots$                           |
| $\vdots$   | $\vdots$         | $\vdots$         | $\vdots$ | $\vdots$         | $\vdots$                           |
| $f(\cdot)$ | $\text{Be}(.71)$ | $\text{Be}(.71)$ |          | $\text{Be}(.71)$ | $\frac{\text{Bin}(625, .71)}{625}$ |
| $E(\cdot)$ | .71              | .71              |          | .71              | .71                                |
| $V(\cdot)$ | $.71(1-.71)$     | $.71(1-.71)$     |          | $.71(1-.71)$     | $\frac{625(.71)(1-.71)}{625^2}$    |

Definition:

Standard deviation:  $\sigma(X) = \sqrt{V(X)}$ .

Definition: mean squared error (MSE) about  $c$ :

$$E \left[ (X - c)^2 \right] = V(X) + (c - E[X])^2$$

What is the minimum MSE choice of  $c$ ?



Definition: mean absolute error (MAE) about  $c$ :

$$E [|X - c|]$$

What is the minimum MAE choice of  $c$ ?

Hint: Remember your Hotelling/Downs.

Defining an expectation over multiple R.V.s.

Let  $Z = h(X, Y)$  be a scalar function of the random vector  $(X, Y)$  as we proceed.

Discrete:

$$E[Z] = \sum_{x \in \text{Supp}(X)} \sum_{y \in \text{Supp}(Y)} h(x, y) f(x, y)$$

Continuous:

$$E[Z] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f(x, y) dy dx$$

So this is just a straightforward generalization (working with joint PMFs/PDFs).

Generalization of variance: covariance.

$$C[X, Y] = E [(X - E[X]) (Y - E[Y])] = E[XY] - E[X]E[Y].$$

Note:

$$C[X, X] = E[X^2] - E[X]^2 = V[X]$$

$$C[a, X] = 0$$

## Covariance rule

$$C[aX + bY, cU + dV] = \\ acC[X, U] + adC[X, V] + bcC[Y, U] + bdC[Y, V]$$

Variance rule in the multivariate case.

$$V[aX + bY] = a^2 V[X] + b^2 V[Y] + 2abC[X, Y].$$

Derive this from the previous rules.

What does this mean for  $V[X - Y]$ ?

Definition: correlation.

$$\rho[X, Y] = \frac{C[X, Y]}{\sqrt{V[X]}\sqrt{V[Y]}}$$

Correlation measures linear dependence and is bounded in  $[-1, 1]$ . Importantly, for positive  $b, d$ :

$$\rho[a + bX, c + dX] = 1$$

$$\rho[a + bX, c - dX] = -1.$$

Conditional PMF/PDF transforms a multivariate PMF/PDF

$$f(x, y)$$

to a univariate PMF/PDF

$$f_{Y|X}(y|x)$$

by conditioning on a value of  $X = x$ .

Discrete:

$$E[Z|x] = \sum_{y \in \text{Supp}(Y)} h(x, y) f_{Y|X}(y|x)$$

Continuous:

$$E[Z|x] = \int_{-\infty}^{\infty} h(x, y) f_{Y|X}(y|x) dy$$



Everything else carries through, just as though you were operating on a univariate PMF/PDF

E.g.,

$$V[Z|x] = E[h(x, y)^2|x] - E[h(x, y)|x]^2.$$

Some other rules. Moving from conditional expectations to unconditional expectations.

The *law of iterated expectations*:

$$E[Z] = E_X[E[Z|X]],$$

where  $E_X$  refers to the expectation over  $X$ .

Proof sketch (with example on board), discrete:

$$\begin{aligned} E[Z] &= \sum_{x \in \text{Supp}(X)} \sum_{y \in \text{Supp}(Y)} h(x, y) f(x, y) \\ &= \sum_{x \in \text{Supp}(X)} \sum_{y \in \text{Supp}(Y)} h(x, y) f_X(x) f_{Y|X}(y|x) \\ &= \sum_{x \in \text{Supp}(X)} \left[ \sum_{y \in \text{Supp}(Y)} h(x, y) f_{Y|X}(y|x) \right] f_X(x) \\ &= \sum_{x \in \text{Supp}(X)} E[Z|x] f_X(x) \\ &= E_X[E[Z|x]]. \end{aligned}$$

Law of iterated expectations directly yields *law of total variance*:

$$V[Y] = E_X [V[Y|X]] + V_X [E[Y|X]] .$$

Practically speaking, we can decompose the variability of a R.V.  $Y$  into the average variability “within” covariate values of  $X$  and the average variability “across” covariate values of  $X$ .

The conditional expectation function and the best [linear] predictor thereof.

For the random vector  $(X, Y)$ , the conditional expectation function (CEF) is a function that takes as an input  $x$  and returns the conditional expectation of  $Y$  given  $x$ .

Discrete:

$$E[Y|x] = \sum_{y \in \text{Supp}(Y)} y f_{Y|X}(y|x)$$

Continuous:

$$E[Y|x] = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy$$

## Properties of deviations from the CEF.

Define the R.V.  $\epsilon = Y - E[Y|X]$ .

- ▶  $E[\epsilon] = 0$
- ▶  $C(h(X), \epsilon) = 0$
- ▶  $V[\epsilon] = E[V[Y|X]]$ .

The CEF is the best (MMSE) predictor of  $Y$  given  $X$ .



I.e., suppose you knew the full joint CDF of  $Y$  and  $X$ . And then someone gave you a randomly drawn value of  $X$ . What would be our guess of  $Y$  that would have the lowest MSE?

The answer is given by the CEF. The function  $g(X)$  that best approximates  $Y$  is the CEF.

But what if we were to restrict ourselves *only* to linear functions? I.e., let  $g(X) = a + bX$ .

By choosing the  $a$  and  $b$  that minimize MSE, we are using the best *linear* predictor (BLP) of  $Y$  given  $X$ .

The BLP, among all functions  $g(X) = a + bX$  is given by the constant

$$a = E[Y] - C[X, Y]/V[X]E[X]$$

and the slope

$$b = C[X, Y]/V[X].$$

Important corollaries:

- ▶ The BLP is also the best linear approximation to the CEF.  
(i.e., the optimal  $a$  and  $b$  minimizes  
$$E \left[ [E[Y|X] - (a + bX)]^2 \right].$$
- ▶ If the CEF *is* linear, the CEF is the BLP.

Visualizing the BLP.

[See whiteboard.]

Chebyshev's inequality. (Bounds how much of the probability mass can be more than a few standard deviations away from the expected value, even if we don't know anything else about the R.V.)

Given R.V.  $X$  with finite  $E[X]$  and finite  $V(X) > 0$ . Then for any  $k > 0$ ,

$$\Pr\left(|X - E[X]| \geq k\sqrt{V(X)}\right) \leq \frac{1}{k^2}$$

or, equivalently,

$$\Pr(|X - E[X]| \geq k) \leq \frac{V(X)}{k^2}.$$

Simple proof in Goldberger (1991), p. 31, via Markov's inequality.

Independence of  $X$  and  $Y$ .

$X$  and  $Y$  are *independent* if and only if  $f(x, y) = f_X(x)f_Y(y)$ , for all  $(x, y)$ .

Implications of independence. If  $X$  and  $Y$  are independent, then:

- ▶ Events are independent:

$$\Pr(A \cap B) = \Pr(A) \Pr(B),$$

for an event  $A$  defined in terms of  $X$  only and  $B$  defined in terms of  $Y$  only.

- ▶ Covariance is zero:

$$C(X, Y) = 0.$$

- ▶ Variances are additive:

$$V(X + Y) = V(X) + V(Y).$$



What can we learn from a random sample?

Suppose we have an *independent and identically distributed* (i.i.d) random sample of  $n$  observations all drawn from some R.V.  $Y$ .

I.e., we draw from the R.V.  $Y$ . Then we draw from  $Y$  again, in such a manner that our 2nd draw does not depend on the first draw. And then repeat this process until we have  $n$  draws.

Note that if we take a random sample from a large population, and we randomly assign half of the units to treatment and half to control, we will approximately have two i.i.d. samples.

$$(Y_1(1), Y_2(1), \dots, Y_{n/2}(1)) \sim_{i.i.d.} F_{Y(1)}$$
$$(Y_{n/2+1}(0), Y_{n/2+2}(0), \dots, Y_n(0)) \sim_{i.i.d.} F_{Y(0)}$$

For the rest of this, when you see  $Y$ , you can also think  $Y(1)$ .

So, we have  $n$  random variables that generate  $n$  values.

We can subscript each of these  $n$  random variables, as well as aggregate them.

The random vector  $\mathbf{Y} = (Y_1, \dots, Y_n)'$ .

Under the assumption of *random sampling* from  $F_Y$ , the collection of values that you see are each produced by an identical random process.

The actual values that we observe are denoted with  $\mathbf{y} = (y_1, \dots, y_n)'$ .

A sample statistic *summarizes* those  $n$  values. I'm going to focus on the sample mean,

$$\bar{Y} = \frac{\sum_{i=1}^n Y_i}{n} = \frac{1}{n} (Y_1 + \dots Y_n).$$

## Sampling distribution of the sample mean

| $i$<br>$s$  | 1<br>$Y_1$ | 2<br>$Y_2$ | ...      | 625<br>$Y_{625}$ | $\bar{Y}_{625}$ |
|-------------|------------|------------|----------|------------------|-----------------|
| 1           | 35         | 50         | ...      | 22               | 45              |
| 2           | 28         | 19         | ...      | 65               | 47              |
| 3           | 33         | 27         | ...      | 45               | 42              |
| 4           | $\vdots$   | $\vdots$   | ...      | $\vdots$         | $\vdots$        |
| 5           | $\vdots$   | $\vdots$   | ...      | $\vdots$         | $\vdots$        |
| $\vdots$    | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$         | $\vdots$        |
| 1 mil       | 41         | 53         | ...      | 21               | 46              |
| $\vdots$    | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$         | $\vdots$        |
| $\vdots$    | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$         | $\vdots$        |
| $f.(\cdot)$ | ?          | ?          |          | ?                | ?               |
| $E.(\cdot)$ | 45         | 45         |          | 45               | ?               |
| $V.(\cdot)$ | 25         | 25         |          | 25               | ?               |

Property #1: *The sample mean is unbiased for the population mean.* The expected value of the sample mean,  $E[\bar{Y}] = E[Y]$ .

Proof:

$$\begin{aligned} E[\bar{Y}] &= E\left[\frac{1}{n}(Y_1 + \dots + Y_n)\right] \\ &= \\ &= \\ &= \\ &= E[Y]. \end{aligned}$$



## Sampling distribution of the sample mean

| $i$<br>$s$ | 1<br>$Y_1$ | 2<br>$Y_2$ | ... | 625<br>$Y_{625}$ | $\bar{Y}_{625}$ |
|------------|------------|------------|-----|------------------|-----------------|
| 1          | 35         | 50         | ... | 22               | 45              |
| 2          | 28         | 19         | ... | 65               | 47              |
| 3          | 33         | 27         | ... | 45               | 42              |
| 4          | :          | :          | ... | :                | :               |
| 5          | :          | :          | ... | :                | :               |
| :          | :          | :          | :   | :                | :               |
| :          | :          | :          | :   | :                | :               |
| 1 mil      | 41         | 53         | ... | 21               | 46              |
| :          | :          | :          | :   | :                | :               |
| :          | :          | :          | :   | :                | :               |
| $f.(.)$    | ?          | ?          |     | ?                | ?               |
| $E.(.)$    | 45         | 45         |     | 45               | 45              |
| $V.(.)$    | 25         | 25         |     | 25               | ?               |

Property #2: The *sampling variance* of the sample mean,

$$V[\bar{Y}] = \frac{V[Y]}{n}.$$

Proof:

$$\begin{aligned} V[\bar{Y}] &= V \left[ \left( \frac{1}{n} Y_1 + \dots + \frac{1}{n} Y_n \right) \right] \\ &= \\ &= \\ &= \\ &= \\ &= \frac{1}{n} V[Y]. \end{aligned}$$

## Sampling distribution of the sample mean

| $i$        | 1        | 2        | ...      | 625       |                  |
|------------|----------|----------|----------|-----------|------------------|
| $s$        | $Y_1$    | $Y_2$    | ...      | $Y_{625}$ | $\bar{Y}_{625}$  |
| 1          | 35       | 50       | ...      | 22        | 45               |
| 2          | 28       | 19       | ...      | 65        | 47               |
| 3          | 33       | 27       | ...      | 45        | 42               |
| 4          | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$         |
| 5          | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$         |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$         |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$         |
| 1 mil      | 41       | 53       | ...      | 21        | 46               |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$         |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$         |
| $f(\cdot)$ | ?        | ?        |          | ?         | ?                |
| $E(\cdot)$ | 45       | 45       |          | 45        | 45               |
| $V(\cdot)$ | 25       | 25       |          | 25        | $\frac{25}{625}$ |

Property # 3: Given finite  $E[Y]$  and finite  $V(Y) > 0$ , then for any  $k > 0$ ,

$$\Pr(|\bar{Y} - E[Y]| \geq k) \leq \frac{V[Y]}{k^2 n}.$$

Proof follows directly from Chebyshev's inequality.

Property # 4. [Stated loosely.] *The sample mean is consistent for the population mean.* Given finite  $E[Y]$  and finite  $V(Y) > 0$ , then as  $n \rightarrow \infty$ ,

$$\Pr(|\bar{Y} - E[Y]| \geq k) \rightarrow 0.$$

Or, equivalently,

$$\bar{Y} \xrightarrow{p} E[Y],$$

where  $\xrightarrow{p}$  is read as “convergence in probability.” This is a form of the *weak law of large numbers* (WLLN).

So random samples tell us a lot about means, but suppose we wanted to estimate the entire distribution.

Remember the CDF is defined in terms of

$$\Pr(Y \leq c).$$

for all values of  $c$ .

Define a random variable  $Z = \mathbb{I}[Y \leq c]$ , where  $\mathbb{I}[\cdot]$  is the indicator function (takes the value 1 if true, 0 if false). Then

$$E[Z] = \Pr(Y \leq c).$$

Proof.

$$\Pr(Z = 1) = \Pr(Y \leq c),$$

Then, by the definition of the expected value,

$$E[Z] = \Pr(Y \leq c) \times 1 + \Pr(Y > c) \times 0 = \Pr(Y \leq c).$$

Hence, learning about means can tell us about the whole distribution.

By property #4 then,

$$\overline{Z} \xrightarrow{p} \Pr(Y \leq c).$$



We could do this for any or every value of  $c$ . Which is to say, we can always approximate the CDF to arbitrary precision. If  $n$  is very large, the probability that we're wrong for a given point of the CDF by much is very small.

Stronger results are possible under weaker assumptions.  
Strong LLN and Glivenko-Cantelli.

This is why so-called identification results work directly off the probability distribution of  $Y$ . Even if we don't *know*  $Y$ , we can learn everything about  $Y$  (with diminishing uncertainty) just by drawing repeated values from  $Y$ .

## Sampling distribution of the sample mean

| $i$     | 1               | 2               | ... | 625                     |                  |                      |
|---------|-----------------|-----------------|-----|-------------------------|------------------|----------------------|
| $s$     | $Y_1, Z_1^{40}$ | $Y_2, Z_2^{40}$ | ... | $Y_{625}, Z_{625}^{40}$ | $\bar{Y}_{625}$  | $\bar{Z}_{625}^{40}$ |
| 1       | 35,1            | 50,0            | ... | 22,1                    | 45               | .45                  |
| 2       | 28,1            | 19,1            | ... | 65,0                    | 47               | .38                  |
| 3       | 33,1            | 27,1            | ... | 45,0                    | 42               | .56                  |
| 4       | :               | :               | ... | :                       | :                |                      |
| 5       | :               | :               | ... | :                       | :                |                      |
| :       | :               | :               | :   | :                       | :                |                      |
| 1 mil   | 41,0            | 53,0            | ... | 21,1                    | 46               | .41                  |
| :       | :               | :               | :   | :                       | :                |                      |
| :       | :               | :               | :   | :                       | :                |                      |
| $f.(.)$ | ?               | ?               |     | ?                       | ?                |                      |
| $E.(.)$ | 45              | 45              |     | 45                      | 45               | $E[Z^{40}]$          |
| $V.(.)$ | 25              | 25              |     | 25                      | $\frac{25}{625}$ |                      |

## Sampling distribution of the sample mean

| $i$        | 1                         | 2                         | ... | 625                                   |                  |                      |                      |
|------------|---------------------------|---------------------------|-----|---------------------------------------|------------------|----------------------|----------------------|
| $s$        | $Y_1, Z_1^{40}, Z_1^{41}$ | $Y_2, Z_2^{40}, Z_2^{41}$ | ... | $Y_{625}, Z_{625}^{40}, Z_{625}^{41}$ | $\bar{Y}_{625}$  | $\bar{Z}_{625}^{40}$ | $\bar{Z}_{625}^{41}$ |
| 1          | 35,1,1                    | 50,0,0                    | ... | 22,1,1                                | 45               | .45                  | .46                  |
| 2          | 28,1,1                    | 19,1,1                    | ... | 65,0,0                                | 47               | .38                  | .38                  |
| 3          | 33,1,1                    | 27,1,1                    | ... | 45,0,0                                | 42               | .56                  | .56                  |
| 4          | :                         | :                         | ... | :                                     | :                |                      |                      |
| 5          | :                         | :                         | ... | :                                     | :                |                      |                      |
| :          | :                         | :                         | :   | :                                     | :                |                      |                      |
| 1 mil      | 41,0,1                    | 53,0,0                    | ... | 21,1,1                                | 46               | .41                  | .42                  |
| :          | :                         | :                         | :   | :                                     | :                |                      |                      |
| :          | :                         | :                         | :   | :                                     | :                |                      |                      |
| $f(\cdot)$ | ?                         | ?                         |     | ?                                     | ?                |                      |                      |
| $E(\cdot)$ | 45                        | 45                        |     | 45                                    | 45               | $E[Z^{40}]$          | $E[Z^{41}]$          |
| $V(\cdot)$ | 25                        | 25                        |     | 25                                    | $\frac{25}{625}$ |                      |                      |

So we can estimate the CDF with large samples, but we want to know how accurate we are for our sample with fixed size  $n$ .

- ▶ need to estimate variance
- ▶ use estimated variance for confidence intervals

(A consequence of the) Continuous Mapping Theorem, stated loosely. Continuous functions are limit-preserving.

For continuous  $f(\cdot)$ , then if  $Y \xrightarrow{P} c$ , then  $f(Y) \xrightarrow{P} f(c)$ .

Many Variances: population, sample and sampling.

Recall that the “population” variance of a random variable  $Y$ :

$$V(Y) = E \left[ Y^2 \right] - E[Y]^2$$



Can we propose an estimator of  $V(Y)$ ?

Put another way:

- ▶ Can we propose an estimator of  $E[Y^2]$ ?
- ▶ Can we propose an estimator of  $E[Y]^2$ ?

- ▶ Can we propose an estimator of  $E[Y^2]$ ?
  - ▶ Plug-in estimator:

$$\overline{Y^2} = \frac{\sum_{i=1}^n Y_i^2}{n}.$$

- ▶  $E[\overline{Y^2}] = E[Y^2]$ .
- ▶  $\overline{Y^2} \xrightarrow{p} E[Y^2]$ .

- ▶ Can we propose an estimator of  $E[Y]^2$ ?
  - ▶ Plug-in estimator:

$$\bar{Y}^2 = \left[ \frac{\sum_{i=1}^n Y_i}{n} \right]^2.$$

- ▶  $E[\bar{Y}^2] = E[Y]^2 + \frac{V[Y]}{n}$
- ▶  $\bar{Y}^2 \xrightarrow{p} E[Y]^2$  (follows from CMT)

Combining terms: a plug-in estimator of  $V(Y)$ , the plug-in *sample variance*

$$\overline{Y^2} - \overline{Y}^2.$$

with

- ▶  $E[\overline{Y^2} - \overline{Y}^2] = V(Y) - \frac{V[Y]}{n} = (1 - \frac{1}{n})V(Y) = \frac{n-1}{n} V(Y)$
- ▶  $\overline{Y^2} - \overline{Y}^2 \xrightarrow{p} V(Y).$

Combining terms: a plug-in estimator of  $V(Y)$ , the *unbiased sample variance*

$$\hat{V}(Y) = \frac{n}{n-1} \left[ \overline{Y^2} - \bar{Y}^2 \right].$$

with

- ▶  $E[\hat{V}(Y)] = V(Y)$
- ▶  $\hat{V}(Y) \xrightarrow{P} V(Y)$ .

This is what R computes when you type `var( . )`.

## Sampling distribution of the sample variance

| $i$<br>$s$ | 1<br>$Y_1$ | 2<br>$Y_2$ | ...      | 625<br>$Y_{625}$ | $\bar{Y}_{625}$  | $\hat{V}(Y)$ |
|------------|------------|------------|----------|------------------|------------------|--------------|
| 1          | 35         | 50         | ...      | 22               | 45               | 20           |
| 2          | 28         | 19         | ...      | 65               | 47               | 17           |
| 3          | 33         | 27         | ...      | 45               | 42               | 38           |
| 4          | $\vdots$   | $\vdots$   | ...      | $\vdots$         | $\vdots$         |              |
| 5          | $\vdots$   | $\vdots$   | ...      | $\vdots$         | $\vdots$         |              |
| $\vdots$   | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$         | $\vdots$         |              |
| 1 mil      | 41         | 53         | ...      | 21               | 46               | 32           |
| $\vdots$   | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$         | $\vdots$         |              |
| $f(\cdot)$ | ?          | ?          |          | ?                | ?                |              |
| $E(\cdot)$ | 45         | 45         |          | 45               | 45               | 25           |
| $V(\cdot)$ | 25         | 25         |          | 25               | $\frac{25}{625}$ |              |

So we have:

- ▶ Population variance,  $V(Y)$
- ▶ (Unbiased) sample variance,  $\hat{V}(Y)$
- ▶ Sampling variance of the sample mean,  $V(\bar{Y}) = V(Y)/n$

What about an estimator of the sampling variance of the sample mean?



Another plug in estimator.

We want to estimate  $V(Y)/n$ . Why not use  $\hat{V}(Y)/n$ ?

This is an *unbiased estimator* of the sampling variance of the sample mean.

Important definition. The *standard error* of the sample mean:

$$\sqrt{V(\bar{Y})}.$$

(This is literally the standard deviation of the sampling distribution of the sample mean.)

An *estimate* of the standard error of the sample mean:

$$\sqrt{\hat{V}(\bar{Y})}.$$

Not unbiased (due to Jensen's inequality), though it is "consistent."

When you see "standard error" in a paper, this is what they mean (an estimate thereof). In theory, it is possible to estimate the sampling variance of the standard error of the sample mean, though.

It is therefore possible to estimate the average of a random variable (the population mean) and estimate our uncertainty (using an estimated SE).

However, we may want to know how likely it would be (across different random samples) for an interval of  $\bar{Y} \pm 2 \text{ SEs}$  to contain  $E[Y]$ .

The central limit theorem provides an approach.

The central limit theorem (CLT) for the sample mean.

In words: if  $n$  is large, the sampling distribution of the sample mean ( $\bar{Y}$ ) is approximately normal, regardless of the distribution of  $Y$ .

In sampling  $n$  times from any R.V.  $Y$  with finite  $E[Y] = \mu$  and finite  $V(Y) = \sigma^2$ , then the standardized sample mean

$$Z = \sqrt{n}(\bar{Y} - \mu)/\sigma$$

converges in distribution as  $n \rightarrow \infty$  to

$$\mathcal{N}(0, 1),$$

where  $\mathcal{N}$  denotes the normal distribution.

I am not going to offer a proof here. The simplest “almost-proof” is given by the de Moivre–Laplace theorem (see the Wikipedia page).

A confidence interval for  $E[Y]$ , loosely speaking, is an *interval estimate* that covers the true  $E[Y]$  with a given probability.

I.e., Suppose we were to repeatedly draw from  $F_{\overline{Y}}$  and compute a (valid) 95% confidence interval for  $E[Y]$ . 95% percent of the time, this interval will encompass  $E[Y]$ .

Suppose we have:

- ▶ An asymptotically normal estimator of  $E[Y]$ :  $\bar{Y}$
- ▶ A consistent estimator of  $nV(\bar{Y})$ :  $n\hat{V}(\bar{Y})$

An approximate 95% confidence interval for  $E[Y]$  would be:

$$\left( \bar{Y} - 1.96\sqrt{\hat{V}(\bar{Y})}, \bar{Y} + 1.96\sqrt{\hat{V}(\bar{Y})} \right).$$



An approximate 95% confidence interval for  $E[Y]$  would be:

$$\left( \bar{Y} - 1.96\sqrt{\frac{\hat{V}(Y)}{n}}, \bar{Y} + 1.96\sqrt{\frac{\hat{V}(Y)}{n}} \right).$$

A loose proof for the asymptotic validity of confidence intervals under a normal approximation.

Suppose  $n$  is large.

Define the R.V.

$$Z = \frac{\bar{Y} - E[Y]}{\sqrt{V(\bar{Y})}},$$

where  $Z$  is distributed  $\mathcal{N}(0, 1)$  (by asymptotic normality).

Plugging in the *estimated* standard error.

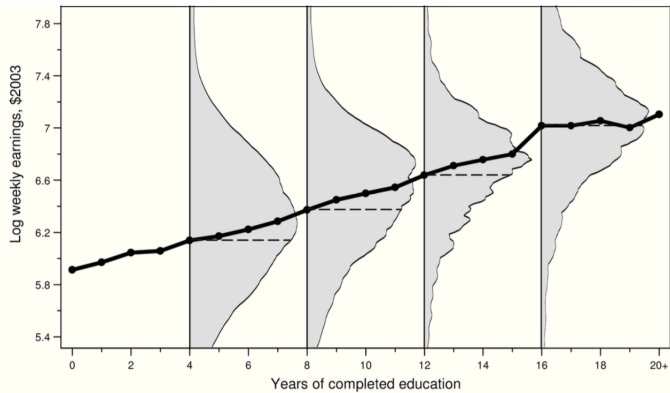
Define the R.V.

$$Z' = \frac{\bar{Y} - E[Y]}{\sqrt{V(\bar{Y})}} \frac{\sqrt{nV(\bar{Y})}}{\sqrt{n\hat{V}(\bar{Y})}} = \frac{\bar{Y} - E[Y]}{\sqrt{\hat{V}(\bar{Y})}}$$

where  $Z'$  is distributed  $\mathcal{N}(0, 1)$  (by asymptotic normality, the consistency of variance estimator, and Slutsky's Theorem).

As we discussed earlier, the entire distribution of  $Y$  can be estimated with means of different types, so this result applies to many things. Next week we will see that something analogous is true for multivariate distributions and conditional expectation functions.

Linear regression can be seen a way of estimating the BLP using the data in your sample.



The BLP of  $Y$  given  $X$  is  $g(X) = a + bX$ , where

$$a = E[Y] - \frac{C[X, Y]}{V[X]}E[X]$$

and the slope

$$b = \frac{C[X, Y]}{V[X]}.$$



The BLP of  $Y$  given  $X$  is  $g(X) = a + bX$ , where

$$a = E[Y] - \frac{E[XY] - E[X]E[Y]}{E[X^2] - E[X]^2}E[X]$$

and the slope

$$b = \frac{E[XY] - E[X]E[Y]}{E[X^2] - E[X]^2}$$

A natural *estimator* of the BLP:

$$\hat{a} = \bar{Y} - \frac{\overline{XY} - \bar{X} \cdot \bar{Y}}{\overline{X^2} - \bar{X}^2} \bar{X}$$

and the slope

$$\hat{b} = \frac{\overline{XY} - \bar{X} \cdot \bar{Y}}{\overline{X^2} - \bar{X}^2}.$$

You will show in the problem set that  $\hat{b} \rightarrow_p b$ . Using this fact, show that

$$\hat{a} = \bar{Y} - \hat{b}\bar{X}$$

converges in probability to

$$a = E[Y] - bE[X]$$

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Note the assumptions that convergence required.

1. Random sampling allows descriptive inference to a larger population.

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2. Randomized treatment assignment allows causal inference to a counterfactual population.

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Important to Note:

- ▶ Many econometric presentations attempt both inferences simultaneously.

1. Random sampling allows descriptive inference to a larger population.
2. Randomized treatment assignment allows causal inference to a counterfactual population.

Important to Note:

- ▶ Many econometric presentations attempt both inferences simultaneously.
- ▶ We start with descriptive inference and finish with causal inference.



## Epstein and Mershon SCOTUS data

- ▶ Data on 27 justices from the Warren, Burger, and Rehnquist courts (can be interpreted as a census)

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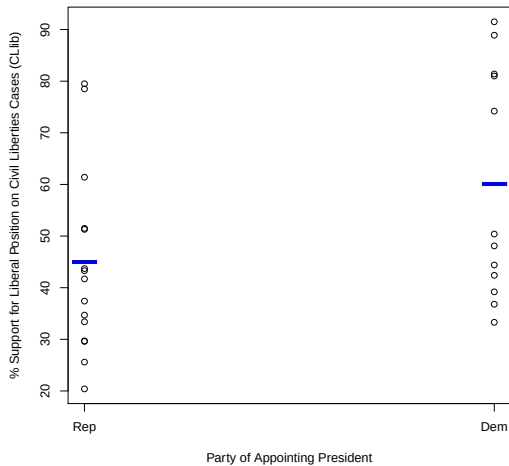
## Epstein and Mershon SCOTUS data

- ▶ Data on 27 justices from the Warren, Burger, and Rehnquist courts (can be interpreted as a census)
- ▶ Percentage of votes in liberal direction for each justice in a number of issue areas
- ▶ Segal-Cover scores for each justice

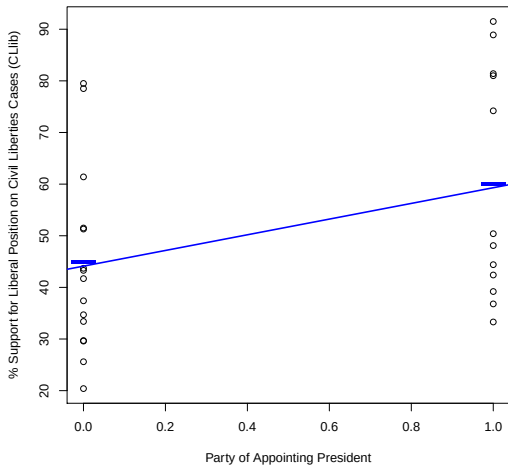
## Epstein and Mershon SCOTUS data

- ▶ Data on 27 justices from the Warren, Burger, and Rehnquist courts (can be interpreted as a census)
- ▶ Percentage of votes in liberal direction for each justice in a number of issue areas
- ▶ Segal-Cover scores for each justice
- ▶ Party of appointing president

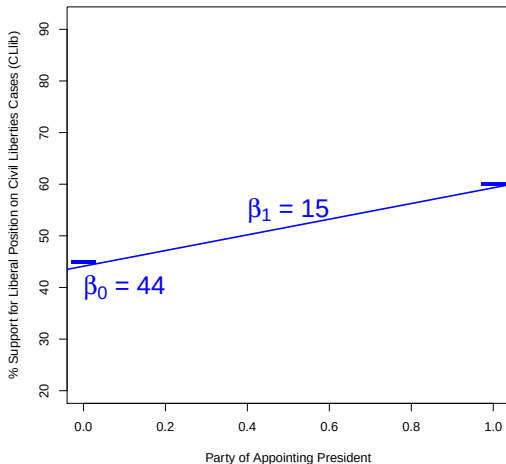
## CLlib Conditional on Party



## Linear Conditional Mean Function

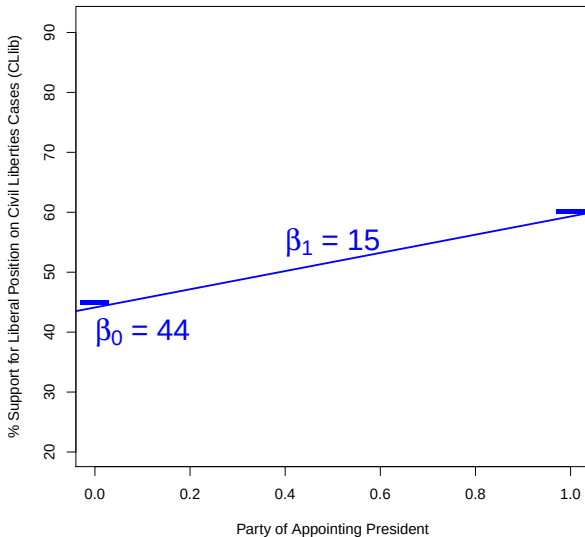


## Linear Conditional Mean Function



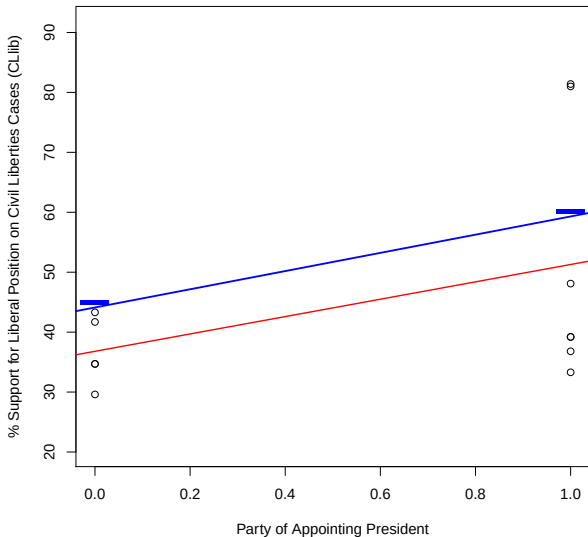
How to interpret the intercept and slope descriptively?

# Linear Conditional Expectation Function with Samples

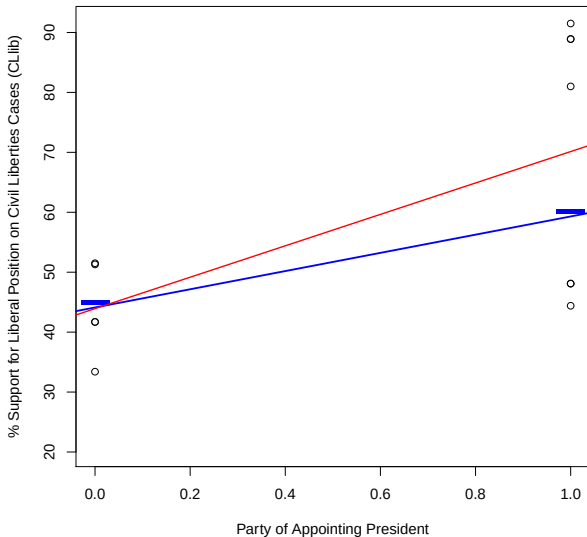




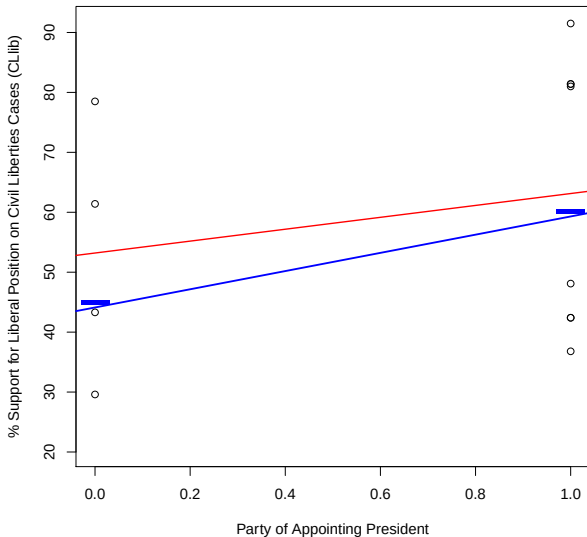
# Linear Conditional Expectation Function with Samples



# Linear Conditional Expectation Function with Samples



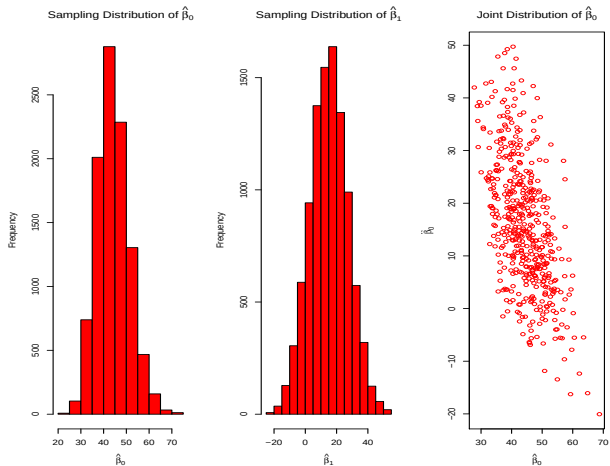
# Linear Conditional Expectation Function with Samples



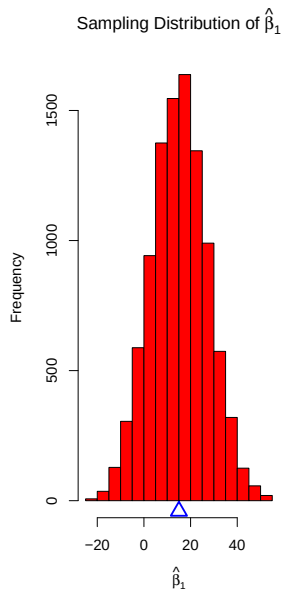
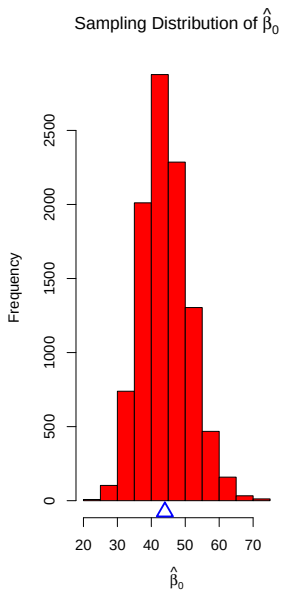
## Sampling distribution of the sample slope

| $i$      | 1          | 2          | ...      | 12               |                 |
|----------|------------|------------|----------|------------------|-----------------|
| $s$      | $Y_1, X_1$ | $Y_2, X_2$ | ...      | $Y_{12}, X_{12}$ | $\hat{\beta}_1$ |
| 1        | 35,1       | 50,0       | ...      | 22,1             | 14.49           |
| 2        | 28,1       | 19,1       | ...      | 65,0             | 26.21           |
| 3        | 33,1       | 27,1       | ...      | 45,0             | 9.93            |
| 4        | $\vdots$   | $\vdots$   | ...      | $\vdots$         | $\vdots$        |
| 5        | $\vdots$   | $\vdots$   | ...      | $\vdots$         | $\vdots$        |
| $\vdots$ | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$         | $\vdots$        |
| $\vdots$ | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$         | $\vdots$        |
| E.(.)    | 50.9, 0.44 | 50.9, 0.44 | ...      | 50.9, 0.44       | 15              |

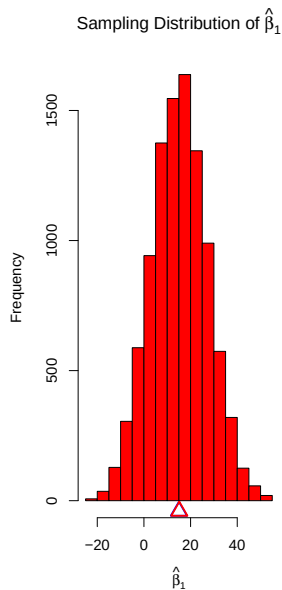
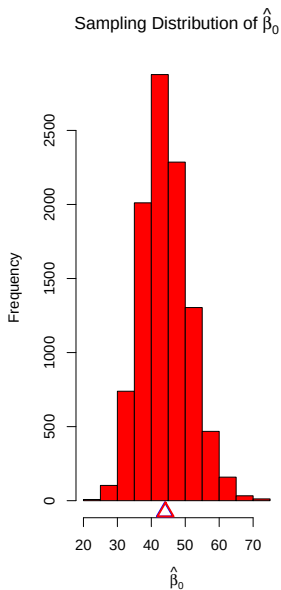
# Sampling Distributions for $\hat{\beta}_0$ and $\hat{\beta}_1$



# Bias



# Bias



# Assumptions for Unbiasedness

What assumptions did we need for  $E[\hat{\beta}_0] = \beta_0$  and  $E[\hat{\beta}_1] = \beta_1$ ?

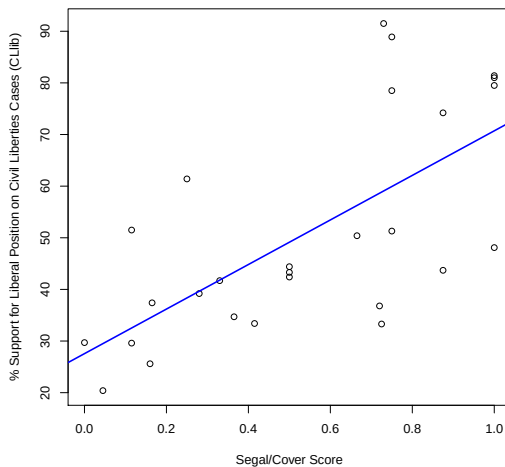


# Assumptions for Unbiasedness

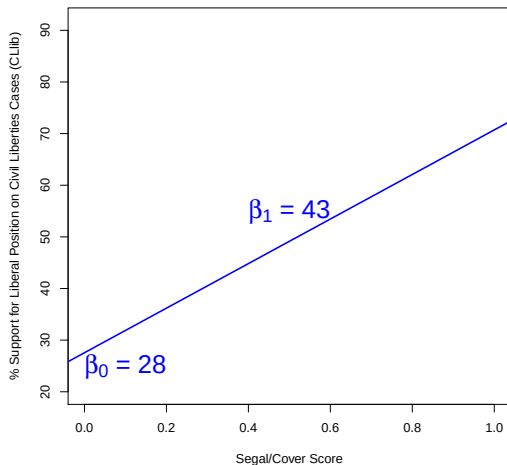
What assumptions did we need for  $E[\hat{\beta}_0] = \beta_0$  and  $E[\hat{\beta}_1] = \beta_1$ ?

Just random sampling!

# BLP for CLlib Conditional on SCscore

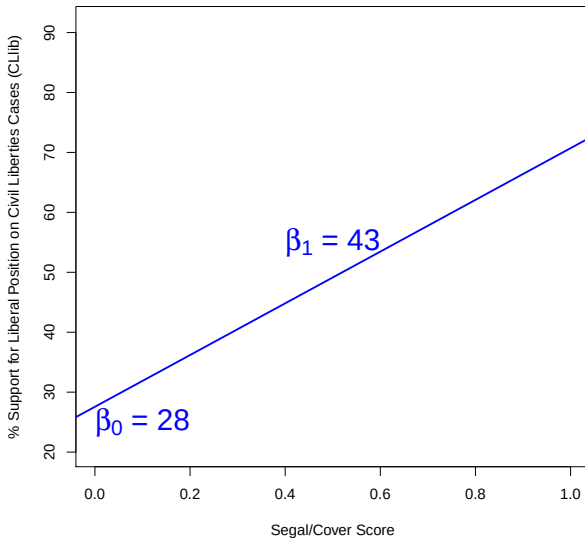


# BLP for CLlib Conditional on SCscore

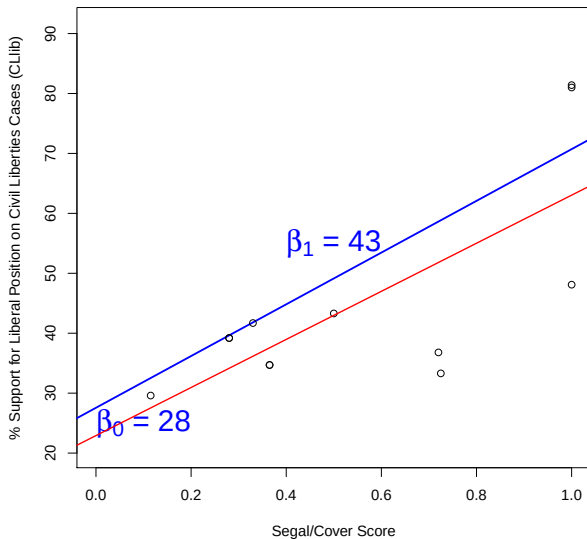


How to interpret the intercept and slope descriptively?

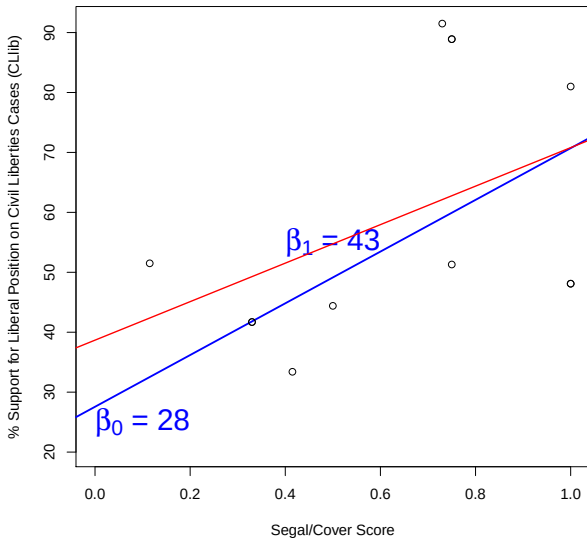
# Linear Conditional Expectation Function with Samples



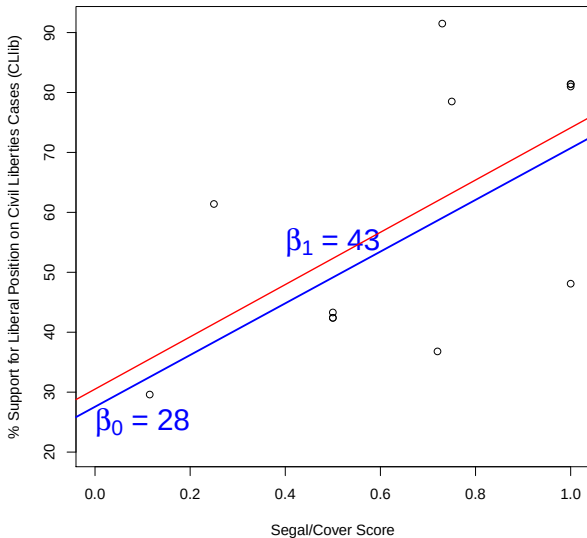
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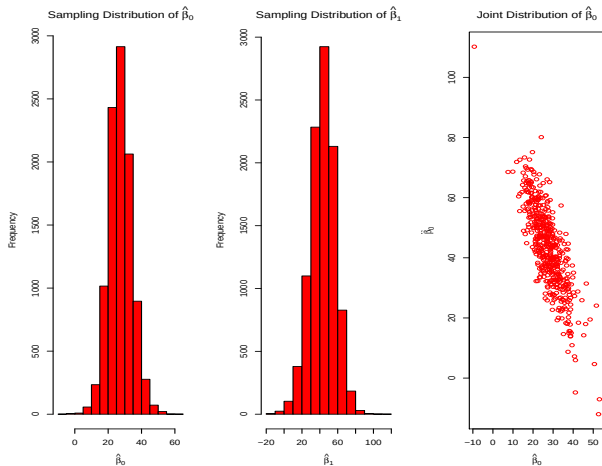
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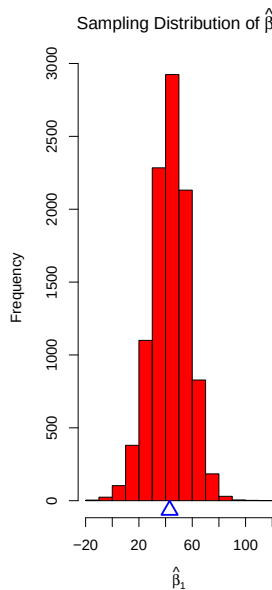
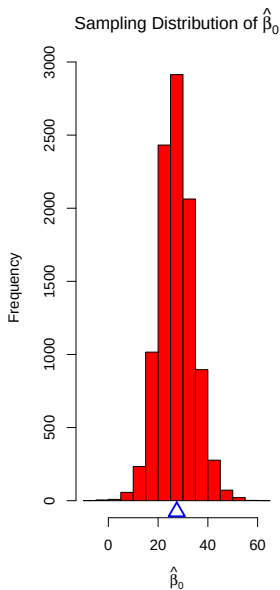


# Sampling Distributions for $\hat{\beta}_0$ and $\hat{\beta}_1$

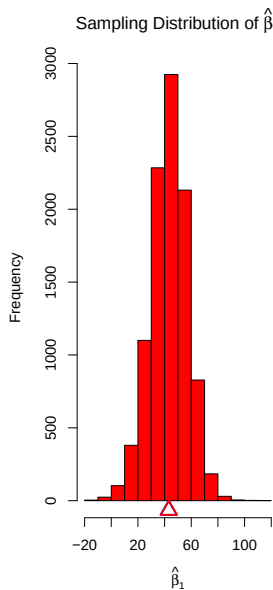
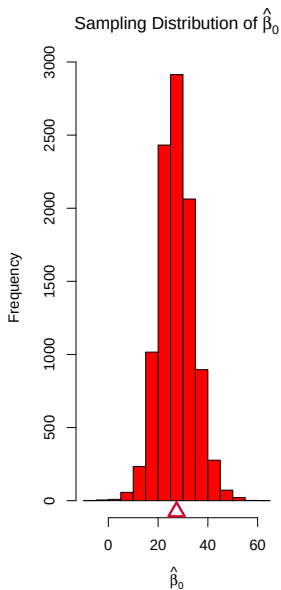




# Bias

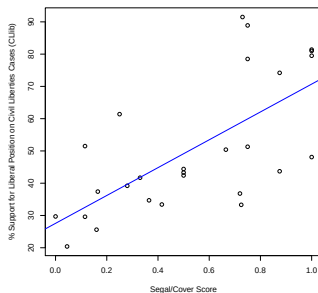


# Bias



# Assumptions for Unbiasedness

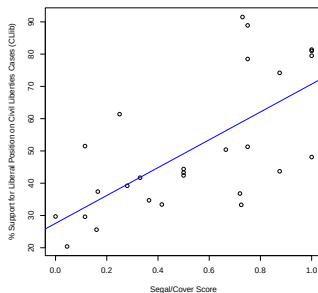
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# Assumptions for Unbiasedness

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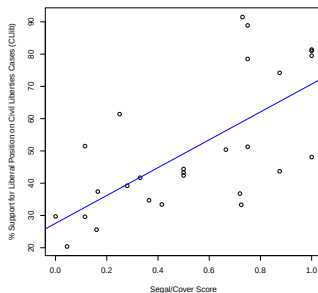
Just random sampling!



# Assumptions for Unbiasedness

What assumptions did we need for  $E[\hat{\beta}_0] = \beta_0$  and  $E[\hat{\beta}_1] = \beta_1$ ?

Just random sampling!



Unless we really wanted to estimate the CEF and not the BLP.

The *multivariate* conditional expectation function, best linear predictor and regression estimator.

Thus far we have detailed analytic results for the bivariate case.  
Suppose now that we have  $K$  explanatory variables:  
 $X_1, X_2, \dots, X_K$ .

CEF of  $Y$  given  $X_1, X_2, \dots, X_K$ :

$$E[Y|X_1, X_2, \dots, X_K]$$



The CEF is the *still* the best (MMSE) predictor of  $Y$  given  $X_1, X_2, \dots, X_K$ .

MMSE predictor: find the  $g(X_1, X_2, \dots, X_K)$  that minimizes the following expression:

$$E \left\{ [Y - g(X_1, X_2, \dots, X_K)]^2 \right\}$$

I.e., suppose you knew the full joint CDF of  $Y$  and  $X_1, X_2, \dots, X_K$ . And then someone gave you a randomly drawn value of  $(X_1, X_2, \dots, X_K)$ . What would be our guess of  $Y$  that would have the lowest MSE?

The answer is given by the CEF. The function  $g(X_1, X_2, \dots, X_K)$  that best approximates  $Y$  is the CEF.

Proof exactly as in the bivariate case.

The BLP is very similar as before, except we don't have, e.g., the very simple  $b = C(X, Y) / V(X)$  formula.

Remember: the BLP is the linear function that minimizes MSE.

BLP: find the  $\beta_0, \beta_1, \beta_2, \dots, \beta_K$  that minimize the following expression:

$$E \left\{ [Y - (\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots \beta_K X_K)]^2 \right\}$$

I.e., find the *linear*  $g(X_1, X_2, \dots, X_K)$  that minimizes MSE. This is the *population regression* function.

Get example. Draw picture on board.

Interpretation of  $\beta_1$

$$\frac{\partial [g(X_1, X_2, \dots, X_K)]}{\partial X_1} = \frac{\partial (\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots \beta_K X_K)}{\partial X_1} = \beta_1$$

or

$$\begin{aligned} g(x+1, X_2, \dots, X_K) - g(x, X_2, \dots, X_K) = \\ \beta_0 + \beta_1(x+1) + \beta_2 X_2 + \dots \beta_K X_K \\ - (\beta_0 + \beta_1(x) + \beta_2 X_2 + \dots \beta_K X_K) = \beta_1 \end{aligned}$$

Linear regression estimator: find the  $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_K$  that minimize the following expression:

$$\frac{1}{n} \sum_{i=1}^n \left\{ [Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \dots \hat{\beta}_K X_{Ki})]^2 \right\}$$

This minimizes the mean of squared *residuals*. (And therefore, also minimizes the sum of squared residuals.)



Linear regression estimator: find the  $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_K$  that minimize the following expression:

$$\overline{[Y - (\hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \dots \hat{\beta}_K X_K)]^2}$$

This is also known as *ordinary least squares* (OLS), since it finds the fit that the minimizes the mean of squared residuals. (And therefore, also minimizes the sum of squared residuals.)

Error: How far is the observation from the population predictor?

Residual: Sample analogue of the error. How far is the observation from our *estimate* of the population predictor?

Informal intuition: since the sample mean converges to the expected value, the solution to the problem with sample data converges to the solution to the problem for the population.

This general principle “usually” works for “most” statistical problems. (Write things in terms of the population problem, estimate it using sample data.)

Given regularity conditions,

$$\arg \min_{\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_K} \sum_{i=1}^n \left\{ [Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \dots + \hat{\beta}_K X_{Ki})]^2 \right\} \xrightarrow{P}$$

$$\arg \min_{\beta_0, \beta_1, \beta_2, \dots, \beta_K} E \left\{ [Y - (\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_K X_K)]^2 \right\}$$

Or, put another way, the linear regression estimate is consistent for the BLP.

More on regression. An informal exposition using linear algebra.

Again suppose that we have i.i.d. draws from some R.V. characterized by  $(Y, X_1, X_2, \dots, X_K)$ . Each observation  $i$  is denoted by  $(Y_i, X_{1i}, X_{2i}, \dots, X_{Ki})$

It is always possible to decompose the R.V. as follows:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_K X_K + \epsilon,$$

where we have just defined

$$\epsilon = Y - (\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_K X_K).$$

$\epsilon$  is the error.

It is always possible to decompose the R.V. as follows:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_K X_K + \epsilon,$$

where we have just defined

$$\epsilon = Y - (\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_K X_K).$$

$\epsilon$  is the error. The  $\beta$ s that minimize  $E[\epsilon^2]$  by definition characterize the BLP.



Define

$$\mathbf{X} = \begin{pmatrix} 1 & X_{11} & X_{21} & \dots & X_{K1} \\ 1 & X_{12} & X_{22} & \dots & X_{K2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{1n} & X_{2n} & \dots & X_{Kn} \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \quad \epsilon = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix},$$

$$\text{and } \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_K \end{pmatrix}.$$

Then it is equivalent to write the following:

$$Y_1 = \beta_0 + \beta_1 X_{11} + \beta_2 X_{21} + \dots \beta_K X_{K1} + \epsilon_1$$

$$Y_2 = \beta_0 + \beta_1 X_{12} + \beta_2 X_{22} + \dots \beta_K X_{K2} + \epsilon_2$$

...

$$Y_n = \beta_0 + \beta_1 X_{1n} + \beta_2 X_{2n} + \dots \beta_K X_{Kn} + \epsilon_n$$

Then it is equivalent to write the following:

$$\begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} = \begin{pmatrix} 1 & X_{11} & X_{21} & \dots & X_{K1} \\ 1 & X_{12} & X_{22} & \dots & X_{K2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{1n} & X_{2n} & \dots & X_{Kn} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_K \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}.$$

Then it is equivalent to write the following:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \epsilon.$$

We observe **Y** and **X**. How can we estimate the  $\beta$ s for the BLP?

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon$$

$$\mathbf{X}^T \mathbf{Y} = \mathbf{X}^T \mathbf{X} \beta + \mathbf{X}^T \epsilon$$

$$\mathbb{E} [\mathbf{X}^T \mathbf{Y}] = \mathbb{E} [\mathbf{X}^T \mathbf{X} \beta + \mathbf{X}^T \epsilon]$$

$$\mathbb{E} [\mathbf{X}^T \mathbf{Y}] = \mathbb{E} [\mathbf{X}^T \mathbf{X} \beta] + \mathbb{E} [\mathbf{X}^T \epsilon]$$

$$\mathbb{E} [\mathbf{X}^T \mathbf{Y}] = \mathbb{E} [\mathbf{X}^T \mathbf{X} \beta] + 0$$

\*\*

$$\mathbb{E} [\mathbf{X}^T \mathbf{Y}] = \mathbb{E} [\mathbf{X}^T \mathbf{X} \beta]$$

$$\mathbb{E} [\mathbf{X}^T \mathbf{Y}] = \mathbb{E} [\mathbf{X}^T \mathbf{X}] \beta$$

$$\left[ \mathbb{E} [\mathbf{X}^T \mathbf{X}] \right]^{-1} \mathbb{E} [\mathbf{X}^T \mathbf{Y}] = \beta$$

\*\* This arises from the first-order condition in solving:

$$\arg \min_{\beta} E [(\mathbf{Y} - \mathbf{X}\beta)^2]$$

Via FOC:

$$\begin{aligned} 2E [\mathbf{X}^T (\mathbf{Y} - \mathbf{X}\beta)] &= 0 \\ E [\mathbf{X}^T \epsilon] &= 0 \end{aligned}$$

Note that the errors are, in expectation, zero, and the covariance between the covariates and errors is also zero.

Estimating with sample data?

$$\left[ E[\mathbf{X}^T \mathbf{X}] \right]^{-1} E[\mathbf{X}^T \mathbf{Y}] = \beta$$



Estimating with sample data?

$$\left[\mathbf{X}^T \mathbf{X}\right]^{-1} \mathbf{X}^T \mathbf{Y} = \hat{\beta}$$

By the LLN, given the usual regularity conditions,  $\hat{\beta}$  is consistent for  $\beta$ .

The FWL theorem (AM), regression anatomy (AP), and the meaning of "controlling" for an additional variable.

# The Social Pressure Experiment

## Social Pressure and Voter Turnout: Evidence from a Large-Scale Field Experiment

ALAN S. GERBER *Yale University*

DONALD P. GREEN *Yale University*

CHRISTOPHER W. LARIMER *University of Northern Iowa*

**V**oter turnout theories based on rational self-interested behavior generally fail to predict significant turnout unless they account for the utility that citizens receive from performing their civic duty. We distinguish between two aspects of this type of utility, intrinsic satisfaction from behaving in accordance with a norm and extrinsic incentives to comply, and test the effects of priming intrinsic motives and applying varying degrees of extrinsic pressure. A large-scale field experiment involving several hundred thousand registered voters used a series of mailings to gauge these effects. Substantially higher turnout was observed among those who received mailings promising to publicize their turnout to their household or their neighbors. These findings demonstrate the profound importance of social pressure as an inducement to political participation.

# The Civic Duty Treatment

Dear Registered Voter:

DO YOUR CIVIC DUTY AND VOTE!

Why do so many people fail to vote? We've been talking about this problem for years, but it only seems to get worse.

The whole point of democracy is that citizens are active participants in government; that we have a voice in government. Your voice starts with your vote. On August 8, remember your rights and responsibilities as a citizen. Remember to vote.

DO YOUR CIVIC DUTY — VOTE!

# The Hawthorne Treatment

Dear Registered Voter:

YOU ARE BEING STUDIED!

Why do so many people fail to vote? We've been talking about this problem for years, but it only seems to get worse.

This year, we're trying to figure out why people do or do not vote. We'll be studying voter turnout in the August 8 primary election.

Our analysis will be based on public records, so you will not be contacted again or disturbed in any way. Anything we learn about your voting or not voting will remain confidential and will not be disclosed to anyone else.

DO YOUR CIVIC DUTY — VOTE!

# The Self Treatment

Dear Registered Voter:

WHO VOTES IS PUBLIC INFORMATION!

Why do so many people fail to vote? We've been talking about the problem for years, but it only seems to get worse.

This year, we're taking a different approach. We are reminding people that who votes is a matter of public record.

The chart shows your name from the list of registered voters, showing past votes, as well as an empty box which we will fill in to show whether you vote in the August 8 primary election. We intend to mail you an updated chart when we have that information.

We will leave the box blank if you do not vote.

DO YOUR CIVIC DUTY—VOTE!

-----  
OAK ST  
9999 ROBERT WAYNE  
9999 LAURA WAYNE

| Aug 04 | Nov 04 | Aug 06 |
|--------|--------|--------|
|        | Voted  | _____  |
| Voted  | Voted  | _____  |

# The Neighbors Treatment

Dear Registered Voter:

## WHAT IF YOUR NEIGHBORS KNEW WHETHER YOU VOTED?

Why do so many people fail to vote? We've been talking about the problem for years, but it only seems to get worse. This year, we're taking a new approach. We're sending this mailing to you and your neighbors to publicize who does and does not vote.

The chart shows the names of some of your neighbors, showing which have voted in the past. After the August 8 election, we intend to mail an updated chart. You and your neighbors will all know who voted and who did not.

## DO YOUR CIVIC DUTY — VOTE!

---

| MAPLE DR                   | Aug 04 | Nov 04 | Aug 06 |
|----------------------------|--------|--------|--------|
| 9995 JOSEPH JAMES SMITH    | Voted  | Voted  | _____  |
| 9995 JENNIFER KAY SMITH    |        | Voted  | _____  |
| 9997 RICHARD B JACKSON     |        | Voted  | _____  |
| 9999 KATHY MARIE JACKSON   |        | Voted  | _____  |
| 9999 BRIAN JOSEPH JACKSON  |        | Voted  | _____  |
| 9991 JENNIFER KAY THOMPSON |        | Voted  | _____  |
| 9991 BOB R THOMPSON        |        | Voted  | _____  |
| 9993 BILL S SMITH          |        |        | _____  |
| 9989 WILLIAM LUKE CASPER   |        | Voted  | _____  |
| 9989 JENNIFER SUE CASPER   |        | Voted  | _____  |
| 9987 MARIA S JOHNSON       | Voted  | Voted  | _____  |
| 9987 TOM JACK JOHNSON      | Voted  | Voted  | _____  |
| 9987 RICHARD TOM JOHNSON   |        | Voted  | _____  |
| 9985 ROSEMARY S SUE        |        | Voted  | _____  |
| 9985 KATHRYN L SUE         |        | Voted  | _____  |
| 9985 HOWARD BEN SUE        |        | Voted  | _____  |

# Balance on Pre-treatment Variables in the Gerber, Green, and Larimer Experiment

**TABLE 1. Relationship between Treatment Group Assignment and Covariates (Household-Level Data)**

|                | Control | Civic Duty | Hawthorne | Self   | Neighbors |
|----------------|---------|------------|-----------|--------|-----------|
|                | Mean    | Mean       | Mean      | Mean   | Mean      |
| Household size | 1.91    | 1.91       | 1.91      | 1.91   | 1.91      |
| Nov 2002       | .83     | .84        | .84       | .84    | .84       |
| Nov 2000       | .87     | .87        | .87       | .86    | .87       |
| Aug 2004       | .42     | .42        | .42       | .42    | .42       |
| Aug 2002       | .41     | .41        | .41       | .41    | .41       |
| Aug 2000       | .26     | .27        | .26       | .26    | .26       |
| Female         | .50     | .50        | .50       | .50    | .50       |
| Age (in years) | 51.98   | 51.85      | 51.87     | 51.91  | 52.01     |
| N =            | 99,999  | 20,001     | 20,002    | 20,000 | 20,000    |

*Note:* Only registered voters who voted in November 2004 were selected for our sample. Although not included in the table, there were no significant differences between treatment group assignment and covariates measuring race and ethnicity.



# Including Covariates in the Gerber, Green, and Larimer Experiment

**TABLE 3. OLS Regression Estimates of the Effects of Four Mail Treatments on Voter Turnout in the August 2006 Primary Election**

|                                                       | Model Specifications |              |              |
|-------------------------------------------------------|----------------------|--------------|--------------|
|                                                       | (a)                  | (b)          | (c)          |
| Civic Duty Treatment (Robust cluster standard errors) | .018* (.003)         | .018* (.003) | .018* (.003) |
| Hawthorne Treatment (Robust cluster standard errors)  | .026* (.003)         | .026* (.003) | .025* (.003) |
| Self-Treatment (Robust cluster standard errors)       | .049* (.003)         | .049* (.003) | .048* (.003) |
| Neighbors Treatment (Robust cluster standard errors)  | .081* (.003)         | .082* (.003) | .081* (.003) |
| N of individuals                                      | 344,084              | 344,084      | 344,084      |
| Covariates**                                          | No                   | No           | Yes          |
| Block-level fixed effects                             | No                   | Yes          | Yes          |

*Note:* Blocks refer to clusters of neighboring voters within which random assignment occurred. Robust cluster standard errors account for the clustering of individuals within household, which was the unit of random assignment.

\*  $p < .001$ .

\*\* Covariates are dummy variables for voting in general elections in November 2002 and 2000, primary elections in August 2004, 2002, and 2000.

# Regression Anatomy Result

Result:

$$\hat{\beta}_X^{Y \sim X+Z} = \hat{\beta}_{r_{X \sim Z}}^{Y \sim r_{X \sim Z}}$$

# Regression Anatomy in Pictures

# Regression Anatomy and FWL Results

Results:

$$\hat{\beta}_X^{Y \sim X+Z} = \hat{\beta}_{r_{X \sim Z}}^{Y \sim r_{X \sim Z}}$$

$$\hat{\beta}_X^{Y \sim X+Z} = \hat{\beta}_{r_{X \sim Z}}^{r_{Y \sim Z} \sim r_{X \sim Z}}$$

$$r_{Y \sim X+Z} = r_{r_{Y \sim Z} \sim r_{X \sim Z}}$$

# Simulating Linear Model Data

```
> n <- 10000
> set.seed(1234)
> z <- rnorm(n)
> x <- z + rnorm(n)
> y <- x + z + rnorm(n)
> cor(x, z)
[1] 0.7044271
```

# Linear Regression Output

```
> lm(y ~ x)
```

Coefficients:

| (Intercept) | x       |
|-------------|---------|
| 0.01099     | 1.48301 |

```
> lm(y ~ x + z)
```

Coefficients:

| (Intercept) | x       | z       |
|-------------|---------|---------|
| 0.00417     | 0.99901 | 0.98608 |

## Added Variable “Plot” (Partial Regression “Plot”)

```
> r_yz <- residuals(lm(y ~ z))  
> r_xz <- residuals(lm(x ~ z))  
> r_yxz <- residuals(lm(y ~ x+z))  
> lm(r_yz ~ r_xz)
```

Coefficients:

| (Intercept) | r_xz      |
|-------------|-----------|
| -3.619e-16  | 9.990e-01 |

```
> range(r_yxz - residuals(lm(r_yz ~ r_xz)))  
[1] -2.211564e-13  4.019007e-14
```

## Just the x

```
> lm(y ~ r_xz)
```

Coefficients:

| (Intercept) | r_xz     |
|-------------|----------|
| 0.008571    | 0.999006 |

```
> range(r_yxz - residuals(lm(y ~ r_xz)))  
[1] -7.209635  6.790846
```



What if  $x$  and  $z$  perfectly correlated?

## What if $x$ and $z$ are uncorrelated?

```
> z <- rnorm(n)
> x <- rnorm(n) # no z
> cor(x,z)
[1] 0.01075281
> y <- x + z + rnorm(n)
> lm(y ~ x)
Coefficients:
(Intercept)                x
    0.01027         1.00940

> lm(y ~ x + z)
Coefficients:
(Intercept)                x                z
    0.00417         0.99901         0.98509
```

# FWL with three explanatory variables

$$y \sim x + z + w$$

1st Approach:

1. residuals from  $y \sim z + w$
2. residuals from  $x \sim z + w$
3. regress  $r_{y \sim z + w}$  on  $r_{x \sim z + w}$

# FWL with three explanatory variables

$$y \sim x + z + w$$

2nd Approach:

1. residuals from  $y \sim w$
2. residuals from  $x \sim w$
3. residuals from  $z \sim w$
4. residuals from  $r_{y \sim w} \sim r_{z \sim w}$
5. residuals from  $r_{x \sim w} \sim r_{z \sim w}$
6. regress  $r_{r_{y \sim w} \sim r_{z \sim w}}$  on  $r_{r_{x \sim w} \sim r_{z \sim w}}$

## Controlling in Logistic Regression

# Simulating Logistic Model Data

```
> n <- 1000000
> set.seed(1234)
> z <- runif(n=n,min=5,max = 100)
> x <- sample(0:1,size = n, replace=TRUE) # no z
cor(x,z)
[1] -0.0002453252
> xb <- exp(-4+.1*z+1*x) / (1+exp(-4+.1*z+1*x))
> y <- rbinom(n=n,size=1,prob = xb)
> dat <- data.frame(y,x,z)
```

## What if $x$ and $z$ are uncorrelated?

```
> log_yx <- glm(y~x,family=binomial,data=dat)
> log_yxz <- glm(y~x+z,family=binomial,data=dat)
> log_yx
```

Coefficients:

| (Intercept) | x      |
|-------------|--------|
| 0.5260      | 0.4598 |

```
> log_yxz
```

Coefficients:

| (Intercept) | x     | z     |
|-------------|-------|-------|
| -4.002      | 1.001 | 0.100 |

## Data with $x = 1$ and $x = 0$

```
> trt_dat <- dat  
> trt_dat$x <- 1  
> ctrl_dat <- dat  
> ctrl_dat$x <- 0
```



## x and z uncorrelated (continued)

```
> trt_resp <- predict(log_yx,type="response",  
newdata = trt_dat)  
> ctrl_resp <- predict(log_yx,type="response",  
newdata = ctrl_dat)  
> mean(trt_resp - ctrl_resp)  
[1] 0.0997118  
>  
> trt_resp <- predict(log_yxz,type="response",  
newdata = trt_dat)  
> ctrl_resp <- predict(log_yxz,type="response",  
newdata = ctrl_dat)  
> mean(trt_resp - ctrl_resp)  
[1] 0.09995198
```

## x and z uncorrelated (continued)

```
> trt_resp <- predict(log_yx,type="response",
newdata = trt_dat)
> ctrl_resp <- predict(log_yx,type="response",
newdata = ctrl_dat)
> mean(trt_resp)
[1] 0.7282527
> mean(ctrl_resp)
[1] 0.6283474
>
> trt_resp <- predict(log_yxz,type="response",
newdata = trt_dat)
> ctrl_resp <- predict(log_yxz,type="response",
newdata = ctrl_dat)
> mean(trt_resp)
[1] 0.728593
> mean(ctrl_resp)
[1] 0.6280122
```

What about non-binary  $x$ ?

## x and z uncorrelated (x non-binary)

```
> z <- runif(n=n,min=5,max = 100)
> x <- rnorm(n)
> cor(x,z)
[1] 0.0007293003
> xb <- exp(-4+.1*z+1*x) / (1+exp(-4+.1*z+1*x))
> y <- rbinom(n=n,size=1,prob = xb)
> dat <- data.frame(y,x,z)
> log_yx <- glm(y~x,family=binomial,data=dat)
> log_yxz <- glm(y~x+z,family=binomial,data=dat)
```

## x and z uncorrelated (x non-binary)

```
> x_dat_half <- dat
> x_dat_half$x <- .5
>
> x_half_resp <- predict(log_yx,type="response",
newdata = x_dat_half)
> mean(x_half_resp)
[1] 0.6830625
> x_half_resp <- predict(log_yxz,type="response",
newdata = x_dat_half)
> mean(x_half_resp)
[1] 0.6791526
```

What about an interactive linear regression?

## What if $x$ and $z$ are uncorrelated in interactive model?

```
> n <- 1000000  
> set.seed(1234)  
> z <- 2 + rnorm(n)  
> x <- rnorm(n) # no z  
> cor(x,z)  
[1] -0.0007890214  
>  
> y <- x + z + x*z + rnorm(n)
```

## What if $x$ and $z$ are uncorrelated in interactive model?

```
> lm(y ~ x)
```

| (Intercept) | x     |
|-------------|-------|
| 1.999       | 2.998 |

```
> lm(y ~ x + z)
```

| (Intercept) | x        | z        |
|-------------|----------|----------|
| 0.002337    | 2.999022 | 0.998214 |

```
> lm(y ~ x * z)
```

| (Intercept) | x         | z         | x:z       |
|-------------|-----------|-----------|-----------|
| -0.0001929  | 0.9984175 | 0.9998744 | 1.0003144 |



## What if $x$ and $z$ are uncorrelated in interactive model?

```
> lm(y ~ x)
```

| (Intercept) | x     |
|-------------|-------|
| 1.999       | 2.998 |

```
> lm(y ~ x + z)
```

| (Intercept) | x        | z        |
|-------------|----------|----------|
| 0.002337    | 2.999022 | 0.998214 |

```
> lm(y ~ x * z)
```

| (Intercept) | x         | z         | x:z       |
|-------------|-----------|-----------|-----------|
| -0.0001929  | 0.9984175 | 0.9998744 | 1.0003144 |

```
> lm(y ~ x * I(z-mean(z)))
```

| (Intercept) | x      | I(z - mean(z)) | x:I(z - me |
|-------------|--------|----------------|------------|
| 2.0000      | 2.9995 | 0.9999         | 1.0003     |

What about nonparametric with  $x$  and  $z$  uncorrelated?

# Omitted Variable Bias

- ▶ Suppose the model you've estimated is

$$y_i = \beta_0 + \beta_1 x_{i1} + u_i$$

when the model you wanted was

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i$$

- ▶ Then your OLS estimator,  $\hat{\beta}_1$  is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) y_i}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}$$

- ▶ Use the rules of conditional expectation to derive the bias.

- Recall that the model we wanted was  $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$ . Replace  $y_i$ . Then,

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) y_i}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}$$

=

=

- Recall that the model we wanted was  $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$ . Replace  $y_i$ . Then,

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) y_i}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} \\ &= \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i)}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}\end{aligned}$$

=

- Recall that the model we wanted was  $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i$ . Replace  $y_i$ . Then,

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) y_i}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} \\&= \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i)}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} \\&= \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) \beta_0}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} + \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) \beta_1 x_{i1}}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} + \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) \beta_2 x_{i2}}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} \\&\quad + \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) u_i}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}\end{aligned}$$

► Therefore we have

$$\hat{\beta}_1 = \beta_1 + \beta_2 \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) x_{i2}}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} + \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) u_i}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}$$

Take expectation conditional on the all the  $x_{i1}, x_{i2}, i = \dots, n$ ,

$$E \left[ \hat{\beta}_1 | x_{i1}, x_{i2}, i = 1, \dots, n \right]$$

=

=



► Therefore we have

$$\hat{\beta}_1 = \beta_1 + \beta_2 \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) x_{i2}}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} + \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) u_i}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}$$

Take expectation conditional on the all the  $x_{i1}, x_{i2}, i = \dots, n$ ,

$$\begin{aligned} & E \left[ \hat{\beta}_1 | x_{i1}, x_{i2}, i = 1, \dots, n \right] \\ &= \beta_1 + \beta_2 \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) x_{i2}}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} + \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) E[u_i | x_{i1}, x_{i2}, i = 1, \dots, n]}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} \end{aligned}$$

=

► Therefore we have

$$\hat{\beta}_1 = \beta_1 + \beta_2 \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) x_{i2}}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} + \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) u_i}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}$$

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► Hence the bias is

$$E \left[ \hat{\beta}_1 | x_{i1}, x_{i2}, i = 1, \dots, n \right] - \beta_1 = \beta_2 \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) x_{i2}}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}$$

# Why do we care about SEs?

- ▶ If  $x$  has been randomized, what do we know about the slope in the  $x$  direction for OLS with  $x$  and OLS with  $x$  and  $z$ ?

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- ▶ Write the estimated classical SE formulas for OLS with  $x$  and OLS with  $x$  and  $z$ , where  $x$  has been randomized.
- ▶ Efficiency
- ▶ Inference (e.g., confidence intervals)

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1. Normal Approximation



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- 1.4 do the CI algebra

2. Bootstrap

We can decompose the  $\hat{\beta}$  into the expected value and the sampling error.

$$[\mathbf{X}^T \mathbf{X}]^{-1} \mathbf{X}^T \mathbf{Y} = \hat{\beta}$$

$$[\mathbf{X}^T \mathbf{X}]^{-1} \mathbf{X}^T [\mathbf{X}\beta + \epsilon] = \hat{\beta}$$

$$[\mathbf{X}^T \mathbf{X}]^{-1} \mathbf{X}^T \mathbf{X}\beta + [\mathbf{X}^T \mathbf{X}]^{-1} \mathbf{X}^T \epsilon = \hat{\beta}$$

$$\beta + [\mathbf{X}^T \mathbf{X}]^{-1} \mathbf{X}^T \epsilon = \hat{\beta}$$

Thus:

$$\begin{aligned} V[\hat{\beta}] &= V\left[\beta + [\mathbf{X}^T \mathbf{X}]^{-1} \mathbf{X}^T \epsilon\right] \\ &= V\left[[\mathbf{X}^T \mathbf{X}]^{-1} \mathbf{X}^T \epsilon\right] \end{aligned}$$

Estimation: plug in residuals. This is the “sandwich estimator.”

$$\hat{V} [\hat{\beta}] = [\mathbf{X}^T \mathbf{X}]^{-1} \mathbf{X}^T \text{diag} [(\mathbf{Y} - \mathbf{X}\hat{\beta})^2] \mathbf{X} [\mathbf{X}^T \mathbf{X}]^{-1}.$$

This uses the residuals to *estimate* the relationship between the  $X$  variables and the errors.



One note: there are many variants of the variance estimators, that involve premultiplying by, e.g.,  $\frac{n}{n-K}$  or  $\frac{n}{n-K-2}$ , etc. These are analogous to the unbiased sample variance vs. plug-in sample variance. It's not so simple with the robust variance estimators (many variants: HC0, HC1, HC2) but they all have the same probability limit.

There is an alternative. "Classical" variance estimation. If we *assume* that  $V[\epsilon|X_1, X_2, \dots, X_K] = \sigma^2$  for all values of  $X_1, X_2, \dots, X_K$ : homoskedasticity. Then:

$$\begin{aligned}
 nV[\hat{\beta}] &\xrightarrow{p} n \left[ E[\mathbf{X}^T \mathbf{X}] \right]^{-1} E \left[ \mathbf{X}^T \text{diag}[\epsilon^2] \mathbf{X} \right] \left[ E[\mathbf{X}^T \mathbf{X}] \right]^{-1} \\
 &= n\sigma^2 \left[ E[\mathbf{X}^T \mathbf{X}] \right]^{-1} E \left[ \mathbf{X}^T \mathbf{I} \mathbf{X} \right] \left[ E[\mathbf{X}^T \mathbf{X}] \right]^{-1} \\
 &= n\sigma^2 \left[ E[\mathbf{X}^T \mathbf{X}] \right]^{-1} E \left[ \mathbf{X}^T \mathbf{X} \right] \left[ E[\mathbf{X}^T \mathbf{X}] \right]^{-1} \\
 &= n\sigma^2 \left[ E[\mathbf{X}^T \mathbf{X}] \right]^{-1}
 \end{aligned}$$

Where  $\sigma^2 = V[\epsilon] = E[\epsilon^2]$ .

Classical standard errors make an additional assumption: the magnitude of the errors (how far off the BLP is from the true value) is *unrelated* to the values of the  $X$ s.

This assumption is unlikely to hold. Usually – in real life – the errors will be “heteroskedastic,” because in real life, data is messy. If homoskedasticity *does* hold, then in large samples, the classical and “robust” variance estimates will agree.

With large  $n$  and i.i.d. sampling, always use robust standard errors to characterize the uncertainty of your regression fit.

What do constant treatment effects imply about homo/heteroskedasticity? What kind of test does the contrapositive allow for?

If  $Y$  and  $X$  have been centered and  $x$  is a nonstochastic scalar (everything works basically the same if we relax non-stochastic assumption)

$$\begin{aligned} V[\hat{\beta}] &= V\left[\beta + \left[\frac{\sum_i x_i \epsilon_i}{\sum_i x_i^2}\right]\right] \\ &= \frac{V[\sum_i x_i \epsilon_i]}{[\sum_i x_i^2]^2} \end{aligned}$$

How would you estimate this?

How does this simplify if iid?

How does this further simplify if homoskedasticity?

## Bootstrapping

1. Take a sample of size  $n$  with replacement from your sample
2. Calculate  $\beta^{(s=1)}$  from that bootstrap sample
3. Repeat for  $s = 1, \dots, S$  where  $S$  is big
4. Do one of the following:
  - ▶ Calculate bootstrapped variance  $\frac{1}{S-1} \sum_{s=1}^S (\beta^{(s)} - \overline{\beta^{(s)}})^2$
  - ▶ Put  $\{\beta^{(s)} : s = 1, \dots, S\}$  in order and report the  $\alpha/2$  and  $1 - \alpha/2$  percentiles
  - ▶ ...

Takes care of heteroskedasticity automatically.

# Clustering

What is clustering?

If  $Y$  and  $X$  have been centered and  $x$  is a nonstochastic scalar (everything works basically the same if we relax non-stochastic assumption)

$$\begin{aligned} V[\hat{\beta}] &= V\left[\beta + \left[\frac{\sum_i x_i \epsilon_i}{\sum_i x_i^2}\right]\right] \\ &= \frac{V[\sum_i x_i \epsilon_i]}{[\sum_i x_i^2]^2} \end{aligned}$$

What if the assumptions don't hold, but we distribute the variance operator?

What is the plug-in estimator for this?

What restrictions might we need for this to work?

How should we change the bootstrapping procedure?



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$$\frac{\hat{\sigma}^2}{\sum x_i^2}$$

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$$\tau_k \approx 1 + \rho_{xk}\rho_{\epsilon}(\bar{N}_g - 1)$$

$$\tau \approx 1 + \rho_x\rho_{\epsilon}(\bar{N}_g - 1)$$

where  $\rho_x$  is the within cluster correlation of  $x$ ,  $\rho_{\epsilon}$  is the within cluster correlation of the errors, and  $\bar{N}_g$  is average group size

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where  $\rho_x$  is the within cluster correlation of  $x$ ,  $\rho_{\epsilon}$  is the within cluster correlation of the errors, and  $\bar{N}_g$  is average group size

Note: if treatment is assigned at the classroom (cluster) level,  $\rho_x = 1$

How might we fix this?

# Linear Models: what are they good for?

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1. Easy to fit
2. Easy to interpret
3. Nice properties (e.g., FWL)
4. Easy SE formulas
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1. Easy to fit? (yep, computers)
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## Non-linear Models

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3. Nice properties (some, BDC later)
4. Easy SE formulas? (no, but bootstrapping)
5. Easy sensitivity analysis? (later)
6. Robust? (yep, computers and cross-validation)

What if the CEF is nonlinear?

I.e., what if  $E[Y|X] \neq a + bX$ ?

Can we still use linear regression?



Suppose the data were generated:

$$Y = 10X^2 + U,$$

where  $X$  is distributed *Uniform*(0, 1) and  $U$  is distributed *Normal*(0, 1).

The CEF is nonlinear  $E[Y|X] = 10X^2$ . But what would the BLP be?

Draw on board

BLP is a good, but not great approximation to the CEF. Can we do better with OLS?

Suppose now we created a second variable:  $X^2$ . Then we want to estimate:

$$\arg \min_{\beta_0, \beta_1, \beta_2} E (Y - \beta_0 - \beta_1 X - \beta_2 X^2)^2.$$

The *best linear predictor* with respect to  $X$  and  $X^2$  is the CEF here!

$$\text{BLP} = \text{CEF} = \text{E}[Y|X, X^2] = \text{E}[Y|X] = \beta_0 + \beta_1 X + \beta_2 X^2.$$

$$\text{BLP} = \text{CEF} = E[Y|X^2] = E[Y|X] = 0 + 0X + 10X^2.$$

When the CEF is linear in  $X$ ,  $X^2$ ,

$$E[Y|X] = \beta_0 + \beta_1 X + \beta_2 X^2.$$



When the CEF is not linear in  $X$ ,  $X^2$ ,

$$E[Y|X] \approx \beta_0 + \beta_1 X + \beta_2 X^2.$$

How to interpret slopes?

Before: linear estimate of slope of CEF was  $\beta_1$ .

How can we generalize?

When the CEF is linear in  $X$ ,  $X^2$ ,

$$\frac{\partial E[Y|X]}{\partial X} = \frac{\partial(\beta_0 + \beta_1 X + \beta_2 X^2)}{\partial X}$$

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The slope of the CEF depends on the value of  $X$ .

When the CEF is linear in  $X$ ,  $X^2$ ,

$$\frac{\partial E[Y|X]}{\partial X} = \beta_1 + 2\beta_2 X$$

what is

$$g(x + 1/2) - g(x - 1/2)$$

Can we develop a general result for approximating the CEF with polynomials? (Yes!)

The Weierstrass approximation theorem:

Any continuous function defined on a closed interval can be uniformly approximated to arbitrary precision using a polynomial function.



Let's visualize this. Let's take a complex CEF:

$$E[Y|X = x] = \begin{cases} 0 & : x \in [-1, 0] \\ x^2 & : x \in (0, 1) \\ 2 - x & : x \in [1, 3] \end{cases}$$

figures from book

See overfitting example from book, but what about a more formal criteria.

# Bias/Var Tradeoff

Recall the estimators:

$$\frac{n}{n-1} [\overline{Y^2} - \bar{Y}^2]$$

Compare the two on bias and variance.

Go to regression bias/variance tradeoff example

Remaining “non-linear” issues with a single  $X$ :

- ▶ How to choose the degree of the polynomial?
- ▶ What if not linear in  $\beta$ ?
- ▶ statistical/machine learning approaches
- ▶ what if  $X$  is ordinal?

## How to choose the degree of the polynomial?

- ▶ Sieve Estimation:

Choose the degree of the polynomial to be  $K_n = \lceil n^{1/9} \rceil$

- ▶ Cross-validation:

1. Separate data into  $G$  groups
2. Leave one group out as test data
3. fit model using remaining groups
4. calculate prediction error on test data
5. repeat for all  $G$  groups
6. repeat whole process for many different polynomials and choose best based on test data prediction error

## Leave one out CV

- ▶  $G = n$  (each observation is a group)
- ▶ test error easy to calculate for OLS (includes polynomial models)

$$\frac{1}{n} \sum_{i=1}^n \frac{(y_i - \hat{y}_i)^2}{(1 - h_i)^2}$$



Examples of nonlinear in  $\beta$

$$g(X) = e^{\beta_0 + \beta_1 X}$$

$$g(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

How to fit?

How to interpret?

statistical/machine learning solutions (go to ISLR slides)

What about multiple  $X$ :

- ▶ Additive, Interactive, and Saturated Models
- ▶ with polynomials
- ▶ What if not linear in  $\beta$ ?
- ▶ statistical/machine learning approaches

What happens when the conditional slope of the CEF with respect to  $X_1$  depends on the value of  $X_2$ ?

The BLP is still the BLP, but we might be missing out on an interesting *interaction* between the variables  $X_1$  and  $X_2$ .

So, to formalize a bit, what if the CEF is otherwise linear, but includes an *interaction*?

$$E[Y|X_1, X_2] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2$$

Much as before, we could derive the slope of the CEF with respect to a given variable.

$$\frac{\partial E[Y|X_1, X_2]}{\partial X_1} = \frac{\partial \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2}{\partial X_1}.$$

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$$\frac{\partial E[Y|X_1, X_2]}{\partial X_1} = \beta_1 + \beta_3 X_2.$$

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The conditional slope of the CEF with respect to  $X_1$  *depends on the value of*  $X_2$ .



Partial derivatives, first differences, and interpretation.

What to do about  $X_2$ ?

What if we had a binary regressor,  $X$ ? (I.e., not a binary  $Y$ .)

I.e.,  $\text{Supp}(X) = \{0, 1\}$

Then considering the bivariate CEF, by definition:

$$E[Y|X = x] = \begin{cases} E[Y|X = 0] & : x = 0 \\ E[Y|X = 1] & : x = 1 \end{cases}$$

Then we can *definitionally* write  $E[Y|X]$  as a linear function of  $X$ .

$$E[Y|X] = \beta_0 + \beta_1 X,$$

where

- ▶  $\beta_0 = E[Y|X = 0]$
- ▶  $\beta_1 = E[Y|X = 1] - E[Y|X = 0]$ .

We could define an alternative specification that is logically equivalent. Consider the regression *without an intercept*:

$$E[Y|X] = \gamma_1(1 - X) + \gamma_2 X,$$

where

- ▶  $\gamma_1 = E[Y|X = 0]$
- ▶  $\gamma_2 = E[Y|X = 1]$ .

When  $X$  is binary, the CEF *must be* the BLP.

The regression estimates of the  $\beta$ s can be written *just using* conditional means.

The idea generalizes to the multivariate case.

Consider binary  $X_1, X_2$ :

$$E[Y|X_1 = x_1, X_2 = x_2] = \begin{cases} E[Y|X_1 = 0, X_2 = 0] & : x_1 = 0, x_2 = 0 \\ E[Y|X_1 = 1, X_2 = 0] & : x_1 = 1, x_2 = 0 \\ E[Y|X_1 = 0, X_2 = 1] & : x_1 = 0, x_2 = 1 \\ E[Y|X_1 = 1, X_2 = 1] & : x_1 = 1, x_2 = 1 \end{cases}$$



Then we can *definitionally* write  $E[Y|X_1, X_2]$  as a linear function of  $X_1$ ,  $X_2$  and  $X_1X_2$ .

$$E[Y|X_1, X_2] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2$$

where

- ▶  $\beta_0 = E[Y|X_1 = 0, X_2 = 0]$
- ▶  $\beta_1 = E[Y|X_1 = 1, X_2 = 0] - E[Y|X_1 = 0, X_2 = 0]$
- ▶  $\beta_2 = E[Y|X_1 = 0, X_2 = 1] - E[Y|X_1 = 0, X_2 = 0]$
- ▶  $\beta_3 = E[Y|X_1 = 1, X_2 = 1] - E[Y|X_1 = 1, X_2 = 0] - E[Y|X_1 = 0, X_2 = 1] + E[Y|X_1 = 0, X_2 = 0]$

Logically equivalent, using indicator or “dummy” variables:

$$E[Y|X_1, X_2] = \gamma_1 \mathbf{I}[X_1 = 0, X_2 = 0] + \gamma_2 \mathbf{I}[X_1 = 1, X_2 = 0] + \\ \gamma_3 \mathbf{I}[X_1 = 0, X_2 = 1] + \gamma_4 \mathbf{I}[X_1 = 1, X_2 = 1]$$

where

- ▶  $\gamma_1 = E[Y|X_1 = 0, X_2 = 0]$
- ▶  $\gamma_2 = E[Y|X_1 = 1, X_2 = 0]$
- ▶  $\gamma_3 = E[Y|X_1 = 0, X_2 = 1]$
- ▶  $\gamma_4 = E[Y|X_1 = 1, X_2 = 1]$

These are cases of *saturated models*: when every unique combination of values of the explanatory variables is included in a fully flexible way. This is only possible with *discrete* regressors ( $X$  variables).

This means including dummy variables for every possible value of each variable, and all possible (including multiway) interactions. Or just dummy variables for each unique combination, and excluding the intercept.

With a saturated model, the CEF is, by definition, the BLP.

What about interactive models with polynomials?

What if not linear in  $\beta$ ?

Statistical/machine learning approaches (go to nonlinear notes)

Let's start by considering a simple but common problem: missing data.

What can we learn about the distribution of a variable of interest when some values are missing from, e.g., nonresponse.

E.g., if we ran a survey asking people who they voted for, some people might not respond to the vote choice question. We want to know the population distribution of outcomes to the question, but we don't observe the answers for some people.

Suppose that we were interested in estimating the expected value and/or full distribution of some random variable characterized by the distribution of  $Y_i$ . However, we *do not* observe  $Y_i$  for any units such that units do not respond.

We will primarily focus on *identification* results – results assuming that we knew the joint distribution of all observable data, and that nothing needs to be estimated. Given i.i.d. sampling, then estimation proceeds via simple plug-in estimation (using sample analogues to population quantities).



Suppose that we had knowledge of the full joint distribution of a random vector generated by sampling values of  $(Y_i^*, R_i)$ .

$R_i$  denotes whether or not unit  $i$  has data available on  $Y_i$  (i.e., unit  $i$  responded), and  $Y_i^*$  is a censored version of  $Y_i$ :

$$Y_i^* = Y_i \cdot R_i + (-99) \cdot (1 - R_i)$$

(This embeds an often-overlooked assumption of stable outcomes – the underlying  $Y_i$  for unit  $i$  is *stable* and it does not depend on how you asked, who you asked or who responded. More on this later.)

Suppose that our inferential target is then the expected value of  $Y_i$ ,  $E[Y_i]$ , which cannot be directly observed. Then the question is: what assumptions yield what information about  $E[Y_i]$ ?

Let's start with a minimal assumption. Without much loss of generality, suppose that  $\text{Supp}[Y_i] \subset [0, 1]$ . (For any bounded outcome variable, we could rescale to  $[0, 1]$ .)

With this assumption, we can derive *sharp bounds* (bounds that cannot be improved upon without further assumptions) on  $E[Y_i]$ .

These bounds are a special case of Manski bounds. See, e.g., Manski (2003).

Manski bounds proceed by imputing the worst-case scenario for all missing data.

For an upper bound, assume that all missing  $Y_i$ s are a 1. For a lower bound, assume that all missing  $Y_i$ s are 0.

For example, we might see the following data:

| Unit | $Y_i^*$ | $R_i$ |
|------|---------|-------|
| 1    | 1       | 1     |
| 2    | -99     | 0     |
| 3    | 1       | 1     |
| 4    | 0       | 1     |
| 5    | 1       | 1     |
| 6    | -99     | 0     |

Given stable outcomes:

| Unit | $Y_i^*$ | $R_i$ |
|------|---------|-------|
| 1    | 1       | 1     |
| 2    | -99     | 0     |
| 3    | 1       | 1     |
| 4    | 0       | 1     |
| 5    | 1       | 1     |
| 6    | -99     | 0     |

| Unit | $Y_i$ | $R_i$ |
|------|-------|-------|
| 1    | 1     | 1     |
| 2    | ?     | 0     |
| 3    | 1     | 1     |
| 4    | 0     | 1     |
| 5    | 1     | 1     |
| 6    | ?     | 0     |

For units 2 and 6, we only know that  $\text{Supp}[Y_i] \subset [0, 1]$ .

Using the sample mean as a plug-in estimator for the expected value, our estimate of a lower bound for  $E[Y_i]$ :

| Unit | $Y_i^*$ | $R_i$ | Unit | $Y_i$ | $R_i$ |
|------|---------|-------|------|-------|-------|
| 1    | 1       | 1     | 1    | 1     | 1     |
| 2    | -99     | 0     | 2    | 0     | 0     |
| 3    | 1       | 1     | 3    | 1     | 1     |
| 4    | 0       | 1     | 4    | 0     | 1     |
| 5    | 1       | 1     | 5    | 1     | 1     |
| 6    | -99     | 0     | 6    | 0     | 0     |

Estimated lower bound for  $E[Y_i]$  is  $1/2$ .

Using the sample mean as a plug-in estimator for the expected value, our estimate of an upper bound for  $E[Y_i]$ :

| Unit | $Y_i^*$ | $R_i$ |
|------|---------|-------|
| 1    | 1       | 1     |
| 2    | -99     | 0     |
| 3    | 1       | 1     |
| 4    | 0       | 1     |
| 5    | 1       | 1     |
| 6    | -99     | 0     |

| Unit | $Y_i$ | $R_i$ |
|------|-------|-------|
| 1    | 1     | 1     |
| 2    | 1     | 0     |
| 3    | 1     | 1     |
| 4    | 0     | 1     |
| 5    | 1     | 1     |
| 6    | 1     | 0     |

Estimated upper bound for  $E[Y_i]$  is  $5/6$ .

Importantly, we can exactly write down *sharp bounds* for  $E[Y_i]$  in terms of the joint distribution of  $(Y_i^*, R_i)$  – which is, in theory, estimable to arbitrary precision.

$$E[Y_i] \in [E[Y_i^* | R_i = 1] \cdot \Pr[R_i = 1] + 0 \cdot \Pr[R_i = 0], \\ E[Y_i^* | R_i = 1] \cdot \Pr[R_i = 1] + 1 \cdot \Pr[R_i = 0]] .$$



A natural estimator here substitutes the sample mean for the expected value. Inference can be conducted using the bootstrap.

$$\hat{E}[Y_i] \in \left[ \hat{E}[Y_i^* | R_i = 1] \cdot \hat{P}_r[R_i = 1] + 0 \cdot \hat{P}_r[R_i = 0], \right. \\ \left. \hat{E}[Y_i^* | R_i = 1] \cdot \hat{P}_r[R_i = 1] + 1 \cdot \hat{P}_r[R_i = 0] \right].$$

Note that this will reproduce our bounds estimate of  $[1/2, 5/6]$  from above.

Without *any* further assumptions, these bounds are the absolute best we can do.

Thus, under these minimal assumptions,  $E[Y_i]$  is *partially identified* (a.k.a., set identified) as opposed to *point identified*.

I.e., even with full knowledge of the joint distribution of  $(Y_i^*, R_i)$ , we can do no better than the above bounds. We are left with an interval of plausible values, none of which can be logically ruled out given the assumptions that we have imposed. We cannot figure out an exact value of  $E[Y_i]$ .

Adding more assumptions definitionally reduces the credibility of our estimates. We might be able to narrow our intervals – perhaps even to a point. But the cost is that our estimates are less believable. Manski refers to this as the “Law of Decreasing Credibility.”

Sometimes, we should not feel comfortable strengthening our assumptions. Bounded support of the outcome might be the only assumption that we’re willing to impose given the data that we have.

In this case – if the bounds are not enough – we need to go out and actually be social *scientists*. New data of some sort are necessary. Perhaps by experimenting, perhaps by getting a set of reasonable variables to condition on, or perhaps by learning more so that we can justify stronger (i.e., more restrictive) assumptions.

Let's consider one of these stronger assumptions. Let's assume that the data are *missing completely at random* (MCAR). This is generally considered to be strongest among the nonparametric assumptions that we could impose.

Suppose that

1.  $Y_i \perp\!\!\!\perp R_i$ . (Independence of the outcome and response)
2.  $\Pr(R_i = 1) > 0$ . (Nonzero probability of response)

Or, put another way, the types of people that respond and the types of people that don't respond are *exactly* the same. (Does this assumption ever make sense?)

Note what these conditions imply. By independence,

$$E[Y_i | R_i = 1] = E[Y_i | R_i = 0] = E[Y_i].$$

But we know that

$$E[Y_i^* | R_i = 1] = E[Y_i | R_i = 1].$$

Therefore, we can write  $E[Y_i]$  in terms of observable characteristics:

$$E[Y_i] = E[Y_i^* | R_i = 1].$$

We could then use a simple plug-in estimator. In the above example, our estimate of  $E[Y_i]$  would be 3/4.

One way to think about MCAR is as follows:

Given large enough  $n$ , we can perfectly “fill in” the conditional distribution of outcomes for nonresponders just by looking at the distribution of outcomes for responders. By assumption, the distribution for those who respond is identical to the distribution for those who do not respond. Thus we can identify the unconditional distribution of outcomes.

| Unit | $Y_i^*$ | $R_i$ |
|------|---------|-------|
| 1    | 1       | 1     |
| 2    | -99     | 0     |
| 3    | 1       | 1     |
| 4    | 0       | 1     |
| 5    | 1       | 1     |
| 6    | -99     | 0     |

| Unit | $Y_i$ | $R_i$ |
|------|-------|-------|
| 1    | 1     | 1     |
| 2    | ?     | 0     |
| 3    | 1     | 1     |
| 4    | 0     | 1     |
| 5    | 1     | 1     |
| 6    | ?     | 0     |

One way to think about MCAR is as follows:

Given large enough  $n$ , we can perfectly “fill in” the conditional distribution of outcomes for nonresponders just by looking at the distribution of outcomes for responders. By assumption, the distribution for those who respond is identical to the distribution for those who do not respond. Thus we can identify the unconditional distribution of outcomes.

For binary  $Y_i$ ,

| Unit | $Y_i^*$ | $R_i$ |
|------|---------|-------|
| 1    | 1       | 1     |
| 2    | -99     | 0     |
| 3    | 1       | 1     |
| 4    | 0       | 1     |
| 5    | 1       | 1     |
| 6    | -99     | 0     |

| Unit | $Y_i$                      | $R_i$ |
|------|----------------------------|-------|
| 1    | 1                          | 1     |
| 2    | $\hat{\Pr}(Y_i = 1) = 3/4$ | 0     |
| 3    | 1                          | 1     |
| 4    | 0                          | 1     |
| 5    | 1                          | 1     |
| 6    | $\hat{\Pr}(Y_i = 1) = 3/4$ | 0     |

Note that, although we can fill in the distribution, we *still don't know which values are associated with which observation*. We just have a distributional result.

If we had additional information on each unit, we could impose a “weaker” version of the MCAR assumption known as *missing at random* (MAR).

It's just like MCAR, except *conditional on covariates*. (So not missing *completely* at random, just missing at random once we condition on covariates.)



Suppose that, associated with each unit  $i$ , we had a vector of covariates  $\mathbf{X}_i$ .

Now the random vector of observables is  $(Y_i^*, R_i, \mathbf{X}_i)$ .

The MAR assumption looks just like MCAR: Suppose that

1.  $Y_i \perp\!\!\!\perp R_i | \mathbf{X}_i = \mathbf{x}_i, \forall \mathbf{x}_i \in \text{Supp}(\mathbf{X}_i)$ . (Independence of the outcome and response, *conditional on covariates*). This is a variant of the “conditional independence assumption” (CIA).
2.  $\Pr(R_i = 1 | \mathbf{X}_i = \mathbf{x}_i) > 0, \forall \mathbf{x}_i \in \text{Supp}(\mathbf{X}_i)$ . (Nonzero probability of response, *conditional on covariates*)

Or, put another way, conditional on their measured background characteristics, the types of people that respond and the types of people that don't respond are *exactly* the same.

Every *common* method of dealing with missing data in applied practice depends on a variant of the MAR assumption: multiple imputation, reweighting, etc.

How does MAR facilitate identification of the distribution of  $Y_i$ ?

For clarity, let's assume that  $\mathbf{X}_i$  can only take on a finite number of discrete values. (We don't need this assumption, but it lets us work with PMFs instead of PDFs.)

Further, let's focus on the expected value of  $Y_i$  as an inferential target.

By the Law of Iterated Expectations:

$$E[Y_i] = \sum_{\mathbf{x}_i \in \text{Supp}(\mathbf{X}_i)} E[Y_i | \mathbf{X}_i = \mathbf{x}_i] \Pr[\mathbf{X}_i = \mathbf{x}_i]$$

By the CIA (and stable outcomes),

$$E[Y_i | \mathbf{X}_i = \mathbf{x}_i] = E[Y_i | \mathbf{X}_i = \mathbf{x}_i, R_i = 1] = E[Y_i^* | \mathbf{X}_i = \mathbf{x}_i, R_i = 1].$$

Therefore, if MAR holds,

$$E[Y_i] = \sum_{\mathbf{x}_i \in \text{Supp}(\mathbf{X}_i)} E[Y_i^* | \mathbf{X}_i = \mathbf{x}_i, R_i = 1] \Pr[\mathbf{X}_i = \mathbf{x}_i]$$

Since we can observe the random vector  $(Y_i^*, R_i, \mathbf{X}_i)$ , we can point identify  $E[Y_i^* | \mathbf{X}_i = \mathbf{x}_i, R_i = 1]$  and  $\Pr[\mathbf{X}_i = \mathbf{x}_i]$ . Under MAR,  $E[Y_i]$  is point identified.

Thus under MAR, a simple plug-in estimator (with a discrete set of covariates) would be:

$$\hat{E}[Y_i] = \sum_{\mathbf{x}_i \in \text{Supp}(\mathbf{X}_i)} \hat{E}[Y_i^* | \mathbf{X}_i = \mathbf{x}_i, R_i = 1] \hat{\text{Pr}}[\mathbf{X}_i = \mathbf{x}_i].$$

Let's see this in action. Let's take our example from above, but now assume that we also observe the gender of every respondent.

| Unit | $Y_i^*$ | $R_i$ | $X_i$ |
|------|---------|-------|-------|
| 1    | 1       | 1     | 0     |
| 2    | -99     | 0     | 0     |
| 3    | 1       | 1     | 0     |
| 4    | 0       | 1     | 0     |
| 5    | 1       | 1     | 1     |
| 6    | -99     | 0     | 1     |

| Unit | $Y_i$ | $R_i$ | $X_i$ |
|------|-------|-------|-------|
| 1    | 1     | 1     | 0     |
| 2    | ?     | 0     | 0     |
| 3    | 1     | 1     | 0     |
| 4    | 0     | 1     | 0     |
| 5    | 1     | 1     | 1     |
| 6    | ?     | 0     | 1     |

| Unit | $Y_i$                               | $R_i$ | $X_i$ |
|------|-------------------------------------|-------|-------|
| 1    | 1                                   | 1     | 0     |
| 2    | $\hat{Pr}[Y_i = 1   X_i = 0] = 2/3$ |       | 0     |
| 3    | 1                                   | 1     | 0     |
| 4    | 0                                   | 1     | 0     |
| 5    | 1                                   | 1     | 1     |
| 6    | $\hat{Pr}[Y_i = 1   X_i = 1] = 1$   |       | 0     |

Assuming MAR, a consistent estimate of  $E[Y_i]$  is  $7/9$ .  
 (Computed from plug-in estimator as  $2/3 \cdot 2/3 + 1 \cdot 1/3$ .) Note  
 that our estimate differs from the MCAR estimate ( $3/4$ ).

This is the same logic behind survey reweighting to try to achieve a representative sample.

Suppose that you have undersampled a certain group such that  $\mathbf{X}_i = \mathbf{x}_i$ , and we know the population distribution of  $\mathbf{X}_i$ . If you assume MAR, you can obtain a consistent estimate using the above plug-in estimator.

Similar methods exist for filling in missing *joint* distributions when you also have missingness in  $\mathbf{X}_i$ , not just outcomes. You just need to invoke MAR with respect to all of the nonmissing variables. When you use, e.g., multiple imputation, MAR is the type of assumption that you are invoking.



Suppose we did an Internet survey, and asked many, many US citizens (i) their height (in inches) and (ii) their gender. Assume that no one lies.

Our goal is to figure out the national average height.

The problem is: people who answer Internet surveys may be unusual, both in ways we can measure and in ways we can't.

Let's assume that our data came back looking like this (where  $X_i$  is an indicator variable for gender; by convention, "men are odd"):

$$\begin{aligned}\hat{E}[Y_i^* | R_i = 1, X_i = 0] &= 64 \text{ in.} & \hat{P}_r[X_i = 0 | R_i = 1] &= .30 \\ \hat{E}[Y_i^* | R_i = 1, X_i = 1] &= 70 \text{ in.} & \hat{P}_r[X_i = 1 | R_i = 1] &= .70\end{aligned}$$

The unadjusted estimate of the national average height is

$$\begin{aligned}&\hat{E}[Y_i^* | R_i = 1, X_i = 0] \cdot \hat{P}_r[X_i = 0 | R_i = 1] + \\ &\hat{E}[Y_i^* | R_i = 1, X_i = 1] \cdot \hat{P}_r[X_i = 1 | R_i = 1] = \\ &68.2 \text{ inches.}\end{aligned}$$

This is the estimate that would obtain if we used a plug-in estimator after assuming MCAR.

Let's suppose that the US population is 50% women, so we knew that  $\Pr[X_i = x_i] = 1/2$ , for  $x_i \in \{0, 1\}$ . Under MAR with respect to  $X_i$ :

$$\begin{aligned} E[Y_i] &= \sum_{\mathbf{x}_i \in \text{Supp}(\mathbf{X}_i)} E[Y_i | X_i = x_i] \Pr[X_i = x_i] \\ &= \sum_{\mathbf{x}_i \in \text{Supp}(\mathbf{X}_i)} E[Y_i^* | X_i = x_i, R_i = 1] \Pr[X_i = x_i] \\ &= E[Y_i^* | X_i = 0, R_i = 1] \cdot 1/2 + E[Y_i^* | X_i = 1, R_i = 1] \cdot 1/2. \end{aligned}$$

Then the associated plug-in MAR estimate of  $E[Y_i]$  is 67 inches.

Go to NA Graphs

Potential outcomes and effects for GG example with villages 1, 3, and 6 removed

| Village | $Y_i(0)$    | $Y_i(1)$    | $\tau_i$    |
|---------|-------------|-------------|-------------|
| 2       | 15          | 15          | 0           |
| 4       | 20          | 15          | -5          |
| 5       | 10          | 20          | 10          |
| 7       | 15          | 30          | 15          |
| AVG     | 15          | 20          | 5           |
|         | $E[Y_i(0)]$ | $E[Y_i(1)]$ | $E[\tau_i]$ |

What assumptions have we already made?

20 can be interpreted as the average of the four “treatment” potential outcomes (15, 15, 20, 30), or it can be interpreted as an expectation of a random draw from the four “treatment” potential outcomes.

$$\begin{aligned} E[Y_i(1)] &= \sum y_i(1) \cdot Pr(y_i(1)) \\ &= \frac{1}{4}15 + \frac{1}{4}15 + \frac{1}{4}20 + \frac{1}{4}30 \\ &= 20 \end{aligned}$$

| $s$ | $P(s)$ | $D_2$ | $Y_2$ | $D_4$ | $Y_4$ | $D_5$ | $Y_5$ | $D_7$ | $Y_7$ | $\frac{\sum D_i Y_i}{\sum D_i}$ | $\frac{\sum (1-D_i) Y_i(1)}{\sum (1-D_i)}$ | $\frac{\sum (1-D_i) Y_i}{\sum (1-D_i)}$ | $\frac{\sum D_i Y_i(0)}{\sum D_i}$ |
|-----|--------|-------|-------|-------|-------|-------|-------|-------|-------|---------------------------------|--------------------------------------------|-----------------------------------------|------------------------------------|
| 1   | 1/6    | 1     | 15    | 1     | 15    | 0     | 10    | 0     | 15    | 15                              | 25                                         | 12.5                                    | 17.5                               |
| 2   | 1/6    | 1     | 15    | 0     | 20    | 1     | 20    | 0     | 15    | 17.5                            | 22.5                                       | 17.5                                    | 12.5                               |
| 3   | 1/6    | 1     | 15    | 0     | 20    | 0     | 10    | 1     | 30    | 22.5                            | 17.5                                       | 15                                      | 15                                 |
| 4   | 1/6    | 0     | 15    | 1     | 15    | 1     | 20    | 0     | 15    | 17.5                            | 22.5                                       | 15                                      | 15                                 |
| 5   | 1/6    | 0     | 15    | 1     | 15    | 0     | 10    | 1     | 30    | 22.5                            | 17.5                                       | 12.5                                    | 17.5                               |
| 6   | 1/6    | 0     | 15    | 0     | 20    | 1     | 20    | 1     | 30    | 25                              | 15                                         | 17.5                                    | 12.5                               |
| Avg |        |       |       |       |       |       |       |       |       | 20                              | 20                                         | 15                                      | 15                                 |
|     |        |       |       |       |       |       |       |       |       | $E[Y_i(1) D_i = 1]$             | $E[Y_i(1) D_i = 0]$                        | $E[Y_i(0) D_i = 0]$                     | $E[Y_i(0) D_i = 1]$                |

| $D_2$ | $Y_2$ | $D_4$ | $Y_4$ | $D_5$ | $Y_5$ | $D_7$ | $Y_7$ | $\frac{\sum D_i Y_i}{\sum D_i}$ | $\frac{\sum (1-D_i) Y_i (1)}{\sum (1-D_i)}$ | $\frac{\sum (1-D_i) Y_i}{\sum (1-D_i)}$ | $\frac{\sum D_i Y_i (0)}{\sum D_i}$ |
|-------|-------|-------|-------|-------|-------|-------|-------|---------------------------------|---------------------------------------------|-----------------------------------------|-------------------------------------|
| 1     | 15    | 1     | 15    | 0     | 10    | 0     | 15    | 15                              | 25                                          | 12.5                                    | 17.5                                |
| 1     | 15    | 0     | 20    | 1     | 20    | 0     | 15    | 17.5                            | 22.5                                        | 17.5                                    | 12.5                                |
| 1     | 15    | 0     | 20    | 0     | 10    | 1     | 30    | 22.5                            | 17.5                                        | 15                                      | 15                                  |
| 0     | 15    | 1     | 15    | 1     | 20    | 0     | 15    | 17.5                            | 22.5                                        | 15                                      | 15                                  |
| 0     | 15    | 1     | 15    | 0     | 10    | 1     | 30    | 22.5                            | 17.5                                        | 12.5                                    | 17.5                                |
| 0     | 15    | 0     | 20    | 1     | 20    | 1     | 30    | 25                              | 15                                          | 17.5                                    | 12.5                                |
|       |       |       |       |       |       |       |       | 20                              | 20                                          | 15                                      | 15                                  |



| Village                                        | $Y_i(0)$ | $1 - D_i$ | $Y_i(1)$ | $D_i$ | $Y_i$ |
|------------------------------------------------|----------|-----------|----------|-------|-------|
| 2                                              | 15       | 1         | 15       | 0     | 15    |
| 4                                              | 20       | 1         | 15       | 0     | 20    |
| 5                                              | 10       | 0         | 20       | 1     | 20    |
| 7                                              | 15       | 0         | 30       | 1     | 30    |
| $\frac{\sum D_i Y_i}{\sum D_i}$                |          |           |          |       | 25    |
| $\frac{\sum (1 - D_i) Y_i(1)}{\sum (1 - D_i)}$ |          |           | 15       |       |       |
| $\frac{\sum (1 - D_i) Y_i}{\sum (1 - D_i)}$    |          |           |          |       | 17.5  |
| $\frac{\sum D_i Y_i(0)}{\sum D_i}$             | 12.5     |           |          |       |       |

$$\begin{aligned}
 E[Y_i(1)|D_i = 1] &= \sum_{s=1}^6 \frac{\sum d_i y_i}{\sum d_i} \cdot Pr(s = 6) \\
 &= \frac{1}{6}15 + \frac{1}{6}17.5 + \frac{1}{6}22.5 + \frac{1}{6}17.5 + \frac{1}{6}22.5 + \frac{1}{6}25 \\
 &= 20
 \end{aligned}$$

From equation 2.15 of GG,

$$\begin{aligned}
 E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 0] &= E[Y_i(1) - Y_i(0)|D_i = 1] \\
 &\quad + \underbrace{E[Y_i(0)|D_i = 1] - E[Y_i(0)|D_i = 0]}_{\text{SelectionBias}}
 \end{aligned}$$

| $s$ | $P(s)$ | $D_2$ | $Y_2$ | $D_4$ | $Y_4$ | $D_5$ | $Y_5$ | $D_7$ | $Y_7$ | $\frac{\sum D_i Y_i}{\sum D_i}$ | $\frac{\sum (1-D_i) Y_i(1)}{\sum (1-D_i)}$ | $\frac{\sum (1-D_i) Y_i}{\sum (1-D_i)}$ | $\frac{\sum D_i Y_i(0)}{\sum D_i}$ |
|-----|--------|-------|-------|-------|-------|-------|-------|-------|-------|---------------------------------|--------------------------------------------|-----------------------------------------|------------------------------------|
| 1   | 0/6    | 1     | 15    | 1     | 15    | 0     | 10    | 0     | 15    | 15                              | 25                                         | 12.5                                    | 17.5                               |
| 2   | 0/6    | 1     | 15    | 0     | 20    | 1     | 20    | 0     | 15    | 17.5                            | 22.5                                       | 17.5                                    | 12.5                               |
| 3   | 0/6    | 1     | 15    | 0     | 20    | 0     | 10    | 1     | 30    | 22.5                            | 17.5                                       | 15                                      | 15                                 |
| 4   | 0/6    | 0     | 15    | 1     | 15    | 1     | 20    | 0     | 15    | 17.5                            | 22.5                                       | 15                                      | 15                                 |
| 5   | 0/6    | 0     | 15    | 1     | 15    | 0     | 10    | 1     | 30    | 22.5                            | 17.5                                       | 12.5                                    | 17.5                               |
| 6   | 6/6    | 0     | 15    | 0     | 20    | 1     | 20    | 1     | 30    | 25                              | 15                                         | 17.5                                    | 12.5                               |
| Avg |        |       |       |       |       |       |       |       |       | 25<br>$E[Y_i(1) D_i = 1]$       | 15<br>$E[Y_i(1) D_i = 0]$                  | 17.5<br>$E[Y_i(0) D_i = 0]$             | 12.5<br>$E[Y_i(0) D_i = 1]$        |

| $D_2$ | $Y_2$ | $D_4$ | $Y_4$ | $D_5$ | $Y_5$ | $D_7$ | $Y_7$ | $\frac{\sum D_i Y_i}{\sum D_i}$ | $\frac{\sum (1-D_i) Y_i(1)}{\sum (1-D_i)}$ | $\frac{\sum (1-D_i) Y_i}{\sum (1-D_i)}$ | $\frac{\sum D_i Y_i(0)}{\sum D_i}$ |
|-------|-------|-------|-------|-------|-------|-------|-------|---------------------------------|--------------------------------------------|-----------------------------------------|------------------------------------|
| 1     | 15    | 1     | 15    | 0     | 10    | 0     | 15    | 15                              | 25                                         | 12.5                                    | 17.5                               |
| 1     | 15    | 0     | 20    | 1     | 20    | 0     | 15    | 17.5                            | 22.5                                       | 17.5                                    | 12.5                               |
| 1     | 15    | 0     | 20    | 0     | 10    | 1     | 30    | 22.5                            | 17.5                                       | 15                                      | 15                                 |
| 0     | 15    | 1     | 15    | 1     | 20    | 0     | 15    | 17.5                            | 22.5                                       | 15                                      | 15                                 |
| 0     | 15    | 1     | 15    | 0     | 10    | 1     | 30    | 22.5                            | 17.5                                       | 12.5                                    | 17.5                               |
| 0     | 15    | 0     | 20    | 1     | 20    | 1     | 30    | 25                              | 15                                         | 17.5                                    | 12.5                               |
|       |       |       |       |       |       |       |       | 20                              | 20                                         | 15                                      | 15                                 |

## Unbiasedness with RCTs

With random treatment assignment in large random samples from large populations, it is straightforward to demonstrate that a difference-in-means estimator is unbiased for the population average treatment effect, because the treated units in the sample are a random sample from  $Y(1)$  in the population, and the control units in the sample are a random sample from  $Y(0)$  in the population.

## Unbiasedness with RCTs

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On Monday, we showed that a difference in means estimator is also unbiased for the sample average treatment effect and the sample average treatment effect on the treated.

## Variance with RCTs

With random treatment assignment in large random samples from large populations, it is straightforward to demonstrate that a standard variance estimator for the difference-in-means estimator is consistent for the population average treatment effect, because the treated units in the sample are a random sample from  $Y(1)$  in the population, and the control units in the sample are a random sample from  $Y(0)$  in the population.

## Variance with RCTs

With random treatment assignment in large random samples from large populations, it is straightforward to demonstrate that a standard variance estimator for the difference-in-means estimator is consistent for the population average treatment effect, because the treated units in the sample are a random sample from  $Y(1)$  in the population, and the control units in the sample are a random sample from  $Y(0)$  in the population.

Today we will show that the standard (stat 101) variance estimator for a difference-in-means estimator is conservative for the sample average treatment effect, and accurate when effects are constant. We will also show that



$$\hat{\tau} = \frac{\sum D_i Y_i}{\sum D_i} - \frac{\sum (1 - D_i) Y_i}{\sum (1 - D_i)}$$

$$V_{Stat101}[\hat{\tau}] = \frac{\sigma_t^2}{n_t} + \frac{\sigma_c^2}{n_c}$$

| Village | $Y_i(0)$ | $Y_i(1)$ | $\tau_i$ |
|---------|----------|----------|----------|
| 2       | 15       | 15       | 0        |
| 4       | 20       | 15       | -5       |
| 5       | 10       | 20       | 10       |
| 7       | 15       | 30       | 15       |
| $E[]$   | 15       | 20       | 5        |
| $V[]$   | 12.5     | 37.5     | 62.5     |

$$\begin{aligned}
 \sigma_t^2 &\equiv V[Y_i(1)] \\
 &= \sum (y_i(1) - E[Y_i(1)])^2 \cdot Pr(y_i(1)) \\
 &= (15 - 20)^2 \cdot \frac{1}{4} + (15 - 20)^2 \cdot \frac{1}{4} + (20 - 20)^2 \cdot \frac{1}{4} + (30 - 20)^2 \cdot \frac{1}{4} \\
 &= 37.5
 \end{aligned}$$

| Village | $Y_i(0)$ | $Y_i(1)$ | $\tau_i$ |
|---------|----------|----------|----------|
| 2       | 15       | 15       | 0        |
| 4       | 20       | 15       | -5       |
| 5       | 10       | 20       | 10       |
| 7       | 15       | 30       | 15       |
| $E[]$   | 15       | 20       | 5        |
| $V[]$   | 12.5     | 37.5     | 62.5     |

$$\begin{aligned}
 \sigma_c^2 &\equiv V[Y_i(0)] \\
 &= \sum (y_i(0) - E[Y_i(0)])^2 \cdot Pr(y_i(0)) \\
 &= (15 - 15)^2 \cdot \frac{1}{4} + (20 - 15)^2 \cdot \frac{1}{4} + (10 - 15)^2 \cdot \frac{1}{4} + (15 - 15)^2 \cdot \frac{1}{4} \\
 &= 12.5
 \end{aligned}$$

| $D_2$ | $Y_2$ | $D_4$ | $Y_4$ | $D_5$ | $Y_5$ | $D_7$ | $Y_7$ | $\frac{\sum D_i Y_i}{\sum D_i}$ | $\frac{\sum (1-D_i) Y_i}{\sum (1-D_i)}$ | $\hat{\tau}$ |
|-------|-------|-------|-------|-------|-------|-------|-------|---------------------------------|-----------------------------------------|--------------|
| 1     | 15    | 1     | 15    | 0     | 10    | 0     | 15    | 15                              | 12.5                                    | 2.5          |
| 1     | 15    | 0     | 20    | 1     | 20    | 0     | 15    | 17.5                            | 17.5                                    | 0            |
| 1     | 15    | 0     | 20    | 0     | 10    | 1     | 30    | 22.5                            | 15                                      | 7.5          |
| 0     | 15    | 1     | 15    | 1     | 20    | 0     | 15    | 17.5                            | 15                                      | 2.5          |
| 0     | 15    | 1     | 15    | 0     | 10    | 1     | 30    | 22.5                            | 12.5                                    | 10           |
| 0     | 15    | 0     | 20    | 1     | 20    | 1     | 30    | 25                              | 17.5                                    | 7.5          |
|       |       |       |       |       |       |       |       | 20                              | 15                                      | 5            |
|       |       |       |       |       |       |       |       | 12.5                            | 4.1667                                  | 12.5         |

$$\begin{aligned}
 V\left[\frac{\sum D_i Y_i}{\sum D_i}\right] &= \frac{\sigma_t^2}{m} \cdot \frac{N-m}{N-1} \\
 &= \frac{37.5}{2} \cdot \frac{4-2}{4-1} \\
 &= 12.5
 \end{aligned}$$

$$\begin{aligned}
 V\left[\frac{\sum (1-D_i) Y_i}{\sum (1-D_i)}\right] &= \frac{\sigma_c^2}{N-m} \cdot \frac{m}{N-1} \\
 &= \frac{12.5}{2} \cdot \frac{2}{4-1} \\
 &= 4.1667
 \end{aligned}$$

$$\begin{aligned}
 V[\hat{\tau}] &= \frac{N\sigma_t^2}{(N-1)m} + \frac{N\sigma_c^2}{(N-1)(N-m)} - \frac{\sigma_\tau^2}{(N-1)} \\
 &= \frac{4 \cdot 37.5}{(4-1) \cdot 2} + \frac{4 \cdot 12.5}{(4-1)(4-2)} - \frac{62.5}{(4-1)} \\
 &= 12.5
 \end{aligned}$$

$$\begin{aligned}
 V[\hat{\tau}] &= \sum (\hat{\tau} - E[\hat{\tau}])^2 \cdot Pr(\hat{\tau}) \\
 &= (2.5 - 5)^2 \cdot \frac{1}{6} + (0 - 5)^2 \cdot \frac{1}{6} + (7.5 - 5)^2 \cdot \frac{1}{6} \\
 &\quad + (2.5 - 5)^2 \cdot \frac{1}{6} + (10 - 5)^2 \cdot \frac{1}{6} + (7.5 - 5)^2 \cdot \frac{1}{6} \\
 &= 12.5
 \end{aligned}$$

$$\begin{aligned}
 V[\hat{\tau}] &= \frac{N\sigma_t^2}{(N-1)m} + \frac{N\sigma_c^2}{(N-1)(N-m)} - \frac{\sigma_\tau^2}{(N-1)} \\
 &\leq \frac{N\sigma_t^2}{(N-1)m} + \frac{N\sigma_c^2}{(N-1)(N-m)}
 \end{aligned}$$

$$\begin{aligned}
 V[\hat{\tau}] &\leq \frac{N\sigma_t^2}{(N-1)m} + \frac{N\sigma_c^2}{(N-1)(N-m)} \\
 &\approx \frac{\sigma_t^2}{m} + \frac{\sigma_c^2}{(N-m)} \\
 &= \frac{\sigma_t^2}{n_t} + \frac{\sigma_c^2}{n_c}
 \end{aligned}$$

$$\hat{\sigma}_t^2 \equiv \hat{V}[Y_i(1)]$$

$$= \frac{1}{m-1} \sum_{i=1}^N D_i \left( Y_i - \frac{\sum D_i Y_i}{\sum D_i} \right)^2$$

$$\hat{\sigma}_c^2 \equiv \hat{V}[Y_i(0)]$$

$$= \frac{1}{(N-m)-1} \sum_{i=1}^N (1-D_i) \left( Y_i - \frac{\sum D_i Y_i}{\sum D_i} \right)^2$$

## Example

Suppose we observe two treated subjects ( $D = 1$ ) and four control subjects ( $D = 0$ ) with outcomes ( $Y$ ) between 0 and 100.

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | ?        | ?                 |
| 2   | 1     | 80    | 80       | ?        | ?                 |
| 3   | 0     | 60    | ?        | 60       | ?                 |
| 4   | 0     | 70    | ?        | 70       | ?                 |
| 5   | 0     | 70    | ?        | 70       | ?                 |
| 6   | 0     | 80    | ?        | 80       | ?                 |



## Example

Suppose we observe two treated subjects ( $D = 1$ ) and four control subjects ( $D = 0$ ) with outcomes ( $Y$ ) between 0 and 100.

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | ?        | ?                 |
| 2   | 1     | 80    | 80       | ?        | ?                 |
| 3   | 0     | 60    | ?        | 60       | ?                 |
| 4   | 0     | 70    | ?        | 70       | ?                 |
| 5   | 0     | 70    | ?        | 70       | ?                 |
| 6   | 0     | 80    | ?        | 80       | ?                 |

What is the “naive” estimate of the average  $Y_i(1) - Y_i(0)$ ?

## Example

Suppose we observe two treated subjects ( $D = 1$ ) and four control subjects ( $D = 0$ ) with outcomes ( $Y$ ) between 0 and 100.

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | ?        | ?                 |
| 2   | 1     | 80    | 80       | ?        | ?                 |
| 3   | 0     | 60    | ?        | 60       | ?                 |
| 4   | 0     | 70    | ?        | 70       | ?                 |
| 5   | 0     | 70    | ?        | 70       | ?                 |
| 6   | 0     | 80    | ?        | 80       | ?                 |

What is the “naive” estimate of the average  $Y_i(1) - Y_i(0)$ ?

$$\frac{70 + 80}{2} - \frac{60 + 70 + 70 + 80}{4} = 75 - 70$$

## Example: Randomized Experiment

Suppose subjects are randomly assigned to  $D = 0$  and  $D = 1$  (also imagine these six subjects actually represent 600 subjects), and we measure a “pre-treatment” covariate (i.e., not affected by  $D$ ).

| $i$ | $X_i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | ?        | ?                 |
| 2   | 1     | 1     | 80    | 80       | ?        | ?                 |
| 3   | 1     | 0     | 60    | ?        | 60       | ?                 |
| 4   | 0     | 0     | 70    | ?        | 70       | ?                 |
| 5   | 0     | 0     | 70    | ?        | 70       | ?                 |
| 6   | 1     | 0     | 80    | ?        | 80       | ?                 |

Randomization implies that  $X$  is independent of  $D$ , which implies that  $E[X|D = 1] = E[X|D = 0]$ .

# Balance on Pre-treatment Variables in the Gerber, Green, and Larimer Experiment

**TABLE 1. Relationship between Treatment Group Assignment and Covariates (Household-Level Data)**

|                | Control | Civic Duty | Hawthorne | Self   | Neighbors |
|----------------|---------|------------|-----------|--------|-----------|
|                | Mean    | Mean       | Mean      | Mean   | Mean      |
| Household size | 1.91    | 1.91       | 1.91      | 1.91   | 1.91      |
| Nov 2002       | .83     | .84        | .84       | .84    | .84       |
| Nov 2000       | .87     | .87        | .87       | .86    | .87       |
| Aug 2004       | .42     | .42        | .42       | .42    | .42       |
| Aug 2002       | .41     | .41        | .41       | .41    | .41       |
| Aug 2000       | .26     | .27        | .26       | .26    | .26       |
| Female         | .50     | .50        | .50       | .50    | .50       |
| Age (in years) | 51.98   | 51.85      | 51.87     | 51.91  | 52.01     |
| N =            | 99,999  | 20,001     | 20,002    | 20,000 | 20,000    |

*Note:* Only registered voters who voted in November 2004 were selected for our sample. Although not included in the table, there were no significant differences between treatment group assignment and covariates measuring race and ethnicity.

## Potential Outcomes are Pre-Treatment Variables

Suppose subjects are randomly assigned to  $D = 0$  and  $D = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$ ?

## Potential Outcomes are Pre-Treatment Variables

Suppose subjects are randomly assigned to  $D = 0$  and  $D = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$ ?

$$\frac{70 + 80 + 70 + 80 + 70 + 80}{6} = 75$$

## Potential Outcomes are Pre-Treatment Variables

Suppose subjects are randomly assigned to  $D = 0$  and  $D = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $D = 1$  subjects?

## Potential Outcomes are Pre-Treatment Variables

Suppose subjects are randomly assigned to  $D = 0$  and  $D = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $D = 1$  subjects?

$$\frac{70 + 80}{2} = 75$$



## Potential Outcomes are Pre-Treatment Variables

Suppose subjects are randomly assigned to  $D = 0$  and  $D = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(0)$ ?

## Potential Outcomes are Pre-Treatment Variables

Suppose subjects are randomly assigned to  $D = 0$  and  $D = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(0)$ ?

$$\frac{70 + 70 + 60 + 70 + 70 + 80}{6} = 70$$

## Potential Outcomes are Pre-Treatment Variables

Suppose subjects are randomly assigned to  $D = 0$  and  $D = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(0)$  among the  $D = 0$  subjects?

## Potential Outcomes are Pre-Treatment Variables

Suppose subjects are randomly assigned to  $D = 0$  and  $D = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(0)$  among the  $D = 0$  subjects?

$$\frac{60 + 70 + 70 + 80}{4} = 70$$

## The ATE is the Difference in Average Potential Outcomes

Suppose subjects are randomly assigned to  $D = 0$  and  $D = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$ ?

# The ATE is the Difference in Average Potential Outcomes

Suppose subjects are randomly assigned to  $D = 0$  and  $D = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$ ?

$$\frac{70 + 80 + 70 + 80 + 70 + 80}{6} = 75$$

# The ATE is the Difference in Average Potential Outcomes

Suppose subjects are randomly assigned to  $D = 0$  and  $D = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(0)$ ?

# The ATE is the Difference in Average Potential Outcomes

Suppose subjects are randomly assigned to  $D = 0$  and  $D = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(0)$ ?

$$\frac{70 + 70 + 60 + 70 + 70 + 80}{6} = 70$$



## The ATE is the Difference in Average Potential Outcomes

Suppose subjects are randomly assigned to  $D = 0$  and  $D = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1) - Y_i(0)$ ?

## The ATE is the Difference in Average Potential Outcomes

Suppose subjects are randomly assigned to  $D = 0$  and  $D = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1) - Y_i(0)$ ?

$$75 - 70 = \frac{0 + 10 + 10 + 10 + 0 + 0}{6} = 5$$

## Exchangeability

With a randomized experiment, the exchangeability/unconfoundedness assumption allows us to relate average potential outcomes to average observed outcomes.

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$$\begin{aligned} E[Y_i(1)] &= E[Y_i(1)|D_i = 1] \\ &= E[Y_i|D_i = 1] \end{aligned}$$

## Exchangeability

With a randomized experiment, the exchangeability/unconfoundedness assumption allows us to relate average potential outcomes to average observed outcomes.

$$\begin{aligned} E[Y_i(1)] &= E[Y_i(1)|D_i = 1] \\ &= E[Y_i|D_i = 1] \end{aligned}$$

$$\begin{aligned} E[Y_i(0)] &= E[Y_i(0)|D_i = 0] \\ &= E[Y_i|D_i = 0] \end{aligned}$$

# Estimation with Randomized Experiments

Recall that with a binary  $D$ , we can always write the conditional expectation as a linear function,

$$E[Y|D = d] = \beta_0 + \beta_1 \cdot d$$

$$E[Y|D = 0] = \beta_0$$

$$E[Y|D = 1] = \beta_0 + \beta_1$$

Therefore, when  $D$  is randomized,  $\beta_1$  from the LCEF can be interpreted as the ATE. This means that the regressions we previously used for descriptive inference with have a causal interpretation in this case.

# Statistical Inference with Randomized Experiments

If treatment is randomized and the descriptive assumptions hold, then the CIs and tests we developed for descriptive inference will be approximately correct.

# Statistical Inference with Randomized Experiments

If treatment is randomized and the descriptive assumptions hold, then the CIs and tests we developed for descriptive inference will be approximately correct.

If instead the sample is the population, then the CIs and tests we developed for descriptive inference will be conservative in the sense that

- ▶  $(1 - \alpha)\%$  CIs will have greater than  $(1 - \alpha)\%$  coverage.
- ▶  $\alpha$  level tests will actually have probabilities of Type I error less than  $\alpha$ .
- ▶ Reported  $p$ -values will be larger than the true  $p$ -values.



# Potential Outcomes and the Classical Econometric Linear Model

$$\begin{aligned}Y_i &= D_i Y_i(1) + (1 - D_i) Y_i(0) \\&= Y_i(0) + (Y_i(1) - Y_i(0)) D_i \\&= \beta_{0i} + \beta_{1i} D_i \\&= \beta_0 + \beta_{1i} D_i + \epsilon_i\end{aligned}$$

If effects are constant across individuals (i.e.,  $\beta_{1i} = \beta_1$  for all  $i$ ), then

$$\begin{aligned}Y_i &= \beta_0 + \beta_{1i} D_i + \epsilon_i \\&= \beta_0 + \beta_1 D_i + \epsilon_i\end{aligned}$$

## Unconfoundedness vs Errors Terms

With the previous definitions and constant effects, we can re-write the potential outcomes as the following:

$$Y_i(1) = \beta_0 + \beta_1 + \epsilon_i$$

$$Y_i(0) = \beta_0 + \epsilon_i$$

This allows us to relate the unconfoundedness assumption to the classical error terms assumption.

$$E[Y_i(1)] = E[Y_i(1)|D_i = 1]$$

$$\beta_0 + \beta_1 + E[\epsilon_i] = \beta_0 + \beta_1 + E[\epsilon_i|D = 1]$$

Therefore we have unconfoundedness when a classical error terms assumption holds (i.e.,  $Cov(D_i, \epsilon_i) = 0$  for all  $i$ )

## Unconfoundedness vs Errors Terms

With the previous definitions and constant effects, we can re-write the potential outcomes as the following:

$$Y_i(1) = \beta_0 + \beta_1 + \epsilon_i$$

$$Y_i(0) = \beta_0 + \epsilon_i$$

This allows us to relate the unconfoundedness assumption to the classical error terms assumption.

$$E[Y_i(0)] = E[Y_i(0)|D_i = 1]$$

$$\beta_0 + E[\epsilon_i] = \beta_0 + E[\epsilon_i|D = 1]$$

Therefore we have unconfoundedness when a classical error terms assumption holds (i.e.,  $Cov(D_i, \epsilon_i) = 0$  for all  $i$ )

Multivalued and continuous treatments

Including covariates in a regression for marginally randomized experiment:

- ▶ the diff-in-means estimator can be expressed as a regression slope
- ▶ DiD estimation (next slide)
- ▶ Use FWL/regression anatomy logic to explain why including covariates in the regression won't affect estimated ATE in large samples.
- ▶ Bias/Var tradeoff in small samples

Including covariates via DiD in marginally randomized experiment

$$\begin{aligned} E(\widehat{ATE}) &= E[Y_i - X_i | D_i = 1] - E[Y_i - X_i | D_i = 0] \\ &= E[Y_i | D_i = 1] - E[Y_i | D_i = 0] \\ &\quad + (E[X_i | D_i = 0] - E[X_i | D_i = 1]) \\ &= E[Y_i(1) | D_i = 1] - E[Y_i(0) | D_i = 0] \\ &= E[Y_i(1)] - E[Y_i(0)] \end{aligned}$$

DiD in marginally randomized experiment has smaller variance when

$$\frac{\text{Cov}(Y_i(0), X_i)}{V(X_i)} + \frac{\text{Cov}(Y_i(1), X_i)}{V(X_i)} > 1$$

Suppose we had randomized within two blocks  $\{2,4\}$  and  $\{5,7\}$ . It corresponds to the removal of rows  $s = 1$  and  $s = 6$  from the table of possible randomizations. Note however that the new  $\hat{\tau}$  is still unbiased.

| Village | $Y_i(0)$ | $Y_i(1)$ | $\tau_i$ |
|---------|----------|----------|----------|
| 2       | 15       | 15       | 0        |
| 4       | 20       | 15       | -5       |
| 5       | 10       | 20       | 10       |
| 7       | 15       | 30       | 15       |
| $E[]$   | 15       | 20       | 5        |



| $s$   | $P(s)$ | $D_2$ | $Y_2$ | $D_4$ | $Y_4$ | $D_5$ | $Y_5$ | $D_7$ | $Y_7$ | $\hat{A}TE_1$ | $\hat{A}TE_2$ | $\hat{A}TE$ |
|-------|--------|-------|-------|-------|-------|-------|-------|-------|-------|---------------|---------------|-------------|
| 2     | 1/6    | 1     | 15    | 0     | 20    | 1     | 20    | 0     | 15    | -5            | 5             | 0           |
| 3     | 1/6    | 1     | 15    | 0     | 20    | 0     | 10    | 1     | 30    | -5            | 20            | 7.5         |
| 4     | 1/6    | 0     | 15    | 1     | 15    | 1     | 20    | 0     | 15    | 0             | 5             | 2.5         |
| 5     | 1/6    | 0     | 15    | 1     | 15    | 0     | 10    | 1     | 30    | 0             | 20            | 10          |
| $E[]$ |        |       |       |       |       |       |       |       |       |               |               | 5           |

## Example: Conditionally Randomized Experiments

If treatments are randomly assigned conditionally on another variable, then the average  $Y_i(1)$  among the  $D_i = 1$  subjects **may not** equal the average  $Y_i(1)$  among the  $D_i = 0$  subjects.

| $i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 60       | 60       | 0                 |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1) - Y_i(0)$ ?

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| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 60       | 60       | 0                 |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1) - Y_i(0)$ ?

$$\begin{aligned}\frac{0 + 10 + 0 + 10 + 0 + 0}{6} &= \frac{20}{6} \\ &\approx 3.3 \\ &\neq 75 - 70\end{aligned}$$

## Example: Conditionally Randomized Experiment

Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

| $i$ | $X_i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | ?        | ?                 |
| 2   | 1     | 1     | 80    | 80       | ?        | ?                 |
| 3   | 0     | 0     | 60    | ?        | 60       | ?                 |
| 4   | 1     | 0     | 70    | ?        | 70       | ?                 |
| 5   | 0     | 0     | 70    | ?        | 70       | ?                 |
| 6   | 0     | 0     | 80    | ?        | 80       | ?                 |

## Example: Conditionally Randomized Experiment

Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

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|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $X = 0$  subjects?

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Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

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| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $X = 0$  subjects?

$$\frac{70 + 60 + 70 + 80}{4} = 70$$

## Example: Conditionally Randomized Experiment

Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

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|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $X = 0, D = 1$  subjects?

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Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

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| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $X = 0, D = 1$  subjects?

$$\frac{70}{1} = 70$$



## Example: Conditionally Randomized Experiment

Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

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|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $X = 1$  subjects?

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| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $X = 1$  subjects?

$$\frac{80 + 80}{2} = 80$$

## Example: Conditionally Randomized Experiment

Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

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|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
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What is the average  $Y_i(1)$  among the  $X = 1, D = 1$  subjects?

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Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

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| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
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| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $X = 1$ ,  $D = 1$  subjects?

$$\frac{80}{1} = 80$$

## Example: Conditionally Randomized Experiment

If we analyze each experiment separately, and average the results by the proportion in each experiment, then we get the ATE.

| $i$ | $X_i$ | $D_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | ?        | ?                 |
| 2   | 1     | 1     | 80    | 80       | ?        | ?                 |
| 3   | 0     | 0     | 60    | ?        | 60       | ?                 |
| 4   | 1     | 0     | 70    | ?        | 70       | ?                 |
| 5   | 0     | 0     | 70    | ?        | 70       | ?                 |
| 6   | 0     | 0     | 80    | ?        | 80       | ?                 |

$$\frac{4}{6} \cdot \left\{ \frac{70}{1} - \frac{60 + 70 + 80}{3} \right\} + \frac{2}{6} \cdot \left\{ \frac{80}{1} - \frac{70}{1} \right\} \\ = \frac{20}{6}$$

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| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
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| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

$$\begin{aligned} \frac{4}{6} \cdot \left\{ \frac{70}{1} - \frac{60 + 70 + 80}{3} \right\} + \frac{2}{6} \cdot \left\{ \frac{80}{1} - \frac{70}{1} \right\} \\ = \frac{20}{6} = \frac{10 + 10}{6} \end{aligned}$$

## Conditional Exchangeability

With a randomized experiment, the conditional exchangeability/unconfoundedness assumption allows us to relate average potential outcomes to average observed outcomes.

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$$\begin{aligned}E[Y_i(0)|X = x] &= E[Y_i(0)|D = 0, X = x] \\&= E[Y|D = 0, X = x]\end{aligned}$$

# Estimation with Conditionally Randomized Experiments

With a binary  $D$  and a binary  $X$ , we can always write the conditional expectation as a linear interactive function,

$$E[Y|D = d, X = x] = \beta_0 + \beta_1 \cdot d + \beta_2 \cdot x + \beta_3 \cdot d \cdot x$$

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$$E[Y|D = 0, X = 1] = \beta_0 + \beta_2$$

$$E[Y|D = 1, X = 1] = \beta_0 + \beta_1 + \beta_2 + \beta_3$$

# Interpretation with Conditionally Randomized Experiments

Therefore, when  $D$  is randomized conditionally on  $X$ ,

$$\begin{aligned} E[Y|D = 1, X = 0] - E[Y|D = 0, X = 0] &= (\beta_0 + \beta_1) - \beta_0 \\ &= \beta_1 \end{aligned}$$

can be interpreted as the ATE among the  $X = 0$  subjects.

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$$\begin{aligned} E[Y|D = 1, X = 1] - E[Y|D = 0, X = 1] &= \\ (\beta_0 + \beta_1 + \beta_2 + \beta_3) - (\beta_0 + \beta_2) &= \\ = \beta_1 + \beta_3 \end{aligned}$$

can be interpreted as the ATE among the  $X = 1$  subjects.



# Interpretation with Conditionally Randomized Experiments

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can be interpreted as the ATE among the  $X = 1$  subjects.

$$\begin{aligned} E[\beta_1 + \beta_3 \cdot X] &= \beta_1 + \beta_3 \cdot E[X] \\ &= (1 - E[X])\beta_1 + E[X](\beta_1 + \beta_3) \end{aligned}$$

can be interpreted as the ATE

# Statistical Inference with Randomized Experiments

If  $\beta_3 = 0$ , and the four descriptive assumptions hold, then we can get approximate CIs and tests for the ATE by running the additive regression of  $Y$  on  $D$  and  $X$ .

# Statistical Inference with Randomized Experiments

If  $\beta_3 = 0$ , and the four descriptive assumptions hold, then we can get approximate CIs and tests for the ATE by running the additive regression of  $Y$  on  $D$  and  $X$ .

If  $\beta_3 \neq 0$ , and the four descriptive assumptions hold then we can get approximate CIs and tests for the ATE by subtracting the mean  $E[X]$  off of all  $X$  values, running the interactive regression of  $Y$  on  $D$ ,  $X - E[X]$ , and  $D \cdot (X - E[X])$  and using the estimate and standard error associated with coefficient on  $D$ .

# Statistical Inference with Randomized Experiments (Cont.)

If  $\beta_3 \neq 0$ , and the four descriptive assumptions hold then we can get approximate CIs and tests for the ATE by weighting with  $w_{ij} = \frac{D_i}{p_{ij}} + \frac{1-D_i}{1-p_{ij}}$  where  $p_{ij} = m_j/N_j$ .

# Statistical Inference with Randomized Experiments (Cont.)

If  $\beta_3 \neq 0$ , and the four descriptive assumptions hold then we can get approximate CIs and tests for the ATE by weighting with  $w_{ij} = \frac{D_i}{p_{ij}} + \frac{1-D_i}{1-p_{ij}}$  where  $p_{ij} = m_j/N_j$ .

If  $\beta_3 \neq 0$ , and we fail to weight or use interactions, then ATE in each block is weighted by  $(N_j/N)P_j(1 - P_j)$ .

# Interpreting the Multiple Regression Model

Consider the classical multiple regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + \epsilon_i$$

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- ▶ Should we interpret both  $\beta_1$  and  $\beta_2$  as causal effects?
- ▶ If so, would this interpretation be justified if  $X_1$  were randomized conditionally on  $X_2$ ?

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- ▶ Should we interpret both  $\beta_1$  and  $\beta_2$  as causal effects?
- ▶ If so, would this interpretation be justified if  $X_1$  were randomized conditionally on  $X_2$ ?
- ▶ What randomized experiment would justify this interpretation?

We will start with the case of a binary treatment again. Assume SUTVA as usual.

Suppose that

1.  $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i | \mathbf{X}_i = \mathbf{x}_i, \forall \mathbf{x}_i \in \text{Supp}(\mathbf{X}_i)$ . (Conditional independence assumption)
2.  $0 < \Pr(D_i = 1 | \mathbf{X}_i = \mathbf{x}_i) < 1, \forall \mathbf{x}_i \in \text{Supp}(\mathbf{X}_i)$ . (Positivity, logically equivalent to a population analogue of “common support”)

These assumptions, together, are usually referred to as *strong ignorability*. MAR is also a type of ignorability assumption. Loosely speaking, ignorability implies that, given the information in the sample, we can safely “ignore” missing data by conditioning on a set of other variables.

The conditional independence assumption is to MAR what independence is to MCAR.

It's random assignment, but *only* conditional on the values of the covariates. It states that, once we condition on  $\mathbf{X}_i$ ,  $D_i$  reveals nothing about the distribution of potential outcomes.

For example, if we measured age and gender, we could say "for people who are 40 years old and male, the CIA implies that there's no systematic differences in who gets the treatment and who does not get the treatment. But we still allow for the possibility that 40-year-olds may be more or less likely to get the treatment than are 50-year-olds."

Put another way, the CIA implies that, once we've controlled for observable background characteristics, the treatment reveals absolutely nothing about potential outcomes.

Some immediate implications, by analogy to MAR.

$\forall \mathbf{x}_i \in \text{Supp}(\mathbf{X}_i)$  and for  $d \in \{0, 1\}$ :

$$E[Y_i(d)|\mathbf{X}_i = \mathbf{x}_i] = E[Y_i|D_i = d, \mathbf{X}_i = \mathbf{x}_i].$$

Thus the *conditional average treatment effect* (CATE), or average treatment effect for units conditional on  $\mathbf{X}_i = \mathbf{x}_i$  is point identified:

$$E[\tau_i|\mathbf{X}_i = \mathbf{x}_i] = E[Y_i(1)|\mathbf{X}_i = \mathbf{x}_i] - E[Y_i(0)|\mathbf{X}_i = \mathbf{x}_i].$$

Applying the Law of Iterated Expectations,

$$E[Y_i(d)] = \sum_{\mathbf{x}_i \in \text{Supp}(\mathbf{X}_i)} E[Y_i|D_i = d, \mathbf{X}_i = \mathbf{x}_i] \Pr[\mathbf{X}_i = \mathbf{x}_i].$$

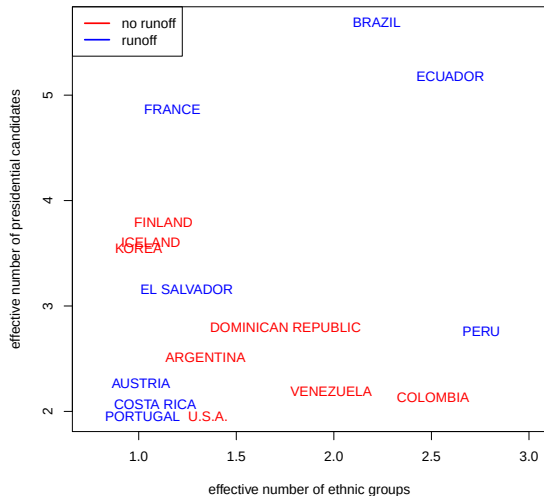
Thus the ATE is also point identified:

$$E[\tau_i] = E[Y_i(1)] - E[Y_i(0)].$$

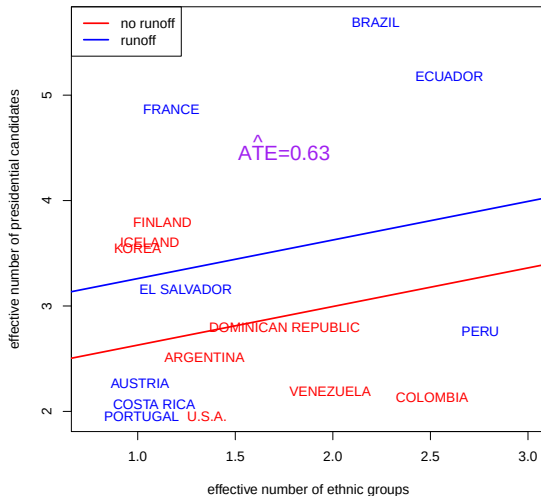
But what should we do when we have continuous  $\mathbf{X}_i$ ? Or when we have many, many covariates?

Plug-in estimators are not straightforward in this setting.

# Neto and Cox (1997) Table 3 Data

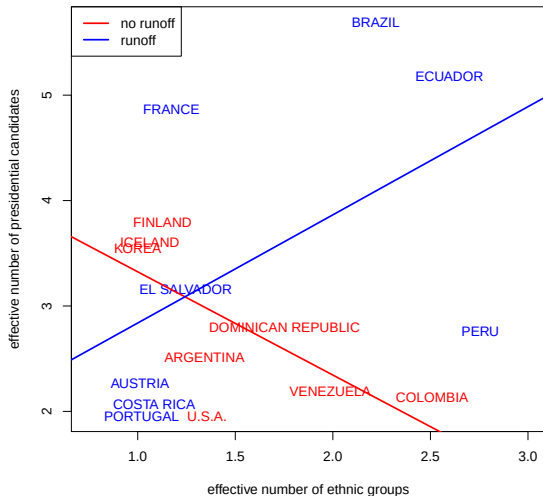


# Average Runoff Effect in Additive Model

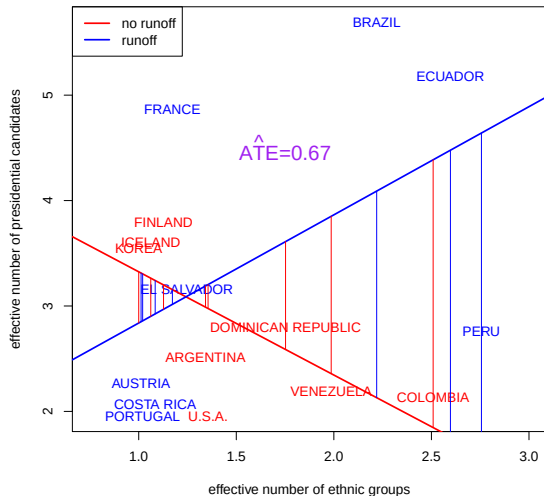




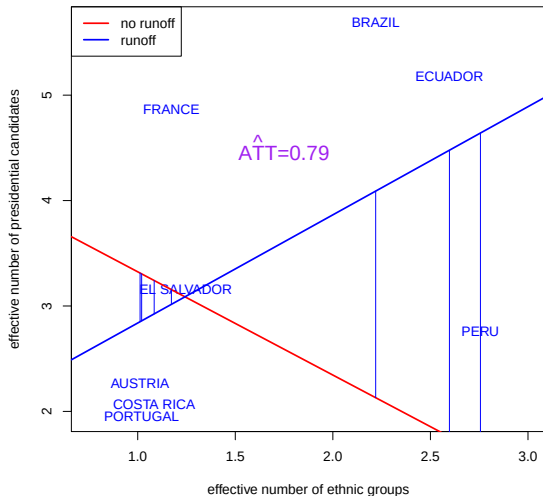
# Average Runoff Effect in Interactive Model



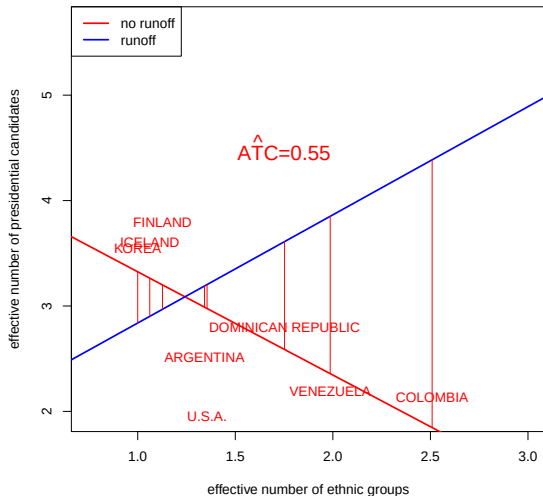
# Average Runoff Effect in Interactive Model



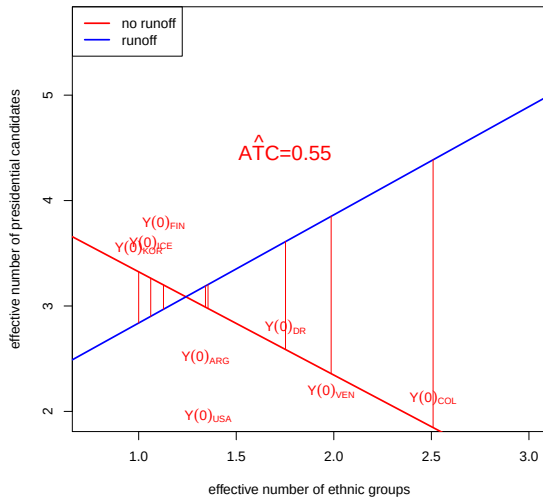
# Average Runoff Effect on the Runoff Countries



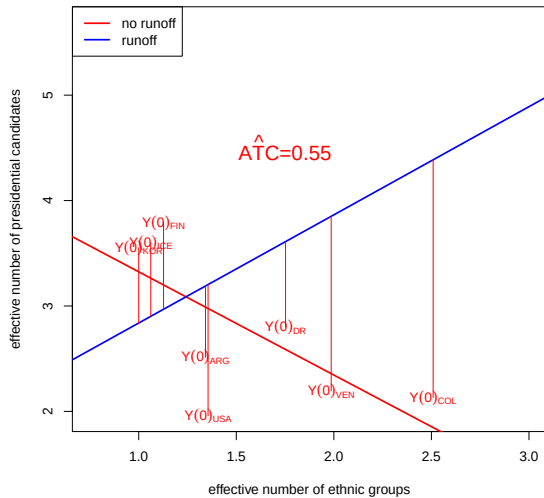
# Average Runoff Effect on the Non-Runoff Countries



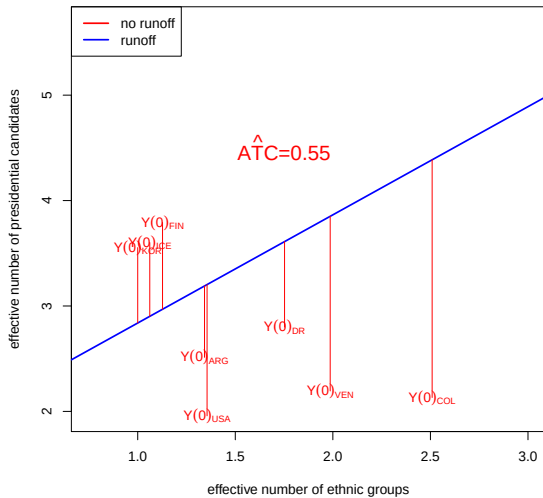
# Translating to Potential Outcomes



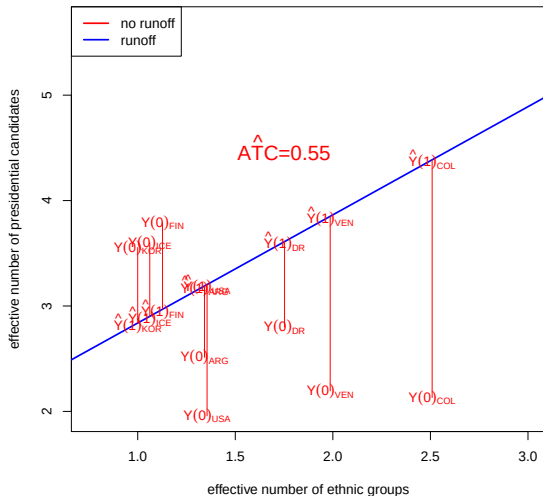
# Translating to Potential Outcomes



# Translating to Potential Outcomes

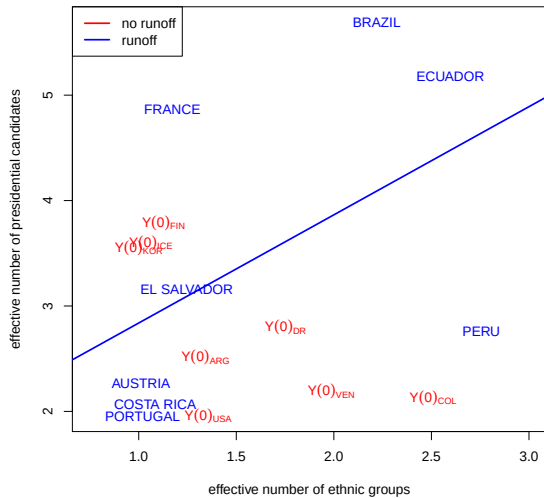


# Regression Imputation of Missing Potential Outcomes

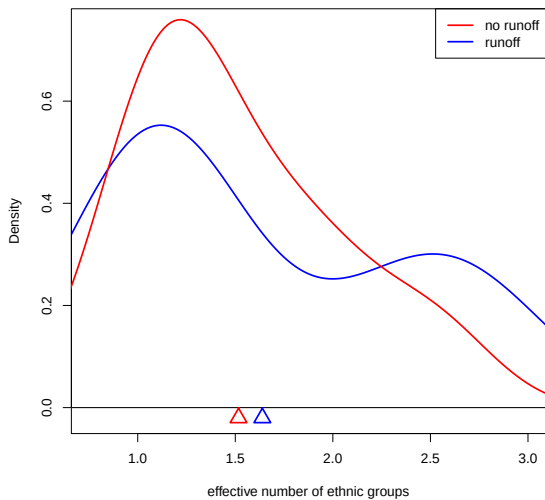




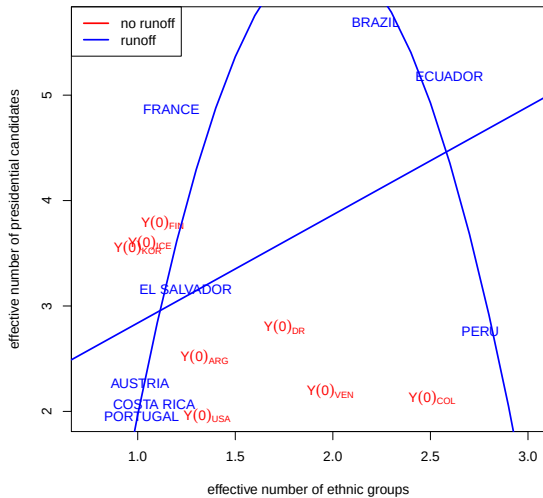
# Diagnostics



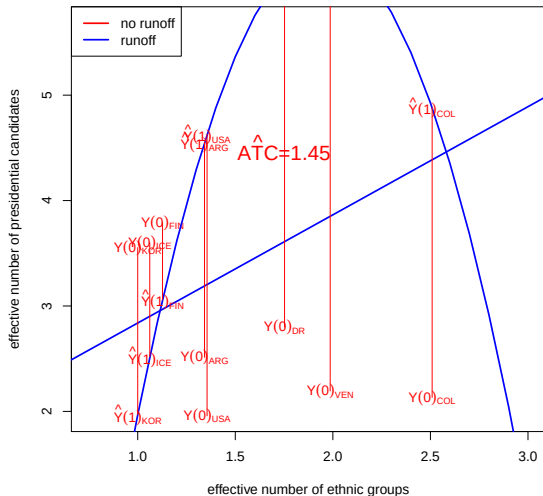
# Balance and Overlap



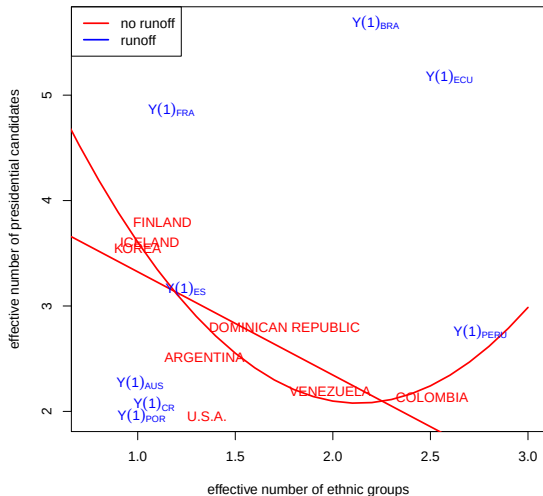
# Nonlinearity (Quadratic)



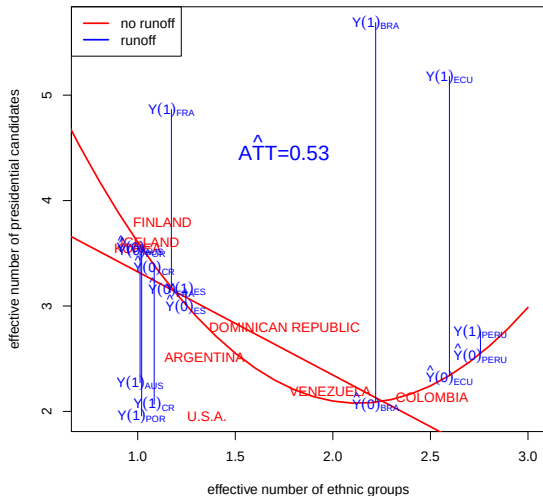
# ATC with Nonlinearity (Quadratic)



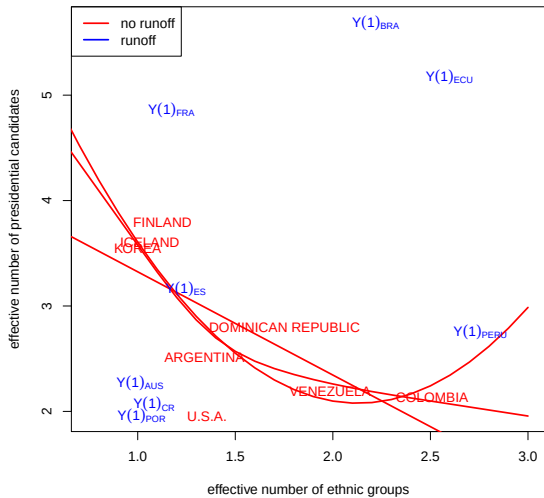
# Quadratic Nonlinearity for the Non-Runoff Countries



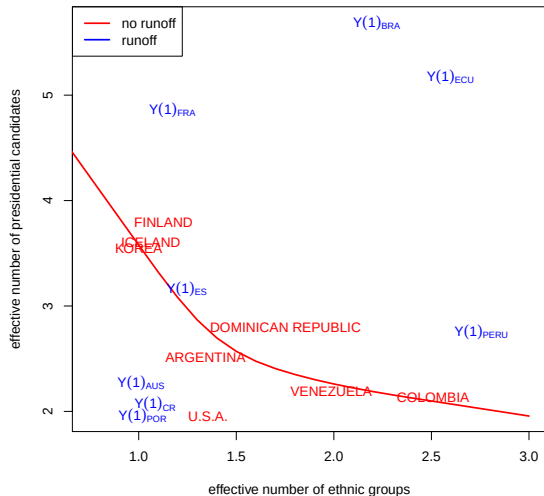
# ATT with Nonlinearity (Quadratic)



# ATT with Nonlinearity (Smoothing)

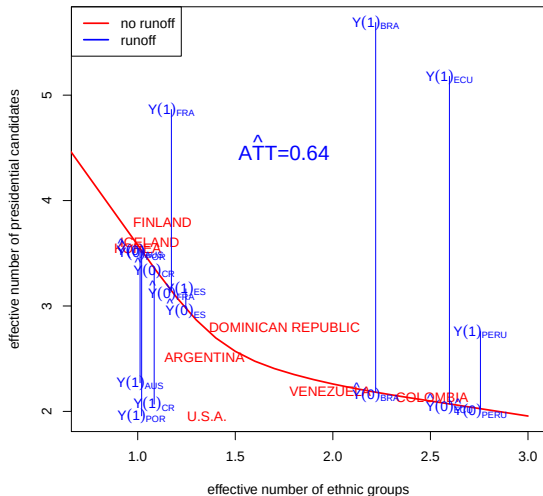


# ATT with Nonlinearity (Smoothing)





# ATT with Nonlinearity (Smoothing)



# Confidence Intervals

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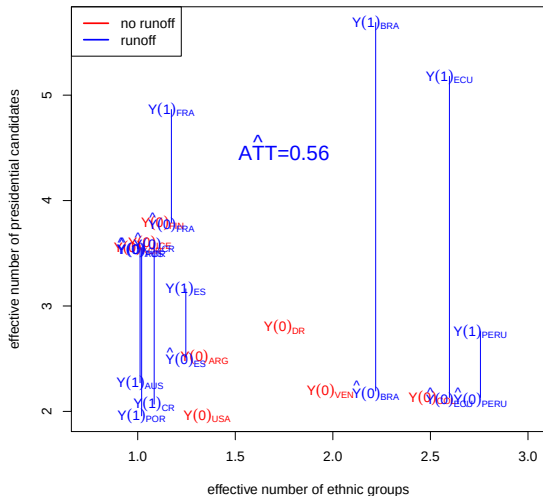
# Confidence Intervals

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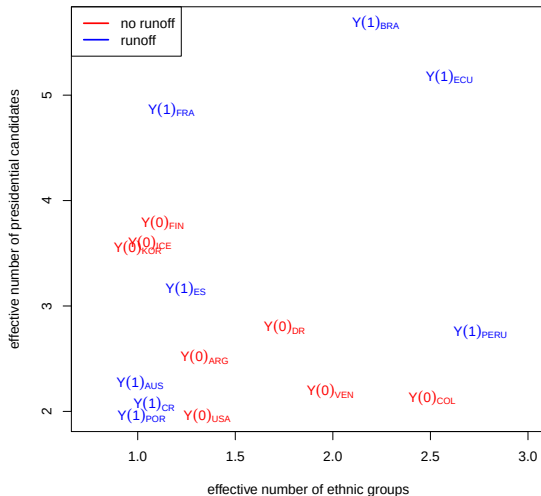
# Confidence Intervals

- ▶ Bootstrapping produces approximate CIs if regression function is smooth enough.
- ▶ These CIs likely to be conservative sample average treatment effects.
- ▶ However, sample size is a concern.
- ▶ Doesn't work for non-smooth estimators.

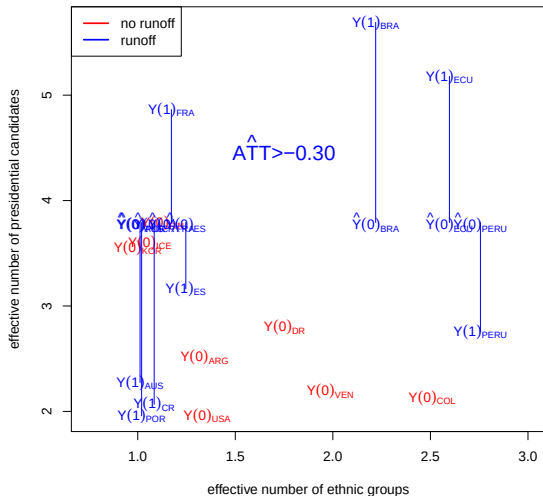
# ATT with One-to-One Matching with Replacement



# Bounding and Sensitivity Analysis for ATT



# Bounding and Sensitivity Analysis for ATT





$E[Y_i|D_i = d, \mathbf{X}_i = \mathbf{x}_i]$  is a conditional expectation!

Given the CIA, the difference between  $E[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i]$  and  $E[Y_i|D_i = 0, \mathbf{X}_i = \mathbf{x}_i]$  is a *causal effect*.

$$\begin{aligned} E[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i] - E[Y_i|D_i = 0, \mathbf{X}_i = \mathbf{x}_i] &= \\ E[Y_i(1)|D_i = 1, \mathbf{X}_i = \mathbf{x}_i] - E[Y_i(0)|D_i = 0, \mathbf{X}_i = \mathbf{x}_i] &= (\text{SUTVA}) \\ E[Y_i(1)|\mathbf{X}_i = \mathbf{x}_i] - E[Y_i(0)|\mathbf{X}_i = \mathbf{x}_i] &= (\text{CIA}) \\ E[Y_i(1) - Y_i(0)|\mathbf{X}_i = \mathbf{x}_i] &= \\ E[\tau_i|\mathbf{X}_i = \mathbf{x}_i] & \end{aligned}$$

When the CIA holds, some elements of the *conditional expectation function* (CEF) are *causal*.

We have an amazing tool for estimating the CEF. OLS regression consistently estimates the best linear predictor (BLP) of the CEF. ("Best" here means minimum-mean-squared-error.)

Given binary  $D_i$  and fitting the parameters in the regression model:

$$E[Y_i|D_i, \mathbf{X}_i] = \beta_0 + \beta_1 D_i + \mathbf{X}_i \beta,$$

then  $\hat{\beta}_1$  is a principled estimate of

$$E[Y_i|D_i = 1, \mathbf{X}_i] - E[Y_i|D_i = 0, \mathbf{X}_i] \text{ (since } E[Y_i|D_i, \mathbf{X}_i] \approx \beta_0 + \beta_1 D_i + \mathbf{X}_i \beta \text{).}$$

Given the CIA,  $E[Y_i|D_i = 1, \mathbf{X}_i] - E[Y_i|D_i = 0, \mathbf{X}_i]$  is a (conditional) ATE. Thus, given the CIA,  $\hat{\beta}_1$  may be interpreted as a good approximation to a causal effect.

There is one additional result on how multiple regression works that may be useful here and in the future.

The OLS estimate  $\hat{\beta}_1$  can be exactly reproduced via a procedure known as *partial regression*. This result is a direct consequence of the famous Frisch-Waugh-Lovell (FWL) Theorem.

In particular, suppose we did the following:

Regress  $Y_i$  on  $\mathbf{X}_i$  with a constant and compute the residuals from that regression:

$$\hat{Y}_i^r = Y_i - \hat{E}[Y_i|\mathbf{X}_i].$$

Regress  $D_i$  on  $\mathbf{X}_i$  with a constant and compute the residuals from that regression:

$$\hat{D}_i^r = D_i - \hat{E}[D_i|\mathbf{X}_i].$$

Then  $\hat{\beta}_1$  can be produced via the regression of  $\hat{Y}_i^r$  on  $\hat{D}_i^r$  either with or without a constant:

$$\begin{aligned}\hat{\beta}_1 &= \frac{\widehat{\text{Cov}}(\hat{Y}_i^r, \hat{D}_i^r)}{\widehat{V}(\hat{D}_i^r)} \\ &= \frac{\hat{E}[\hat{Y}_i^r \hat{D}_i^r]}{\hat{E}[(\hat{D}_i^r)^2]}\end{aligned}$$

(where the latter expression is the bivariate OLS solution from regression of  $\hat{Y}_i^r$  on  $\hat{D}_i^r$  without a constant).

A simplified proof sketch of FWL (adapted from Davidson and MacKinnon, 2004, Section 2.4):

WLOG, consider the OLS estimator,

$$\hat{\beta} = (\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'Y.$$

Let  $\mathbf{W}_1$  refer to the first  $K - 1$  columns of  $\mathbf{W}$ , and let  $D$  refer to the  $K$ th. Then:

- ▶  $\hat{\gamma}_1 = (\mathbf{W}_1'\mathbf{W}_1)^{-1}\mathbf{W}_1'Y$ , with the vector of outcome residuals  $\hat{Y}^r = Y - \mathbf{W}_1\hat{\gamma}_1$ .
- ▶  $\hat{\gamma}_2 = (\mathbf{W}_1'\mathbf{W}_1)^{-1}\mathbf{W}_1'D$  with the vector of treatment residuals  $\hat{D}^r = D - \mathbf{W}_1\hat{\gamma}_2$ .
- ▶ Then the coefficient from the partial regression,  
 $\hat{\gamma}_3 = \frac{\hat{E}[\hat{Y}_i^r \hat{D}_i^r]}{\hat{E}[(\hat{D}_i^r)^2]}$  with associated vector of residuals  
 $\hat{e} = \hat{Y}^r - \hat{D}^r \hat{\gamma}_3$ .

By the definition of residuals:

- ▶  $\hat{E}[\mathbf{W}_{1i}\hat{e}_i] = \hat{E}[\mathbf{W}_{1i}\hat{Y}_i^r] - \hat{E}[\mathbf{W}_{1i}\hat{D}_i^r]\hat{\gamma}_3 = 0.$
- ▶  $\hat{E}[(\hat{D}^r + \mathbf{W}_{1i}\hat{\gamma}_2)\hat{e}_i] = \hat{E}[D_i\hat{e}_i] = 0.$
- ▶ Ergo:  $\hat{E}[\mathbf{W}_i\hat{e}_i] = 0$ , or that the original set of regressors is orthogonal to residuals from the partial regression.

Then:

$$\begin{aligned} Y &= \mathbf{W}_1\hat{\gamma}_1 + \hat{Y}^r \\ &= \mathbf{W}_1\hat{\gamma}_1 + \hat{D}^r\hat{\gamma}_3 + \hat{e} \\ &= \mathbf{W}_1\hat{\gamma}_1 + (D - \mathbf{W}_1\hat{\gamma}_2)\hat{\gamma}_3 + \hat{e} \\ &= \mathbf{W}_1(\hat{\gamma}_1 - \hat{\gamma}_2\hat{\gamma}_3) + D\hat{\gamma}_3 + \hat{e}, \text{ with } \hat{E}[\mathbf{W}\hat{e}] = 0, \end{aligned}$$

so  $\hat{\gamma}_3 = \hat{\beta}_1$ , establishing that the partial regression recovers the coefficient from multiple regression.

Here's a little example:

In general, you can always "partial out" variables by regressing every other variable on them and taking the residuals. You can even do this repeated times. For example, if you have 3 RHS variables  $D_i$ ,  $X_{1i}$ , and  $X_{2i}$ , you could regress  $Y_i$ ,  $D_i$ , and  $X_{1i}$  on  $X_{2i}$ , then regress the residuals  $\hat{Y}_i^r$  and  $\hat{D}_i^r$  on  $\hat{X}_{1i}^r$ , then perform the bivariate regression with the residuals from that regression.

$$\hat{\beta}_1 = \frac{\hat{\text{Cov}}(\hat{Y}_i^r, \hat{D}_i^r)}{\hat{V}(\hat{D}_i^r)}.$$

Note what FWL substantively implies – the “usual” regression estimate (i.e., no treatment-covariate interactions) of the ATE *when controlling for covariates* identifies off of variation in the treatment is that is *not* explained by the covariates.

In general, if you don't include treatment-covariate interactions and everything else is otherwise well-specified, then regression gives you a reweighted causal effect, where the weights are proportional to how much unexplained variance in the treatment there is. (This is hardly a problem!) See Angrist and Krueger (1999), Angrist and Pischke (2009, Ch. 3), or Aronow and Samii (2013) for more details.



We have now established conditions for linear regression to produce principled estimates of causal relationships.

We have established a basis for causal inference with regression without:

- ▶ Making any assumptions about an error distribution (e.g., homoskedasticity, normality)
- ▶ Assuming linearity
- ▶ Discussing “exogeneity,” “endogeneity,” or “omitted variables bias”

We have invoked a nonparametric – and substantive – assumption about the relationship between potential outcomes and treatment. This assumption is strong, but we need not conflate it with other elements of specification or statistical identification. Justifying the CIA is the hard part; estimation is comparatively easy.

Once we see that the CEF is causal under the CIA, we no longer need to think solely in terms of binary treatments.

Consider the following generalization of the earlier potential outcomes model:

$$Y_i = \{ Y_i(d) : D_i = d, \forall d \in \text{Supp}(D_i) \}$$

When  $D_i$  is binary, this generalization reduces to the simpler case. But now, we can accommodate continuous or multivalued treatments. (E.g., estimating the causal effects of varying amounts of cash transfers on body weight.)

Causal effects are then defined as differences in potential outcomes:

$$\tau_i(d, d') = Y_i(d) - Y_i(d'), \text{ for } d, d' \in \text{Supp}(D_i).$$

Then, we need only invoke the equivalent version of the CIA to estimate average causal effects:

$$(Y_i(d)) \perp\!\!\!\perp D_i | \mathbf{X}_i = \mathbf{x}_i, \forall d \in \text{Supp}(D_i), \forall \mathbf{x}_i \in \text{Supp}(\mathbf{X}_i).$$

(Conditional independence assumption)

Like Angrist and Pischke, I will gloss over the requisite positivity assumption (since it is considerably more opaque in the generalized setting), but the assumption basically ensures the identifiability of all salient expected values.

Under the CIA (and the vague positivity assumption),  
 $E[\tau_i(d, d') | \mathbf{X}_i = \mathbf{x}_i] = E[Y_i | D_i = d, \mathbf{X}_i = \mathbf{x}_i] - E[Y_i | D_i = d', \mathbf{X}_i = \mathbf{x}_i].$

Returning to the estimates produced by the regression specification

$$E[Y_i|D_i, \mathbf{X}_i] = \beta_0 + \beta_1 D_i + \mathbf{X}_i\beta,$$

$\hat{\beta}_1$  represents a principled linear approximation to the *average causal effect* on the outcome of moving one “tick” in  $D_i$ . (The correct word here is unit, but I’d like to avoid ambiguity in nomenclature.)

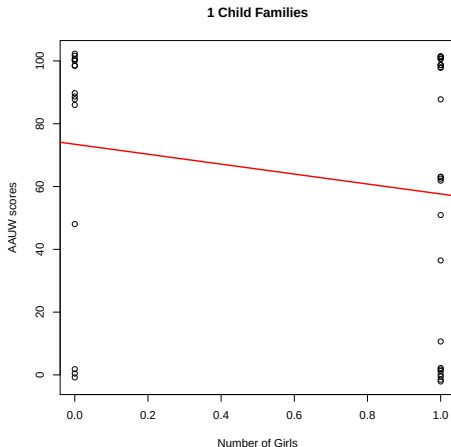
When the CEF is linear and the CIA holds, the partial derivative of the CEF with respect to the treatment is exactly the effect of moving one tick. Otherwise it’s just an approximation.

OLS is usually good enough. And it is indeed mostly harmless. You can't screw it up too badly, and you can write the requisite assumptions for a causal interpretation without worrying too much about functional form or error distributions.

Now, the question may be: what should we do when linearity may be not a good approximation for the CEF?

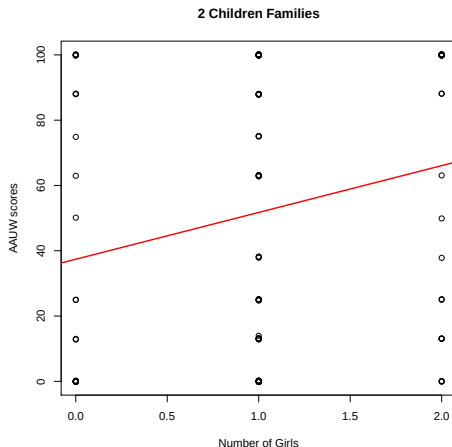
The short answer is that linear regression is still probably better than anything else you could do, unless your  $n$  is very large.

# One Child Legislators in the Washington (2008) Data



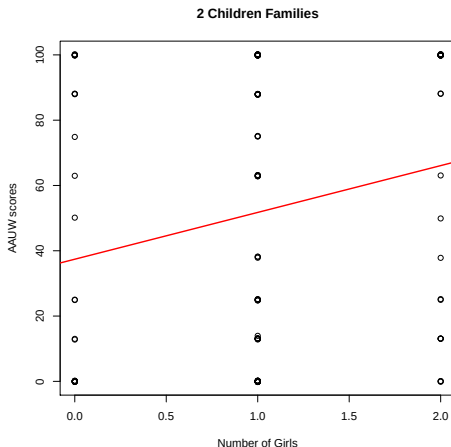
Among the subset of one-child legislators, we could adopt the standard potential outcomes approach.

# Two Children Legislators in the Washington (2008) Data



How would you construct the potential outcomes table for families with 2 children?

# Two Children Legislators in the Washington (2008) Data



How would you construct the potential outcomes table for families with 2 children? What assumption is now implicit in the regression?



Among the subset of 2 children legislators, assume a linear CEF

$$E[Y|T = t] = \beta_0 + \beta_1 \cdot t$$

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When is  $\beta_1$  interpretable as the ATE of a one unit increase?

$$\begin{aligned}\beta_1 &= E[Y|T = t + 1] - E[Y|T = t] \text{ for all } t \\ &= \\ &= \end{aligned}$$

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This linear approach would also be straightforward for 3 children, 4 children, etc.

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This linear approach would also be straightforward for 3 children, 4 children, etc.

What are some ways we could combine these analyses across all legislators?

Assume a linear additive CEF

$$E[Y|T = t, X = x] = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x$$

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$$=$$

$$=$$

$$=$$

$$=$$

$$=$$



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When is  $\beta_1$  interpretable as the ATE of a one unit increase?

$$\beta_1 = E[Y|T = t + 1, X = x] - E[Y|T = t, X = x] \text{ for all } t, x$$

=

=

=

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=

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$$\begin{aligned}\beta_1 &= E[Y|T = t + 1, X = x] - E[Y|T = t, X = x] \text{ for all } t, x \\ &= E_x \{E[Y|T = t + 1, X = x] - E[Y|T = t, X = x]\} \text{ for all } t \\ &= \\ &= \\ &= \\ &= \end{aligned}$$

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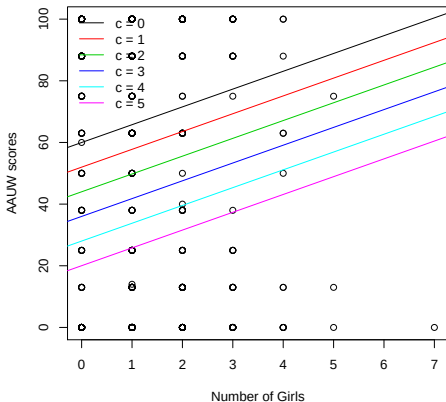
Assume a linear additive CEF

$$E[Y|T = t, X = x] = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x$$

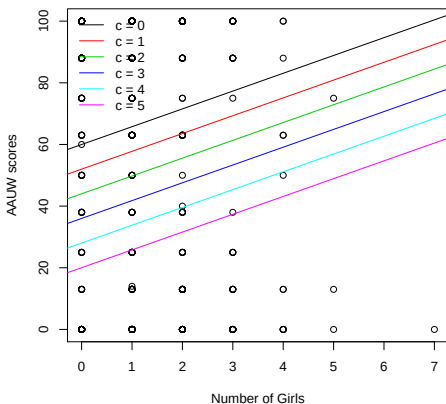
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# The Additive Linear Model



# The Additive Linear Model



How would this picture change if we used the model in Equation 1 of Washington (2008)?



# The Interactive Linear Model

$$E[Y|T = t, X = x] = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x + \beta_3 \cdot t \cdot x$$
$$=$$

# The Interactive Linear Model

$$\begin{aligned}E[Y|T = t, X = x] &= \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x + \beta_3 \cdot t \cdot x \\ &= \beta_0 + (\beta_1 + \beta_3 \cdot x) \cdot t + \beta_2 \cdot x\end{aligned}$$

When is  $(\beta_1 + \beta_3 \cdot x)$  interpretable as the ATE of a one unit increase, conditional on  $X = x$ ?

$$(\beta_1 + \beta_3 \cdot x) =$$

=

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# The Interactive Linear Model

$$\begin{aligned}E[Y|T = t, X = x] &= \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x + \beta_3 \cdot t \cdot x \\ &= \beta_0 + (\beta_1 + \beta_3 \cdot x) \cdot t + \beta_2 \cdot x\end{aligned}$$

When is  $(\beta_1 + \beta_3 \cdot x)$  interpretable as the ATE of a one unit increase, conditional on  $X = x$ ?

$$(\beta_1 + \beta_3 \cdot x) = E[Y|T = t + 1, X = x] - E[Y|T = t, X = x] \text{ for all } t, x$$

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# The Interactive Linear Model

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$$\begin{aligned}(\beta_1 + \beta_3 \cdot x) &= E[Y|T = t + 1, X = x] - E[Y|T = t, X = x] \text{ for all } t, x \\&= E[Y|T = t + 1, X = x] - E[Y|T = t, X = x] \text{ for all } t \\&= \\&= \\&= \end{aligned}$$

# The Interactive Linear Model

$$\begin{aligned}E[Y|T = t, X = x] &= \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x + \beta_3 \cdot t \cdot x \\&= \beta_0 + (\beta_1 + \beta_3 \cdot x) \cdot t + \beta_2 \cdot x\end{aligned}$$

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$$\begin{aligned}(\beta_1 + \beta_3 \cdot x) &= E[Y|T = t + 1, X = x] - E[Y|T = t, X = x] \text{ for all } t, x \\&= E[Y|T = t + 1, X = x] - E[Y|T = t, X = x] \text{ for all } t \\&= E[Y(t + 1)|T = t + 1, X = x] - E[Y(t)|T = t, X = x] \text{ for all } t \\&= \\&= \end{aligned}$$

# The Interactive Linear Model

$$\begin{aligned}E[Y|T = t, X = x] &= \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x + \beta_3 \cdot t \cdot x \\&= \beta_0 + (\beta_1 + \beta_3 \cdot x) \cdot t + \beta_2 \cdot x\end{aligned}$$

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$$\begin{aligned}(\beta_1 + \beta_3 \cdot x) &= E[Y|T = t + 1, X = x] - E[Y|T = t, X = x] \text{ for all } t, x \\&= E[Y|T = t + 1, X = x] - E[Y|T = t, X = x] \text{ for all } t \\&= E[Y(t + 1)|T = t + 1, X = x] - E[Y(t)|T = t, X = x] \text{ for all } t \\&= E[Y(t + 1)|X = x] - E[Y(t)|X = x] \text{ for all } t \\&= \end{aligned}$$

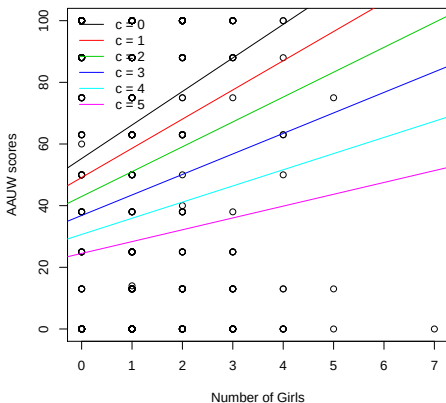
# The Interactive Linear Model

$$\begin{aligned}E[Y|T = t, X = x] &= \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x + \beta_3 \cdot t \cdot x \\&= \beta_0 + (\beta_1 + \beta_3 \cdot x) \cdot t + \beta_2 \cdot x\end{aligned}$$

When is  $(\beta_1 + \beta_3 \cdot x)$  interpretable as the ATE of a one unit increase, conditional on  $X = x$ ?

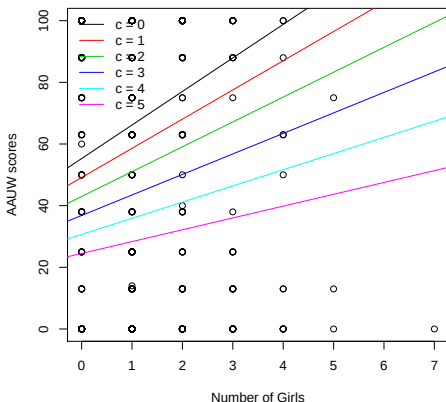
$$\begin{aligned}(\beta_1 + \beta_3 \cdot x) &= E[Y|T = t + 1, X = x] - E[Y|T = t, X = x] \text{ for all } t, x \\&= E[Y|T = t + 1, X = x] - E[Y|T = t, X = x] \text{ for all } t \\&= E[Y(t + 1)|T = t + 1, X = x] - E[Y(t)|T = t, X = x] \text{ for all } t \\&= E[Y(t + 1)|X = x] - E[Y(t)|X = x] \text{ for all } t \\&= E[Y(t + 1) - Y(t)|X = x] \text{ for all } t\end{aligned}$$

# The Interactive Linear Model





# The Interactive Linear Model



What would the picture look like if we used a more general model (for which the interactive model and Equation 1 of Washington (2008) are special cases)?

# Standardization in the Interactive Linear Model

When is  $\beta_1 + \beta_3 \cdot E_X \{X\}$  interpretable as the ATE of a one unit increase?

$$\beta_1 + \beta_3 \cdot E_X \{X\} =$$

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# Standardization in the Interactive Linear Model

When is  $\beta_1 + \beta_3 \cdot E_X \{X\}$  interpretable as the ATE of a one unit increase?

$$\beta_1 + \beta_3 \cdot E_X \{X\} = E_X \{\beta_1 + \beta_3 \cdot X\}$$

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# Standardization in the Interactive Linear Model

When is  $\beta_1 + \beta_3 \cdot E_X \{X\}$  interpretable as the ATE of a one unit increase?

$$\begin{aligned}\beta_1 + \beta_3 \cdot E_X \{X\} &= E_X \{\beta_1 + \beta_3 \cdot X\} \\ &= E_X \{E[Y|T = t + 1, X = x] - E[Y|T = t, X = x]\} \\ &= \\ &= \\ &= \\ &= \end{aligned}$$

# Standardization in the Interactive Linear Model

When is  $\beta_1 + \beta_3 \cdot E_X \{X\}$  interpretable as the ATE of a one unit increase?

$$\begin{aligned}\beta_1 + \beta_3 \cdot E_X \{X\} &= E_X \{\beta_1 + \beta_3 \cdot X\} \\ &= E_X \{E[Y|T = t + 1, X = x] - E[Y|T = t, X = x]\} \\ &= E_X \{E[Y(t + 1)|T = t + 1, X = x] - E[Y(t)|T = t, X = x]\} \\ &= \\ &= \\ &= \end{aligned}$$

# Standardization in the Interactive Linear Model

When is  $\beta_1 + \beta_3 \cdot E_X \{X\}$  interpretable as the ATE of a one unit increase?

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# Standardization in the Interactive Linear Model

When is  $\beta_1 + \beta_3 \cdot E_X \{X\}$  interpretable as the ATE of a one unit increase?

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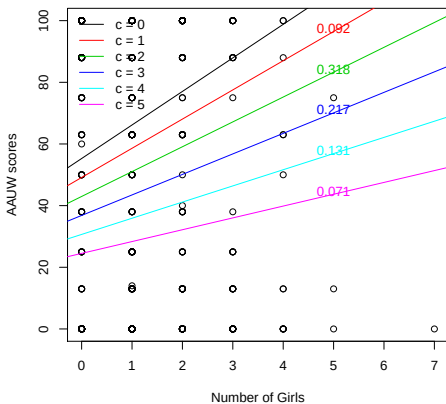
# Standardization in the Interactive Linear Model

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$$\begin{aligned}\beta_1 + \beta_3 \cdot E_X \{X\} &= E_X \{\beta_1 + \beta_3 \cdot X\} \\&= E_X \{E[Y|T = t + 1, X = x] - E[Y|T = t, X = x]\} \\&= E_X \{E[Y(t + 1)|T = t + 1, X = x] - E[Y(t)|T = t, X = x]\} \\&= E_X \{E[Y(t + 1)|X = x] - E[Y(t)|X = x]\} \\&= E_X \{E[Y(t + 1) - Y(t)|X = x]\} \\&= E[Y(t + 1) - Y(t)]\end{aligned}$$



# Standardization in the Interactive Linear Model



The propensity score, matching estimators (as a special case of imputation-based estimators), weighting-based estimators.

Last class, assuming ignorability and discrete  $\mathbf{X}$ :

$$E[Y_i(d)] = \sum_{\mathbf{x}_i \in \text{Supp}(\mathbf{X}_i)} E[Y_i | D_i = d, \mathbf{X}_i = \mathbf{x}_i] \Pr[\mathbf{X}_i = \mathbf{x}_i]$$

with

$$E[\tau_i] = E[Y_i(1)] - E[Y_i(0)].$$

This implies a simple plug-in estimator (using sample analogues to conditional expectations and covariate probabilities).

But what should we do when we have continuous variables in  $\mathbf{X}_i$ ? Or when we have many, many covariates?

One option:

- ▶ The CIA implies a causal interpretation to the CEF.
- ▶ The CEF is “probably” close to linear in  $D_i$  and  $\mathbf{X}_i$ .
- ▶ Thus OLS “probably” yields a good approximation to a causal effect under the CIA.

We sketched this out in the last class. This is one reason why OLS regression is *mostly harmless*. There are alternative ways to justify OLS as a sensible estimator of causal effects (later).

There are, however, alternative (non-OLS) approaches to estimating causal effects given strong ignorability.

This class, we will discuss some of these other estimators. In particular, we will derive and discuss matching estimators (as a special case of imputation estimators) and inverse-probability-weighting-based estimators.

To proceed, we first need to establish one more fundamental result. “The Central Role of the Propensity Score in Observational Studies for Causal Effects” (Rosenbaum and Rubin 1983).

For a binary treatment, if we have strong ignorability (CIA + positivity), then there is a shortcut to estimating causal effects that does not depend on directly conditioning on all of the covariates. Instead, we can condition on a summary measure of the treatment and the covariates.

This summary measure is the propensity score: the *probability of treatment conditional on the covariates*. Note that the propensity score (sometimes abbreviated as p-score) can be generated as a deterministic function of the covariates – if we know  $\mathbf{X}_i = \mathbf{x}_i$  and the full joint distribution of  $(D_i, \mathbf{X}_i)$ , then we know  $\Pr[D_i | \mathbf{X}_i = \mathbf{x}_i]$ .

To formalize this idea, suppose we have strong ignorability (CIA + positivity):

1.  $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i | \mathbf{X}_i = \mathbf{x}_i, \forall \mathbf{x}_i \in \text{Supp}(\mathbf{X}_i)$ . (CIA w.r.t.  $\mathbf{X}_i$ )
2.  $0 < \Pr(D_i = 1 | \mathbf{X}_i = \mathbf{x}_i) < 1, \forall \mathbf{x}_i \in \text{Supp}(\mathbf{X}_i)$ . (Positivity w.r.t.  $\mathbf{X}_i$ )

Define the propensity score as  $P_i = \Pr(D_i = 1 | \mathbf{X}_i)$ .

Then

1.  $D_i \perp\!\!\!\perp \mathbf{X}_i | P_i = p_i, \forall p_i \in \text{Supp}(P_i)$ . (Balance)
2.  $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i | P_i = p_i, \forall p_i \in \text{Supp}(P_i)$ . (CIA w.r.t. p-score)
3.  $0 < \Pr(D_i = 1 | P_i = p_i) < 1, \forall p_i \in \text{Supp}(P_i)$ . (Positivity w.r.t. p-score)

Proof sketch of the “balancing property”, that is that conditioning on the p-score yields balance.

Since  $D_i$  is binary, its (conditional) distribution is fully governed by its mean. Thus

$$D_i \perp\!\!\!\perp \mathbf{X}_i | P_i \Leftrightarrow E[D_i | \mathbf{X}_i, P_i] = E[D_i | P_i]$$

Now let us note that the propensity score can be written as a function of  $\mathbf{X}_i$ , so that

$$E[D_i | \mathbf{X}_i, P_i] = E[D_i | \mathbf{X}_i, f(\mathbf{X}_i)] = E[D_i | \mathbf{X}_i] = P_i.$$

Applying the Law of Iterated Expectations:

$$E[D_i | P_i] = E[E[D_i | \mathbf{X}_i] | P_i] = E[P_i | P_i] = P_i.$$

Since  $E[D_i | \mathbf{X}_i, P_i] = E[D_i | P_i] = P_i$ , the proof is complete.



Note: balance on  $\mathbf{X}_i$  is a *consequence* of conditioning on the p-score. (I.e., after you have conditioned the p-score, the conditional distribution of covariates for treatment units will be the same as the conditional distribution of covariates for control units.)

In the context of observational studies, a balance test (a statistical test of equality of covariate distributions) tests the null hypothesis that we have successfully conditioned on the p-score. If the distributions are significantly different, then we have likely not conditioned well enough on the p-score.

However, balance does *not* imply that the CIA holds. Balance on observable characteristics does not imply balance on unobservable characteristics. Without further assumptions, there exist no general tests of the CIA.

Proof sketch of the CIA w.r.t. to the p-score. Similar proof:

Since  $D_i$  is binary,

$$D_i \perp\!\!\!\perp (Y_i(0), Y_i(1)) | P_i \Leftrightarrow E[D_i | (Y_i(0), Y_i(1)), P_i] = E[D_i | P_i]$$

Law of Iterated Expectations:

$$E[D_i | (Y_i(0), Y_i(1)), P_i] = E[E[D_i | (Y_i(0), Y_i(1)), X_i] | (Y_i(0), Y_i(1)), P_i]$$

Then by ignorability,

$$\begin{aligned} &= E[E[D_i | X_i] | (Y_i(0), Y_i(1)), P_i] \\ &= E[P_i | (Y_i(0), Y_i(1)), P_i] \\ &= P_i = E[D_i | P_i], \end{aligned}$$

completing the proof. (I'll skip over the proof of positivity.)

In layman's terms: conditioning on the propensity score (or conditional probability of treatment) is sufficient for implementing the CIA.

In order to obtain consistent causal estimates, we do not need to condition on all of the covariates as long as we condition on the propensity score.

This result is thought to be useful since it might be difficult to (nonparametrically) condition on a large number of variables (unless we use a simple approximate result like OLS). Think about what the plug-in estimator would look like.

There is a problem here though. With real data, the propensity score *must usually be estimated*.

How would we estimate the propensity score?

By necessity: performing some sort of regression of  $D_i$  on all of the covariates.

Well, that seems to just move the problem from estimating  $E[Y_i|D_i, \mathbf{X}_i]$  to estimating  $E[Y_i|D_i, P_i]$  and  $P_i = E[D_i|\mathbf{X}_i]$ . Nevertheless, this is what must be done when working with propensity scores.

The most common – but certainly not only – propensity score-based method is *propensity score matching*.

Matching is a way to impute the missing distributions of potential outcomes. Given strong ignorability, matching will typically yield a consistent (but otherwise pretty bad) estimator of the ATE.

There are many types of matching, but I will be discussing just one type: 1:1, nearest-neighbor, with-replacement, propensity score matching for the ATE. For the moment, I'm going to ignore the “estimated” propensity score aspect of this and just assume that the p-score is known.

To show this, I'm going to give an illustrative example.

| Unit | $Y_i$ | $D_i$ | $P_i$ | Unit | $Y_i(0)$ | $Y_i(1)$ | $D_i$ | $P_i$ |
|------|-------|-------|-------|------|----------|----------|-------|-------|
| 1    | 2     | 1     | 0.33  | 1    | ?        | 2        | 1     | 0.33  |
| 2    | 5     | 0     | 0.14  | 2    | 5        | ?        | 0     | 0.14  |
| 3    | 3     | 1     | 0.73  | 3    | ?        | 3        | 1     | 0.73  |
| 4    | 10    | 1     | 0.35  | 4    | ?        | 10       | 1     | 0.35  |
| 5    | 2     | 1     | 0.78  | 5    | ?        | 2        | 1     | 0.78  |
| 6    | 0     | 0     | 0.70  | 6    | 0        | ?        | 0     | 0.70  |

Let's assume that strong ignorability holds and we're after the ATE.

Let's impute the expected value of the missing potential outcomes.

| Unit | $Y_i(0)$                     | $Y_i(1)$                     | $D_i$ | $P_i$ |
|------|------------------------------|------------------------------|-------|-------|
| 1    | $E[Y_i D_i = 0, P_i = 0.33]$ | 2                            | 1     | 0.33  |
| 2    | 5                            | $E[Y_i D_i = 1, P_i = 0.14]$ | 0     | 0.14  |
| 3    | $E[Y_i D_i = 0, P_i = 0.73]$ | 3                            | 1     | 0.73  |
| 4    | $E[Y_i D_i = 0, P_i = 0.35]$ | 10                           | 1     | 0.35  |
| 5    | $E[Y_i D_i = 0, P_i = 0.78]$ | 2                            | 1     | 0.78  |
| 6    | 0                            | $E[Y_i D_i = 1, P_i = 0.70]$ | 0     | 0.70  |

Given strong ignorability, if we had consistent estimators for each value of  $E[Y_i|D_i = d_i, P_i = p_i]$ , then the average difference between the columns for  $E[Y_i(1)]$  and  $E[Y_i(0)]$  would be a consistent estimator of the average treatment effect.

There are many imputation-type estimators for the missing potential outcome distributions. (Regression is an imputation estimator!)

Let's talk about what (nearest-neighbor) matching would do.



For each treatment unit, find the control unit that is *closest* on  $P_i$ . Use that unit's outcome to fill in the missing potential outcome. Do the analogous procedure for control units (finding the treatment unit that is closest on  $P_i$ ).

| Unit | $Y_i(0)$                           | $Y_i(1)$                           | $D_i$ | $P_i$ |
|------|------------------------------------|------------------------------------|-------|-------|
| 1    | $\hat{E}[Y_i D_i = 0, P_i = 0.33]$ | 2                                  | 1     | 0.33  |
| 2    | 5                                  | $\hat{E}[Y_i D_i = 1, P_i = 0.14]$ | 0     | 0.14  |
| 3    | $\hat{E}[Y_i D_i = 0, P_i = 0.73]$ | 3                                  | 1     | 0.73  |
| 4    | $\hat{E}[Y_i D_i = 0, P_i = 0.35]$ | 10                                 | 1     | 0.35  |
| 5    | $\hat{E}[Y_i D_i = 0, P_i = 0.78]$ | 2                                  | 1     | 0.78  |
| 6    | 0                                  | $\hat{E}[Y_i D_i = 1, P_i = 0.70]$ | 0     | 0.70  |

For each treatment unit, find the control unit that is *closest* on  $P_i$ . Use that unit's outcome to fill in the missing potential outcome. Do the analogous procedure for control units (finding the treatment unit that is closest on  $P_i$ ).

| Unit | $Y_i(0)$                           | $Y_i(1)$                           | $D_i$ | $P_i$ |
|------|------------------------------------|------------------------------------|-------|-------|
| 1    | 5                                  | 2                                  | 1     | 0.33  |
| 2    | 5                                  | $\hat{E}[Y_i D_i = 1, P_i = 0.14]$ | 0     | 0.14  |
| 3    | $\hat{E}[Y_i D_i = 0, P_i = 0.73]$ | 3                                  | 1     | 0.73  |
| 4    | $\hat{E}[Y_i D_i = 0, P_i = 0.35]$ | 10                                 | 1     | 0.35  |
| 5    | $\hat{E}[Y_i D_i = 0, P_i = 0.78]$ | 2                                  | 1     | 0.78  |
| 6    | 0                                  | $\hat{E}[Y_i D_i = 1, P_i = 0.70]$ | 0     | 0.70  |

For each treatment unit, find the control unit that is *closest* on  $P_i$ . Use that unit's outcome to fill in the missing potential outcome. Do the analogous procedure for control units (finding the treatment unit that is closest on  $P_i$ ).

| Unit | $Y_i(0)$ | $Y_i(1)$ | $D_i$ | $P_i$ |
|------|----------|----------|-------|-------|
| 1    | 5        | 2        | 1     | 0.33  |
| 2    | 5        | 2        | 0     | 0.14  |
| 3    | 0        | 3        | 1     | 0.73  |
| 4    | 5        | 10       | 1     | 0.35  |
| 5    | 0        | 2        | 1     | 0.78  |
| 6    | 0        | 3        | 0     | 0.70  |

The matching estimator of the ATE is: 1.17.

All matching estimators roughly follow this logic. Sometimes people match on some other summary metric of  $\mathbf{X}_i$ , do 1:k matching, choose observations to empirically maximize the balance of the matched groups, or, when possible, choose observations that *exactly* match on all covariates (for discrete  $\mathbf{X}_i$ ). Sometimes people will also attempt to estimate the ATT (average treatment effect on the treated,  $E[\tau_i | D_i = 1]$ ) or ATC (average treatment effect on the controls  $E[\tau_i | D_i = 0]$ ).

Intuitively, why is matching a not-so-great idea *even when we assume strong ignorability*? Can you come up with a better alternative?

To be specific, given the general form of the problem,

| Unit | $Y_i(0)$                           | $Y_i(1)$                           | $D_i$ | $P_i$ |
|------|------------------------------------|------------------------------------|-------|-------|
| 1    | $\hat{E}[Y_i D_i = 0, P_i = 0.33]$ | 2                                  | 1     | 0.33  |
| 2    | 5                                  | $\hat{E}[Y_i D_i = 1, P_i = 0.14]$ | 0     | 0.14  |
| 3    | $\hat{E}[Y_i D_i = 0, P_i = 0.73]$ | 3                                  | 1     | 0.73  |
| 4    | $\hat{E}[Y_i D_i = 0, P_i = 0.35]$ | 10                                 | 1     | 0.35  |
| 5    | $\hat{E}[Y_i D_i = 0, P_i = 0.78]$ | 2                                  | 1     | 0.78  |
| 6    | 0                                  | $\hat{E}[Y_i D_i = 1, P_i = 0.70]$ | 0     | 0.70  |

Why wouldn't you just want to pick the closest unit of a different treatment condition on  $P_i$ ? What would you do to estimate each of the  $\hat{E}[Y_i|D_i = 0, P_i = p_i]$ ?

The problem with propensity score matching gets even worse when you estimate the propensity score.

Suppose you don't know what the functional form of  $E[D_i | \mathbf{X}_i = \mathbf{x}_i]$  is. If you aren't very, very careful to set up a nonparametrically consistent estimator (e.g., some variant of regression with a growing number of polynomials and interactions), then your estimator of  $P_i$  may never converge to the right value. Then you won't estimate the ATE correctly, even under strong ignorability.

Furthermore, even if you did estimate  $P_i$  consistently, there will be problems in that you'll have to account for the fact that you're introducing estimation error in how you construct your matches.

Matching with estimated propensity scores is not generally root- $n$  consistent – i.e., the MSE of the estimator is greater than  $O(1/n)$  – and the usual methods for constructing confidence intervals (*including the bootstrap*) do not generally work. (There is a violation of a necessary smoothness condition for bootstrap validity; see the papers of Abadie and Imbens.)

A few alternatives:

- ▶ Least squares regression, perhaps with polynomials and interactions
- ▶ Nonparametric regression
- ▶ Fully parametric imputation, perhaps using a likelihood function – we will discuss this in the last few weeks of class
- ▶ Weighting-based estimators

The regression estimators are (mostly) straightforward. Let's talk about the weighting-based estimators.

Let's consider a variant of the famous inverse-probability-weighted (IPW) estimator. The earliest such estimator was derived in the case of survey sampling where the probabilities of sample selection were exactly known (Horvitz and Thompson 1952).

Suppose the usual (unreasonable) ignorability assumption (CIA + positivity) holds.

And, to ease our calculations a bit, let's suppose that the propensity score,  $P_i$ , is known.



Let's detail a population weighting result. Suppose that we were to weight every observation by the inverse of the p-score.

$$E \left[ \frac{Y_i \cdot D_i}{P_i} \middle| \mathbf{X}_i = \mathbf{x}_i \right] = E \left[ Y_i(1) \cdot \frac{D_i}{\Pr(D_i = 1 | \mathbf{X}_i = \mathbf{x}_i)} \middle| \mathbf{X}_i = \mathbf{x}_i \right]$$

Then assuming the CIA,

$$E \left[ Y_i(1) \cdot \frac{D_i}{\Pr(D_i = 1 | \mathbf{X}_i = \mathbf{x}_i)} \middle| \mathbf{X}_i = \mathbf{x}_i \right] = E [ Y_i(1) | \mathbf{X}_i = \mathbf{x}_i ] .$$

Thus, invoking the Law of Iterated Expectations:

$$E \left[ \frac{Y_i \cdot D_i}{P_i} \right] = E [ Y_i(1) ] .$$

So the inverse propensity score weighted expectation of treatment observations is the expected value of treatment potential outcomes.

Similarly,

$$E \left[ \frac{Y_i \cdot (1 - D_i)}{(1 - P_i)} \right] = E[Y_i(0)].$$

and thus, by linearity of expectations,

$$E \left[ \frac{Y_i \cdot D_i}{P_i} - \frac{Y_i \cdot (1 - D_i)}{(1 - P_i)} \right] = E[\tau_i].$$

The expression

$$E[\tau_i] = E \left[ \frac{Y_i \cdot D_i}{P_i} - \frac{Y_i \cdot (1 - D_i)}{(1 - P_i)} \right]$$

suggests a pair of natural estimators for  $E[\tau_i]$ .

The first, a straightforward plug-in estimator, is the analogue estimator (substituting the sample mean for the expected value):

$$\hat{E}_{HT}[\tau_i] = \frac{1}{n} \sum_{i=1}^n \left[ \frac{Y_i \cdot D_i}{P_i} - \frac{Y_i \cdot (1 - D_i)}{(1 - P_i)} \right],$$

which is consistent under strong ignorability by the WLLN.

The second estimator is a “ratio estimator” that performs better in practice, and owes to Hajek (1971). In this case, we normalize the weights ( $1/P_i$ ) to sum to  $n$  in each treatment condition. This normalization is useful in case we draw an unusually large number of units in either condition where  $P_i$  is large or  $P_i$  is small.

$$\hat{E}_{Haj}[\tau_i] = \frac{\sum_{i=1}^n \frac{Y_i \cdot D_i}{P_i}}{\sum_{i=1}^n \frac{D_i}{P_i}} - \frac{\sum_{i=1}^n \frac{Y_i \cdot (1-D_i)}{(1-P_i)}}{\sum_{i=1}^n \frac{1-D_i}{1-P_i}},$$

noting that  $E[\sum_{i=1}^n \frac{D_i}{P_i}] = E[\sum_{i=1}^n \frac{1-D_i}{1-P_i}] = n$ .

Consistency under strong ignorability follows by the WLLN and Slutsky's Theorem.

An intuition for the IPW-type estimators follows directly from our logic for survey reweighting.

Suppose that some units are very unlikely to be observed in the treatment condition, because they have a small  $p$ -score. Then we want to upweight these units when they are in the treatment condition. Similarly, since they are likely to be observed in the control condition, we want to downweight these units when they are in the control condition.

In the case where we have a discrete and finite number of values for  $\mathbf{X}_i$ , the IPW estimator is logically equivalent to the plug-in estimator.

## Some notes on IPW estimators:

- ▶ In general, IPW estimators tend to be more asymptotically efficient than matching estimators, but may be highly sensitive to extremely low or extremely high p-scores. (Question: what happens when you observe a unit in the treatment condition with a very low p-score?)
- ▶ We still have to estimate p-scores. But there's a seemingly counterintuitive finding for the weighting estimators. Assuming that you have a consistent estimator for the p-scores, using estimated p-scores can actually be more efficient than using the true p-scores. The reason for this is that using the estimated p-scores does more to balance the treatment and control groups in the covariates.

Remember the nature of the problem:

| Unit | $Y_i(0)$                                        | $Y_i(1)$                                       | $D_i$ | $X_{1i}$ |
|------|-------------------------------------------------|------------------------------------------------|-------|----------|
| 1    | $\hat{E}[Y_i D_i = 0, X_{1i} = 1, X_{2i} = 7]$  | 2                                              | 1     | 1        |
| 2    | 5                                               | $\hat{E}[Y_i D_i = 1, X_{1i} = 8, X_{2i} = 2]$ | 0     | 8        |
| 3    | $\hat{E}[Y_i D_i = 0, X_{1i} = 10, X_{2i} = 3]$ | 3                                              | 1     | 10       |
| 4    | $\hat{E}[Y_i D_i = 0, X_{1i} = 3, X_{2i} = 1]$  | 10                                             | 1     | 3        |
| 5    | $\hat{E}[Y_i D_i = 0, X_{1i} = 5, X_{2i} = 2]$  | 2                                              | 1     | 5        |
| 6    | 0                                               | $\hat{E}[Y_i D_i = 1, X_{1i} = 7, X_{2i} = 0]$ | 0     | 7        |

We want to construct an estimator of all of the missing  $\hat{E}[Y_i|D_i = d_i, X_{1i} = x_{1i}, X_{2i} = x_{2i}]$ . Matching estimators just pick the nearest neighbor, usually based on an estimated propensity score.

Let's attempt to fit the model with regression. Let's start by assuming that, at least to a first approximation,

$$E[Y_i|D_i, X_{1i}, X_{2i}] = \beta_0 + \beta_1 D_i + \beta_2 X_{1i} + \beta_3 X_{2i}.$$

Then we could fit this model with, e.g., OLS, and estimate the ATE with a simple imputation model.

When  $n$  is very small, we likely want to fit a simple model so as to reduce the amount of variability of our imputation estimates.



| Unit | $Y_i(0)$                                                                                 | $Y_i(1)$                                                                                | $D_i$ | $X_{1i}$ | $X_{2i}$ |
|------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------|----------|----------|
| 1    | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 1 + \hat{\beta}_3 \cdot 7$  | 2                                                                                       | 1     | 1        | 7        |
| 2    | 5                                                                                        | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 8 + \hat{\beta}_3 \cdot 2$ | 0     | 8        | 2        |
| 3    | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 10 + \hat{\beta}_3 \cdot 3$ | 3                                                                                       | 1     | 10       | 3        |
| 4    | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 3 + \hat{\beta}_3 \cdot 1$  | 10                                                                                      | 1     | 3        | 1        |
| 5    | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 5 + \hat{\beta}_3 \cdot 2$  | 2                                                                                       | 1     | 5        | 2        |
| 6    | 0                                                                                        | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 7 + \hat{\beta}_3 \cdot 0$ | 0     | 7        | 0        |

[Remember: these are expected values for units with the same covariate distribution, *not* actually filling in the missing potential outcomes.]

Then the empirical estimate  $\hat{E}[Y_i(1)] - \hat{E}[Y_i(0)]$  would be consistent if indeed

$$E[Y_i | D_i, X_{1i}, X_{2i}] = \beta_0 + \beta_1 D_i + \beta_2 X_{1i} + \beta_3 X_{2i}.$$

In the case of OLS without interactions between  $D_i$  and the covariates, this imputation estimate of  $E[\tau_i] = \beta_1$ .

| Unit | $Y_i(0)$                                                                                 | $Y_i(1)$                                                                                 | $D_i$ | $X_{1i}$ | $X_{2i}$ |
|------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------|----------|----------|
| 1    | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 1 + \hat{\beta}_3 \cdot 7$  | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 1 + \hat{\beta}_3 \cdot 7$  | 1     | 1        | 7        |
| 2    | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 8 + \hat{\beta}_3 \cdot 2$  | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 8 + \hat{\beta}_3 \cdot 2$  | 0     | 8        | 2        |
| 3    | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 10 + \hat{\beta}_3 \cdot 3$ | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 10 + \hat{\beta}_3 \cdot 3$ | 1     | 10       | 3        |
| 4    | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 3 + \hat{\beta}_3 \cdot 1$  | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 3 + \hat{\beta}_3 \cdot 1$  | 1     | 3        | 1        |
| 5    | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 5 + \hat{\beta}_3 \cdot 2$  | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 5 + \hat{\beta}_3 \cdot 2$  | 1     | 5        | 2        |
| 6    | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 7 + \hat{\beta}_3 \cdot 0$  | $\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 7 + \hat{\beta}_3 \cdot 0$  | 0     | 7        | 0        |

[Remember: these are expected values for units with the same covariate distribution, *not* actually filling in the missing potential outcomes.]

Then the empirical estimate  $\hat{E}[Y_i(1)] - \hat{E}[Y_i(0)]$  would be consistent if indeed

$$E[Y_i | D_i, X_{1i}, X_{2i}] = \beta_0 + \beta_1 D_i + \beta_2 X_{1i} + \beta_3 X_{2i}.$$

In the case of OLS without interactions between  $D_i$  and the covariates, this imputation estimate of  $E[\tau_i] = \beta_1$ .

You may ask, though, how can we relax the working assumption  $E[Y_i|D_i, X_{1i}, X_{2i}] = \beta_0 + \beta_1 D_i + \beta_2 X_{1i} + \beta_3 X_{2i}$ ?

Interactions, polynomials, splines, etc.

E.g., we could instead impose the following model:

$$E[Y_i|D_i, X_{1i}, X_{2i}] = \beta_0 + \beta_1 D_i + \beta_2 X_{1i} + \beta_3 X_{2i} + \beta_4 D_i X_{1i} + \beta_5 D_i X_{2i} \\ + \beta_6 X_{1i} X_{2i} + \beta_7 D_i X_{1i} X_{2i} + \beta_8 X_{1i}^2 + \beta_9 X_{2i}^2.$$

This less restrictive model is of course preferable with large  $n$  – the worst case scenario is that the extra estimated coefficients are unnecessary. But with small  $n$ , we might not want to estimate so many coefficients.

We could consider a “sieve” estimator, where the number of interactions and polynomials grows large as the  $n$  increases. There are a number of theoretical results on this, including results that the number of polynomial terms should be  $n^{1/9}$ .

When the CEF is continuous over the support of the treatment and covariates – a usually minor technical assumption – flexible regression allows us to approximate the CEF arbitrarily well given data constraints.

In general, to quickly produce estimates of the ATE from any of these nonparametric or semiparametric regression models, it is easiest to take predicted values for all observations given  $D_i = 1$  and  $D_i = 0$  and take the average over all observations. The logic of this is clear as a plug-in estimator.

$$E[Y_i(d)] = E_{\mathbf{x}_i} [E[Y_i | D_i = d, \mathbf{X}_i = \mathbf{x}_i]].$$

for which a natural plug-in estimator is the sample mean of predicted values

$$\hat{E}[Y_i(d)] = \frac{1}{n} \sum_{i=1}^n \hat{E}[Y_i | D_i = d, \mathbf{X}_i = \mathbf{x}_i].$$

It is also possible to combine regression models with weighting models. These approaches are known as “doubly robust” (DR). Assuming strong ignorability, these approaches yield consistent estimates when *either* the regression specification *or* the propensity score specification is correct.

Similar ideas have been proposed using matching – see Ho, Imai, King and Stuart (2007) on matching as “nonparametric preprocessing” (though in reality it is usually semiparametric). However, the statistical theory for the DR property has only been developed in the case of IPW estimation.

To formalize this idea a bit consider the following estimator of  $E[Y_i(1)]$ :

$$\hat{E}_{DR}[Y_i(1)] = \frac{1}{n} \sum_{i=1}^n \frac{D_i(Y_i - \hat{E}[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i])}{\hat{P}_i} + \frac{1}{n} \sum_{i=1}^n \hat{E}[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i].$$

This estimator combines the weighting estimator and the imputation estimator.



$$\hat{E}_{DR}[Y_i(1)] = \frac{1}{n} \sum_{i=1}^n \frac{D_i(Y_i - \hat{E}[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i])}{\hat{P}_i} + \frac{1}{n} \sum_{i=1}^n \hat{E}[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i].$$

One simple way to think about the DR estimator.

$\frac{1}{n} \sum_{i=1}^n \hat{E}[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i]$  is just the standard regression estimator.

$\frac{1}{n} \sum_{i=1}^n \frac{D_i(Y_i - \hat{E}[Y_i|D_i=1, \mathbf{X}_i=\mathbf{x}_i])}{\hat{P}_i}$  is an IPW estimator correcting for any “unusual” deviations of the actual data from the imputation estimate.

Let's give a proof sketch for the DR property, ignoring the requisite regularity conditions. I will show that the estimator is consistent even when one of the two (imputation or weighting) models is wrong.

⇒ Reminder! We're still assuming strong ignorability! ⇐

Suppose that the regression model is wrong, but the weighting model is right.

$$\hat{E}_{DR}[Y_i(1)] =$$

$$\frac{1}{n} \sum_{i=1}^n \frac{D_i(Y_i - \hat{E}[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i])}{\hat{P}_i} + \frac{1}{n} \sum_{i=1}^n \hat{E}[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i].$$

Can break apart the first expression.

$$\frac{1}{n} \sum_{i=1}^n \frac{D_i Y_i}{\hat{P}_i} - \frac{1}{n} \sum_{i=1}^n \frac{D_i(\hat{E}[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i])}{\hat{P}_i} + \frac{1}{n} \sum_{i=1}^n \hat{E}[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i]$$

Let's consider the limiting value of this expression assuming that  $\hat{P}_i \rightarrow P_i$ . Denote the limiting value of the (misspecified)  $\hat{E}[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i]$  as  $E^*[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i]$ .

$$\begin{aligned} & \hat{E}_{DR}[Y_i(1)] \\ & \xrightarrow{P} E \left[ \frac{D_i Y_i}{P_i} \right] - E \left[ \frac{D_i (E^*[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i])}{P_i} \right] + E [E^*[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i]] \\ & = E[Y_i(1)] - E[E^*[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i]] + E[E^*[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i]] \\ & = E[Y_i(1)], \end{aligned}$$

through invoking the Law of Iterated Expectations. In short, because the weighting model is right, the first term is just the (consistent) standard HT estimator and the second term is consistent for the (negative) expectation population distribution of the incorrect  $E^*[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i]$ .

Suppose that the weighting model is wrong, but the regression model is right now.

Let's consider the limiting value of this expression assuming that  $\hat{E}[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i] \rightarrow E[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i]$ . Denote the limiting value of the (misspecified)  $\hat{P}_i$  as  $P_i^*$ .

$$\hat{E}_{DR}[Y_i(1)]$$

$$\begin{aligned} & \xrightarrow{P} E \left[ \frac{D_i(Y_i - E[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i])}{P_i^*} \right] + E[E[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i]] \\ &= E \left[ \frac{D_i(E[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i] - E[Y_i|D_i = 1, \mathbf{X}_i = \mathbf{x}_i])}{P_i^*} \right] + E[Y_i(1)] \\ &= E \left[ \frac{D_i \cdot 0}{P_i^*} \right] + E[Y_i(1)] = E[Y_i(1)]. \end{aligned}$$

again by invoking the Law of Iterated Expectations. If the imputation model is right, the average deviation from the imputation model conditional on  $\mathbf{X}_i$  is zero, leaving us with just the imputation estimate.

Analogous results hold for  $E[Y_i(0)]$  – substituting  $D_i = 0$ ,  $(1 - D_i)$  and  $(1 - P_i)$  as appropriate. Thus, as the difference between two consistent estimators,  $E[\tau_i]$  can be estimated consistently when either estimator is wrong.

It is also straightforward to adapt the DR estimator to the missing data case. Just replace all of the  $D_i$ s with  $R_i$ s! You'll see that everything works out.

Inference for the DR estimators can be achieved via the bootstrap. Just make sure to include the estimation of the propensity scores and regression fits in your imputation process.

Estimation is easy. Identification is hard. No amount of fancy statistics can get around this.

Given a well-identified study, OLS is probably good enough. It's clear, it's hard to screw up, and it gives you sensible estimates.

If you want or need to go high-tech in your estimation approach, use a DR estimator with reasonably flexible specifications for the imputation and weighting models. These are arguably the best estimators around, and you'd be hard pressed to do better. (But it doesn't solve the identification problem!)

# Omitted Variable Bias

Suppose we believe that a two variable linear model is the true model in Washington (2008),

$$Y_i = \beta_0 + \beta_1 \textit{Girls}_i + \beta_2 \textit{Children}_i + \epsilon_i$$

but suppose that we omit the *Children* variable from the analysis

$$Y_i = \beta_0^* + \beta_1^* \textit{Girls}_i + \epsilon_i^*$$



If we write the model for children as the following:

$$\text{Children}_i = \gamma_0 + \gamma_1 \text{Girls}_i + \nu_i$$

then the two explanatory variable model can be re-written as the following:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 \text{Girls}_i + \beta_2 \text{Children}_i + \epsilon_i \\ &= \beta_0 + \beta_1 \text{Girls}_i + \beta_2 (\gamma_0 + \gamma_1 \text{Girls}_i + \nu_i) + \epsilon_i \\ &= \underbrace{\beta_0 + \beta_2 \gamma_0}_{\beta_0^*} + \underbrace{(\beta_1 + \beta_2 \gamma_1)}_{\beta_1^*} \text{Girls}_i + \underbrace{\epsilon_i + \beta_2 \nu_i}_{\epsilon_i^*} \end{aligned}$$

# Points of Potential Confusion

Is  $\beta_1^*$  meant to be causal?

$$Y_i = \beta_0^* + \beta_1^* \textit{Girls}_i + \epsilon_i^*$$

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Is  $\gamma_1$  meant to be causal?

$$\text{Children}_i = \gamma_0 + \gamma_1 \text{Girls}_i + \nu_i$$

If  $\gamma_1$  were meant to be causal would we want to include the omitted variable?

Could we ever go wrong by including a variable like AAUW score for previous MC in each district?

## Some Typical Econometric Advice

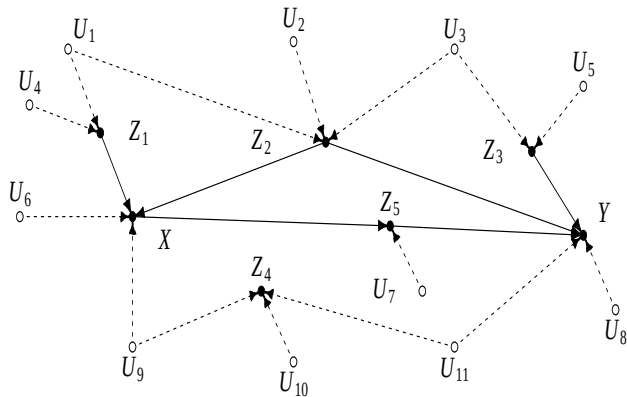
*In sum, if the variable we leave out is positively correlated with the variable we include and also has itself a positive effect on the dependent variable, then our misspecified model will produce an upward-biased estimator of the effect of  $X_1$  on  $Y$ . [Ashenfelter, Levine, and Zimmerman, p. 189]*

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*The short answer to the questions posed above is that the OLS estimates based on the equation containing too many explanatory variables will indeed be unbiased estimates of the true parameters. [Ashenfelter, Levine, and Zimmerman, p. 187]*

# Directed Acyclic Graphs

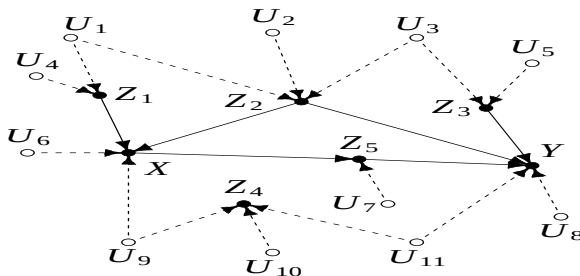




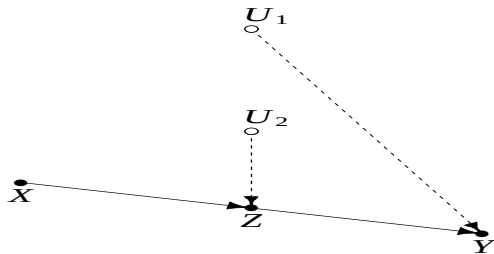
# Choosing Conditioning Variables

Definition (Back-Door Criterion (Pearl (2000, p. 79)))

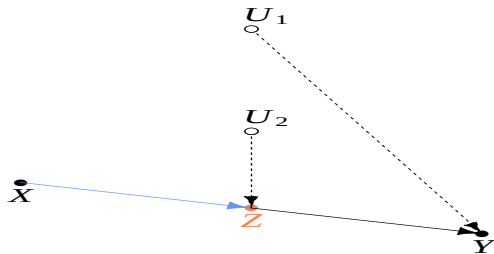
1. none of the conditioning variables is post- $X$ ; and
2.  $X$  is “ $d$ -separated” from  $Y$  by the conditioning variables in the graph formed by deleting all arrows out of  $X$ .



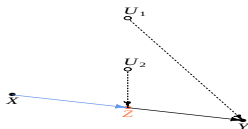
## BD1: Don't condition on post-treatment variables



## BD1: Don't condition on descendants of $X$



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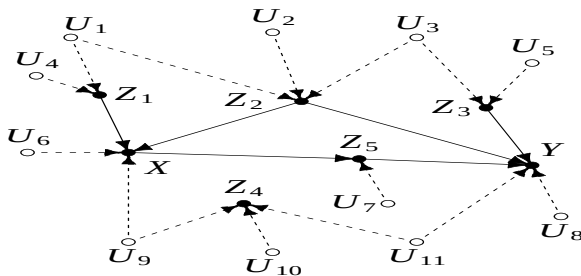
BD2:  $X$  is  $d$ -separated from  $Y$  by the conditioning variables in the graph formed by deleting all arrows out of  $X$ .

Definition ( $d$ -Separation (Pearl (2000, p. 16)))

1.  $X$  is  $d$ -Separated from  $Y$  if all paths are blocked
2. paths are blocked by “unconditioned colliders” or conditioned non-colliders

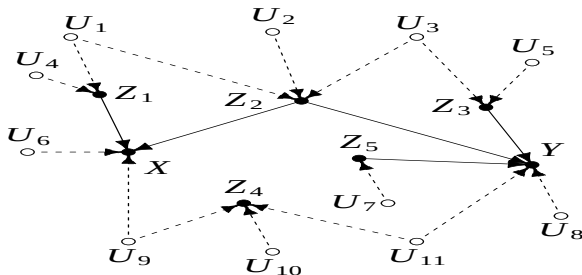
$d$ -Separation allows conditional independence relations to be read from a DAG consistent with a probabilistic model.

## Example: Sufficient Conditioning Sets



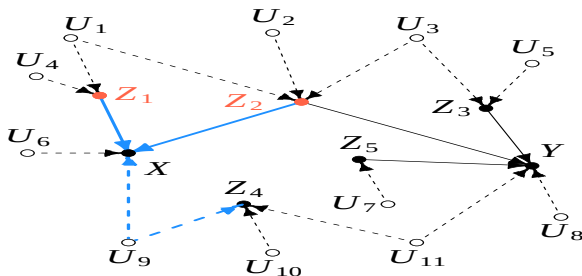
Remove arrows out of  $X$ .

## Example: Sufficient Conditioning Sets



Recall that paths are blocked by “unconditioned colliders” or conditioned non-colliders

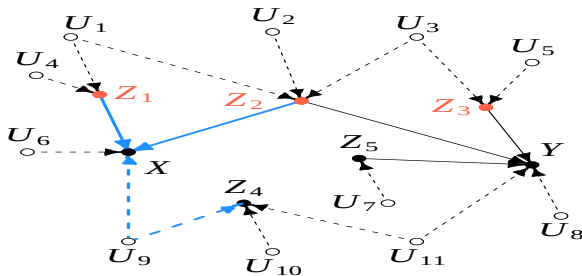
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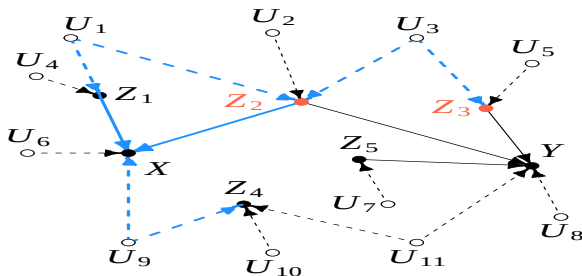


## Example: Sufficient Conditioning Sets



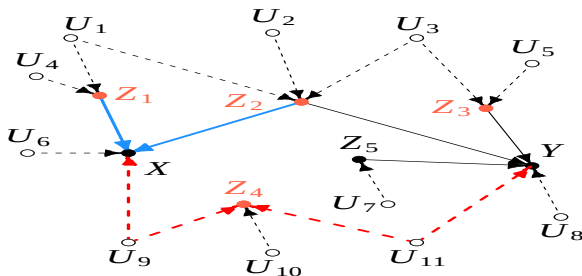
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## Example: Sufficient Conditioning Sets



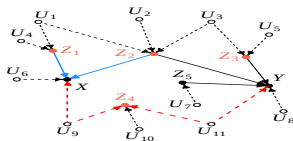
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## Example: Non-sufficient Conditioning Sets



*The short answer to the questions posed above is that the OLS estimates based on the equation containing too many explanatory variables will indeed be unbiased estimates of the true parameters. [Ashenfelter, Levine, and Zimmerman, p. 187]*

SWIGs on board

# Bias Amplification

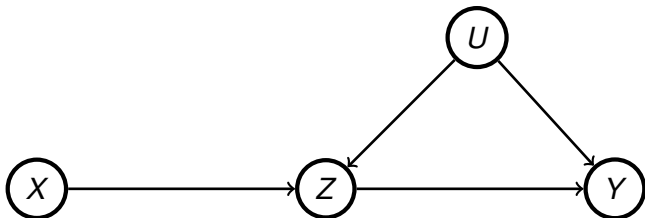
Bias amplification due to the addition of a control variable is subtly different than bias induced by including a variable that violates the backdoor criterion.

# Bias Amplification

Bias amplification due to the addition of a control variable is subtly different than bias induced by including a variable that violates the backdoor criterion.

Bias amplification occurs when there is already some confounding, and we add a control variable that does not fully fix that confounding.

## Sufficient Sets vs Bias Amplification





CAUTION: OLS is not so easy after all...

What causal parameter is OLS targeting under ignorability?

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Weighted average of CATE with weights based on treatment variance within blocks

What does this mean for us?

## CAUTION: OLS is not so easy after all...

What causal parameter is OLS targeting under ignorability?

Not the ATE...

Weighted average of CATE with weights based on treatment variance within blocks

What does this mean for us?

The parameter of interest is changing based on what variables are in the model!